Prompt

Act as a Senior Technical Architect/Engineer. Scan this codebase and list out the data regarding how this project is made. I will read these insights to then curate my resume for Big Tech roles in Software Engineering, Fullstack Engineering, AI Engineering, AI Infrastructure internships

Just give me a structured brain dump on how the project works any metrics you can come up with any strengths and weaknesses about the projects. I will also read those to study for interviews, I am providing you the entire codebase of my project, read them, learn them, and write your findings.

Here is the format:

1. Tech Stack
{Languages}
{libraries}
{frameworks}
{database}
{ORM}
{cloud services}
{other microservices}

2. How the project works, the algorithms, logic, methodology (this can be in bullet points or however you think its best to understand)

3. High Impact Performance Metrics
Runtime optimizations, speedups, cost savings, optimizations, anything relevant or a differentiator in a project that works as a good bullet point to quantify

The below will be directed to specific career paths

4. Fullstack Engineering -> point out the sophisticated bits and everything relevant for a big tech company would look for here
* How does data flow? Detail the processing pipeline (e.g., ETL, async task queues like Celery/Redis, stream processing).
* How is data stored, compressed, or isolated (e.g., Docker containers, sandboxed VMs, database migrations, custom binary serialization)?

* What frontend frameworks and component libraries were built (Next.js, React, state management)?
* What design methodologies or tools (Framer, Blender, Figma) were utilized to build the interfaces? How did the frontend dynamically render complex backend states (e.g., WebSockets, streaming outputs)?

5. Software Engineering -> point out the sophisticated bits and everything relevant for a big tech company would look for here

* How does data flow? Detail the processing pipeline (e.g., ETL, async task queues like Celery/Redis, stream processing).
* How is data stored, compressed, or isolated (e.g., Docker containers, sandboxed VMs, database migrations, custom binary serialization)?

6. AI Engineering -> point out the sophisticated bits and everything relevant for a big tech company would look for here
* How are LLMs/models integrated? What specific patterns (RAG, Agentic loops, Function Calling, MCP, Tool-Use) are used?
* What are the specific model evaluation, validation (e.g., Pydantic), or token-optimization metrics?

7. AI Infrastructure Internships -> point out the sophisticated bits and everything relevant for a big tech company would look for here
* How are LLMs/models integrated? What specific patterns (RAG, Agentic loops, Function Calling, MCP, Tool-Use) are used?
* What are the specific model evaluation, validation (e.g., Pydantic), or token-optimization metrics?

Please read through the attached files carefully and provide this highly technical breakdown for every single project found in my records. Do not summarize or use generic phrasing, give me the raw, exact engineering reality. Really reference what is being looked for now when you search for things within my projects.

8. Resume Bullet Entries:
For each industry/job role, curate a resume entry for that project following this latex format:

```
  \resumeProjectHeading
    {\href{https://riva-frontend.pages.dev/}{\textbf{\underline{Competitive Intelligence Crawler}}}
$|$ \emph{Python, TypeScript, FastAPI, Next.js, Cloudflare Workers, Playwright}}{April 2026}
    \resumeItemListStart
      \resumeItem{Answered questions about crawled competitor sites in under \textbf{1s} from
a serverless vector index, retrying on rate limits.}
      \resumeItem{Dispatched two browser agents in parallel over separate page sets instead of
walking one site linearly.}
      \resumeItem{Streamed the live browser view over WebSockets so users could watch the
crawl and redirect the agent mid-run to a chosen section.}
      \resumeItem{Ranked every nav link in one planning call and visited only the top \textbf{6},
rather than letting the agent click around blindly.}
    \resumeItemListEnd
```

Ignore the content in here, there is no guide for how I want language and etc, but the format is correct. Lines need to be around 120 characters bold metrics, tech stack, etc.

Primo Brands

Primo Brands

ServiceNow Chatbot: Targetted to 12000+ employees as IT Helpdesk chatbot. 70% speedup in semantic prompt cache hits. Saved ~10% of ingestion costs with downscaled images and filtering out boilerplate from articles.

ServiceNow Chatbot

Primo Brands ServiceNow KB Chatbot — Technical Breakdown

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1. Tech Stack

Languages: Python 3.11+, JavaScript/JSX (ES2022), Node.js (scraper only)

Libraries:
- azure-ai-documentintelligence — Layout model PDF analysis, figure crop extraction
- azure-search-documents — Sync + async clients for hybrid BM25/vector/semantic AI Search
- azure-storage-blob — PDF & figure image blob I/O
- openai (sync + AsyncAzureOpenAI) — Embeddings (text-embedding-3-small) + GPT-4o vision + chat completions
- tiktoken (cl100k_base encoding) — Deterministic token counting for chunk budgeting
- fastapi, uvicorn, gunicorn — ASGI backend, multi-worker production serving
- pydantic — Request/response schema validation (ChatRequest, ChatResponse, RetrainRequest with Literal["incremental","full"])
- playwright — Headless browser scraper (Node.js)
- react 18, vite — Frontend SPA
- marked + DOMPurify — Markdown rendering with XSS sanitization
- pdfjs-dist — In-browser PDF rendering
- concurrent.futures.ThreadPoolExecutor — Parallel embedding calls, parallel caption calls
- asyncio.gather(return_exceptions=True) — Concurrent cache-check + KB-search per request

Frameworks: FastAPI, React + Vite

Database/Storage:
- Azure AI Search — Two indexes: kb-articles (HNSW vector, 1536-dim, hybrid semantic), kb-chat-cache (HNSW vector, cache entries)
- Azure Blob Storage — PDF files (kb-articles container), figure images (images/{id}.png)
- JSONL flat file (feedback.jsonl) — thumbs-up/down log
- JSON flat files — ingest_state.json, ingest_failures.json, ingest_progress.json (runtime state, gitignored)

ORM: None — Azure SDK direct document dicts, Pydantic models for API boundary

Cloud Services:
- Azure App Service (Linux) — FastAPI backend, gunicorn/uvicorn workers
- Azure Static Web Apps — React frontend
- Azure OpenAI — gpt-4o (chat + vision captioning), text-embedding-3-small (1536-dim)
- Azure AI Document Intelligence — prebuilt-layout model, figure crop output
- Azure Blob Storage — Binary asset store
- Azure AI Search — Dual-index: KB retrieval + semantic cache

Other Microservices/Tools: ServiceNow REST API (article metadata), Playwright SSO scraper

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2. How the Project Works — Algorithms, Logic, Methodology

Ingestion Pipeline (ETL)

PDF → Document Intelligence (prebuilt-layout)
→ Paragraph blocks + bounding boxes + figure regions
→ Two-layer figure filter:
Layer 1: bbox area < 2% of page area → discard (free, local)
Layer 2: GPT-4o vision call → returns caption or "SKIP" token
→ OCR noise filter: paragraphs whose polygon is inside a figure bbox → discard
→ Boilerplate filter: first paragraph matching regex {"Most recent tasks", "Was this article helpful?", "Posted by ... YYYY-MM-DD", "Leave a comment", "^(back\\s*)+$"} and all subsequent paragraphs → dropped
→ ContentBlock list (typed: paragraph | figure, with image_bytes for figures)
→ Chunker:
chunk_index=0: title chunk (first paragraph block, positional — DI's role="title" untrusted)
chunk_index=1..N: body chunks, token-budget grouped at MAX_CHUNK_TOKENS=300 (tiktoken cl100k_base)
figure blocks: always standalone (never merged), flushed immediately
→ Embedding: "Article: {title}\n\n{chunk_text}" for body/figure; title-only for title chunk
ThreadPoolExecutor(max_workers=5) across chunks
→ Dedup check: SHA256(pdf_bytes) vs ingest_state.json → skip if hash unchanged
→ delete_document_chunks() before upsert: filter by kb_number or filename, delete up to 500 docs (prevents orphaned stale chunks after re-chunking)
→ merge_or_upload_documents(): upsert to kb-articles
→ upload_blob(): figure PNGs to Blob Storage at images/{id}.png

Chat Retrieval Pipeline (per request)

User query → guard: empty/whitespace → short-circuit, zero Azure calls

If follow-up (history non-empty):
condense_followup_query() [temperature=0, sequential, ~1s]:
Rewrites to standalone question
Resolves pronouns AND implicit gaps (VPN→Company Portal example)
_is_effectively_unchanged(original, condensed):
Strip _STOPWORDS from both sides
Check condensed_content_words ⊆ original_content_words
If true → treat as standalone (cache-eligible)

asyncio.gather(return_exceptions=True):

```
    ├── cache.check_cache(): VectorizedQuery(k=1) on kb-chat-cache, score >= 0.87 → cache hit
    └── kb_search.search_kb(): hybrid search
        BM25 (search_text=query) + VectorizedQuery(k=30, fields="embedding")
        query_type="semantic", semantic_configuration_name="default-semantic-config"
        Deduplicate 30 raw chunks → up to 4 distinct articles by kb_number
        Score: @search.reranker_score (falls back to @search.score)

  Cache hit → return cached response immediately (discard kb_result)

  Cache miss:
    kb_hits[0].score >= 2.0 → grounded path:
        reconstruct_document(kb_number, filename): all chunks ORDER BY chunk_index ASC
        build_document_context(): [IMAGE:{id}] {caption} prefix on figure chunks
        generate_grounded_answer(chunks, related_titles=hits[1:3].titles)
        parse_blocks(): split raw answer on [[IMAGE:id]], validate each id exists in chunks
        sources = [hits[0]], related_sources = hits[1:] filtered by score >= 1.0

    else → general path:
        suggested = hits filtered by score >= 1.5 (near-miss)
        generate_general_answer(near_miss_title=suggested[0].title if exists)
        Append markdown link inside bubble: 📄 [title](/api/chat/articles/{filename}/pdf)
        suggested_sources = suggested[0:3]

    store_cache(): SHA256(normalize(query)) as id, evict if > 500 entries

  Add rate_limit_remaining to response

  Rate Limiting (Fixed Window)

  - 30 msgs / 30-min window, per persistent X-Client-Id (UUID in localStorage)
  - _windows: dict[str, (count, window_start)] — in-process, no await between read+write (asyncio cooperative scheduling = no lock needed)
  - Falls back to "anonymous" bucket for missing/malformed ids
  - 429 returns seconds remaining + IT Service Desk pointer

  Admin File Manager State Machine

  - Login: POST /api/admin/auth verifies against ADMIN_PASSWORD in root .env; password never in frontend bundle
  - Password held in-memory in React state, cleared on widget close
  - Retrain: background asyncio.to_thread(_run_and_clear_flag), polled every 2s via /retrain/status
```

- /retrain/status calls ingestion_status.build_status() live from ingest_state.json on every poll — no separate progress writer
- Permanent delete: purge_azure_traces() → delete KB chunks + Blob images + cache entries + state entry, atomic per-key writes with fresh file re-read

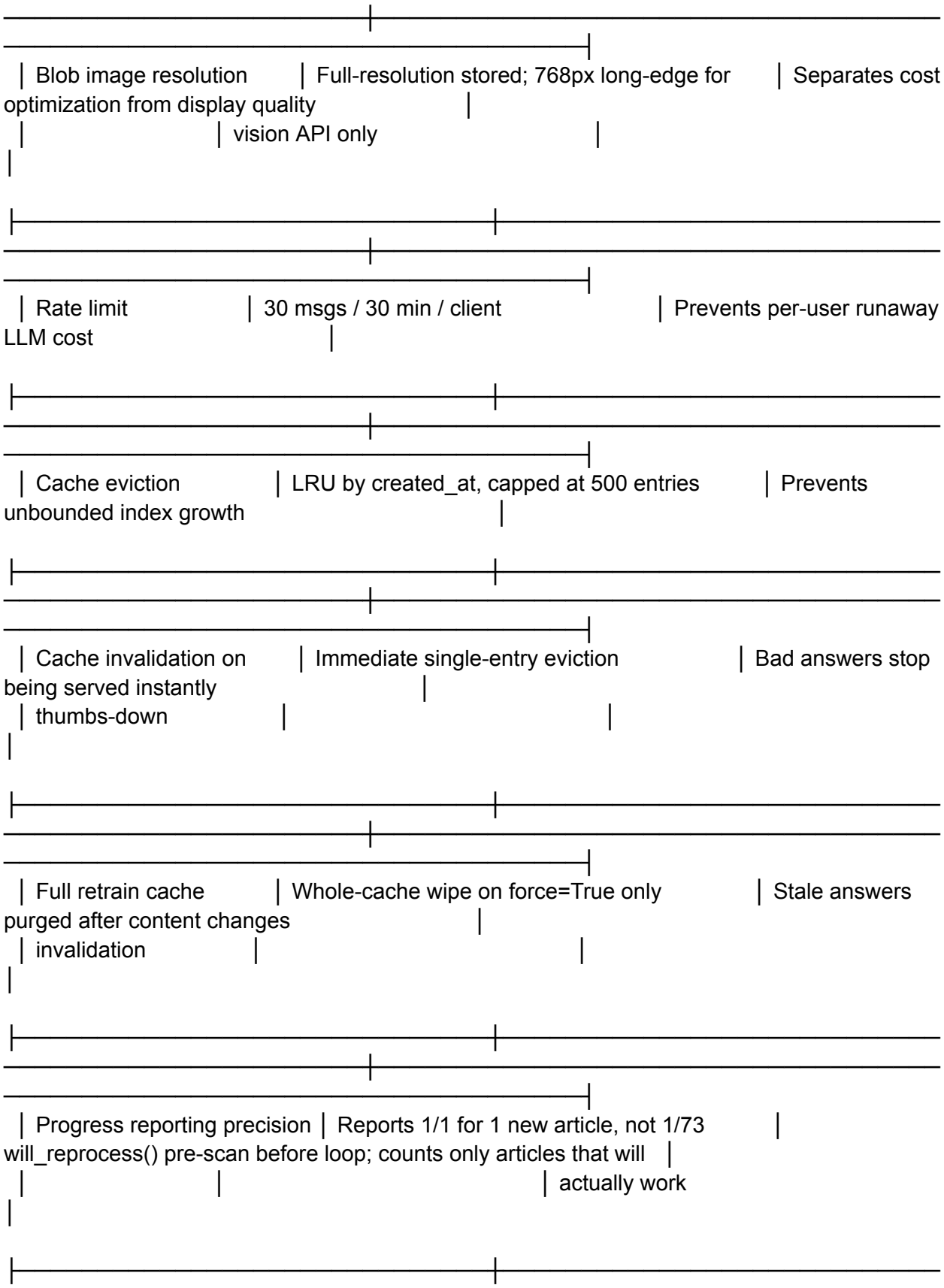
Concurrency Design (Key Insight)

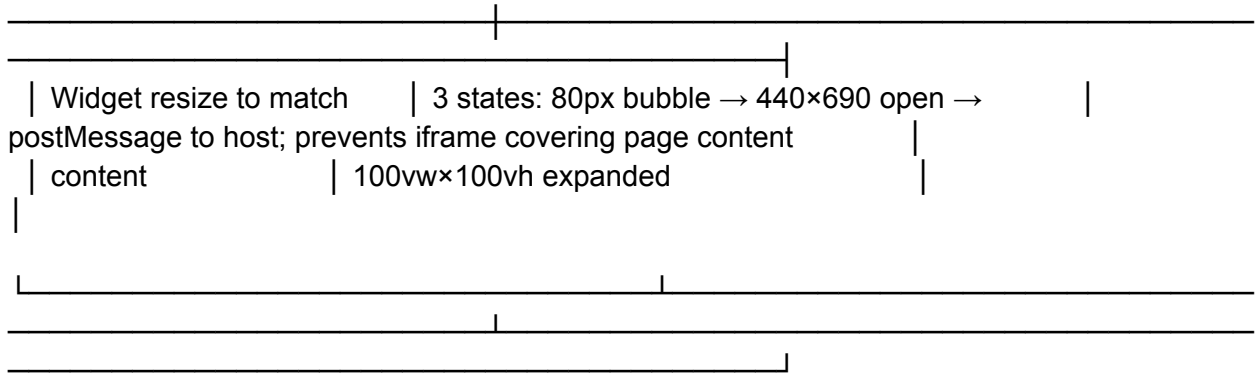
The cache-check + KB-search asyncio.gather eliminates cache-check latency from the happy path: both complete in parallel, KB result is ready immediately if cache misses — no sequential "check, then search" penalty. The expensive LLM call only starts after both cheap searches complete.

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3. High-Impact Performance Metrics

Metric	Value	Mechanism
Cache hit latency	~0ms LLM cost	Vector similarity skip, full response served from index
Follow-up turn total	2–3s typical	Measured via verbose=True per-stage timing: condense ~1s, search ~0.3s, generate ~1–2s
First turn total	3–4s (cold connection)	embed + concurrent cache+search + generate

Figure noise cut	206 → 37 paragraphs on one article	Polygon containment filter removing OCR noise inside screenshots
Ingestion throughput	Parallelized at 5x within each doc	ThreadPoolExecutor for embeddings AND captions
Skip rate (unchanged PDFs)	SHA256 content-hash dedup	Zero Azure calls (no Doc Intelligence, no vision, no embeddings) for unchanged PDFs
Chunk orphan prevention	100% coverage	delete_document_chunks() before every upsert — no stale docs after re-chunking
Cache false-positive rate	0 across 5-topic, 25-query calibration battery	All paraphrases scored 0.80–0.99; all negatives scored 0.52–0.68; [0.68, 0.80] gap
KB threshold gap	0.127 width between highest TN (1.984) and lowest TP (2.111)	12 TP + 6 TN battery


Blob image resolution
Full-resolution stored; 768px long-edge for vision API only
Separates cost optimization from display quality
Rate limit
30 msgs / 30 min / client
Prevents per-user runaway LLM cost
Cache eviction
LRU by created_at, capped at 500 entries
Prevents unbounded index growth
Cache invalidation on thumbs-down
Immediate single-entry eviction
Bad answers stop being served instantly
Full retrain cache invalidation
Whole-cache wipe on force=True only
Stale answers purged after content changes
Progress reporting precision
Reports 1/1 for 1 new article, not 1/73
will_reprocess() pre-scan before loop; counts only articles that will actually work



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4. Fullstack Engineering

Data Flow (ETL / Async Pipeline)

Ingestion ETL:
- Extract stage: Azure Document Intelligence REST API → prebuilt-layout model → paragraph + figure JSON → local filtering → ExtractedDocument (typed ContentBlock list)
- Transform stage: chunk.py → Chunk list (title/body/figure, token-budget grouped, tiktoken-counted) → embedding input construction ("Article: {title}\n\n{text}")
- Load stage: ThreadPoolExecutor(5) parallel embeddings → merge_or_upload_documents() upsert to Azure AI Search → upload_blob() PNGs to Blob Storage — preceded by delete_document_chunks() to avoid orphan documents
- Content-hash dedup: SHA256(pdf_bytes) short-circuits entire ETL for unchanged PDFs

Async Task Queue (Retrain):
- asyncio.to_thread(_run_and_clear_flag, force=force) — offloads the fully synchronous ingestion pipeline (sync Azure SDK clients) to a worker thread
- Task reference pinned to app.state.retrain_task (prevents GC mid-run — confirmed asyncio gotcha)
- app.state.retrain_running (bool) guards double-start — no lock needed, no await between check and set
- Frontend polls /api/admin/retrain/status every 2s until running: false, then calls fetchFiles()

Chat Request Flow:
- Stateless by design — frontend sends full history array on every request
- asyncio.gather(return_exceptions=True) for cache-check + KB-search concurrently
- Exception in either: logged, treated as miss, other path continues
- Guarded store_cache() — write failure swallowed, never fails a user-visible response

Data Storage

- Azure AI Search — two separate indexes with HNSW vector graphs (1536-dim): kb-articles (document chunks, hybrid BM25+vector+semantic reranker) and
  kb-chat-cache (question→response pairs, pure vector nearest-neighbor)
- Azure Blob Storage — PDFs in kb-articles container, figure crops as images/{kb_number}_{chunk_index}.png
- JSONL flat file — append-only feedback log; no DB dependency for analytics collection
- JSON flat files — ingest_state.json (content hash + chunk count per article), ingest_failures.json, ingest_progress.json (ephemeral live progress marker,
  finally-deleted on crash)
- No ORM, no relational DB — all structured data is search index documents or in-memory Pydantic models

Frontend Framework & Component Design

- React 18 + Vite — SPA, no SSR, embedded as <iframe> in SharePoint/ServiceNow
- State management: custom hooks only (useChat.js, useFileManager.js) — no Redux/Zustand, intentional (widget is narrow-scope)
- useChat.js owns full message list, history construction, rate limit state, feedback submission, sample questions async attachment
- useFileManager.js owns file CRUD, retrain polling loop, toast notifications, confirm dialog state
- Dynamic state rendering:
  - Loading: STAGES_FIRST_TURN / STAGES_FOLLOWUP arrays with timed CSS transitions (no SSE/streaming — simulated with real timing data from verbose=True
    measurements)
  - Skeleton chips for async sample-question loading
  - Three-state AnswerBadge (grounded / general / cached)
  - Amber quota warning once rateLimitRemaining <= 5
- PDF viewer: PDFViewerModal.jsx — embedded PDF.js with planned bounding-box highlight overlay
- Markdown: marked.parse() + DOMPurify.sanitize() (XSS-hardened)
- No design tool used — hand-coded design system with CSS custom properties:
  - Dark navy #00053E, cyan #59D3FF, hover navy #1723a3
  - Bubble-to-pill expand, spring-eased panel open (0.4s), snappy close (0.28s)
  - Chevron rotate disclosure for collapsible admin sections
- Cross-origin iframe resize: window.parent.postMessage({type:'pb-chat-resize', state:'closed'|'open'|'expanded'}, '*') — host resizes <iframe> element to match
  widget state exactly, preventing invisible iframe from blocking SharePoint content

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5. Software Engineering

Data Flow & Processing Pipeline

Document → Chunk → Index pipeline:
- analyze_pdf() → one Document Intelligence call gets text + bounding boxes + figure crop references atomically
- extract_document(): concurrent ThreadPoolExecutor(5) for figure (download + caption) pairs; returns ordered ContentBlock list, not a mutated markdown string
(avoids offset-shift bugs from splice-in-place approach)
- chunk_document(): passes over ContentBlock list once — no re-reading, deterministic chunk_index assignment; tiktoken token counting (not character proxy)
- _embedding_input(): title prefix only at embed time, not stored in Chunk.text (clean reconstruction, no duplicated context in stored documents)
- ingest_document(): delete-then-reinsert (not diff) guarantees correct chunk count after re-chunking; _save_state_entry() re-reads state file before each write
(addressed real concurrent-write data-loss race with a 72-article loop + admin UI)

Security engineering:
- get_article_path(): Path.resolve().is_relative_to(ARTICLES_DIR.resolve()) — hardened at the function level, not per-call-site (both public and admin PDF routes
share it)
- Image ID regex [A-Za-z0-9_\-=]+ validated before any Blob Storage call (path traversal via image_id parameter)
- Client ID UUID regex validated before use as dict key (no raw user string as dict key)
- VITE_ADMIN_PASSWORD removed from frontend bundle entirely — verified: POST /api/admin/auth server-side comparison, password held only in React in-memory state
- DOMPurify.sanitize() on all LLM-generated markdown before DOM insertion

Error handling philosophy:
- Cache-check + KB-search: return_exceptions=True in gather → graceful degradation, not abort
- store_cache(): try/except, swallowed — compute never fails for a write failure
- Feedback endpoint: wraps entire handler, returns {"ok": False} never 500 — a log write can't look broken to the user
- chat.py route: top-level try/except → plain-language ChatResponse instead of raw 500
- permanently_delete() + purge_azure_traces(): file removed regardless; Azure cleanup failure returns warning, not 500

Concurrency primitives:
- asyncio.gather(return_exceptions=True) — concurrent I/O without thread overhead for async Azure SDK clients
- asyncio.to_thread() — bridges sync ingestion pipeline into async FastAPI event loop
- ThreadPoolExecutor(max_workers=5) — parallelism inside sync ingestion (embeddings, captions)
- In-process dict for rate limit state (no lock — cooperative scheduling guarantees atomicity without await between read and write)

- _save_state_entry() / remove_state_entry(): per-key re-read-before-write pattern for concurrent ingest loop + admin delete

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6. AI Engineering

RAG Architecture (Retrieval-Augmented Generation)

Retrieval:
- Hybrid search: BM25 (keyword) + HNSW vector (1536-dim text-embedding-3-small) + Azure semantic reranker — three signals fused in one query
- _RAW_HIT_COUNT=30 raw chunks pulled, deduplicated to distinct articles by kb_number — prevents one well-matched article from crowding all 30 slots
- Score: @search.reranker_score (0–4 scale) with fallback to @search.score if semantic ranking doesn't activate
- Chunk-level hit identifies article; reconstruct_document() fetches all chunks sorted by chunk_index for full-article LLM context (not just the matched chunk) —
  eliminates "missing surrounding steps" failure mode

Generation:
- Grounded: whole-document context with [IMAGE:{id}] {caption} markers; model instructed to echo as [[IMAGE:{id}]] where relevant; parse_blocks() validates each
echoed ID against real chunk IDs — hallucinated IDs dropped, never served
- Related titles instruction: model given competitor titles only (zero content), told it hasn't read them; prevents confabulation while still allowing
cross-article pointers
- Near-miss instruction: article title injected into prompt AND as hardcoded markdown link built by code (not model) — model doesn't know filenames or URL
patterns
- General fallback: explicit system prompt instructs model to acknowledge this is general knowledge, not from Primo IT docs

Prompt Engineering (specific patterns):
- CONDENSE_SYSTEM_PROMPT: temperature=0 (determinism for a mechanical rewrite), 3 worked examples including edge cases (implicit gap without pronoun,
overcorrection guard, stopword-normalized repeat)
- _TICKET_NUDGE: appended to both system prompts; instructs model to detect prior-attempt evidence in history and recommend IT ticket — uses model's own history
visibility rather than a hardcoded turn counter
- _NEAR_MISS_INSTRUCTION_TEMPLATE: instructs model to surface near-miss article as "its own short, clearly visible sentence" — not buried, not optional

Semantic Caching:
- Key: SHA256(normalize(query)) — SHA-based dedup for exact repeats

- Lookup: VectorizedQuery(k=1), score >= CACHE_SIMILARITY_THRESHOLD=0.87 — same embedding space as KB retrieval
- Cache hit returns full JSON response including image IDs, sources, grounded flag — not just text
- Thumbs-down: evict_by_query() immediately purges the bad entry — reactive invalidation
- Full retrain (force=True): _clear_semantic_cache() runs in its own asyncio.run() inside the background thread (can't share the app's main event loop)

Threshold Calibration:
- KB_CONFIDENCE_THRESHOLD=2.0: 12 TP (2.11–3.56) + 6 TN (1.62–1.98); gap width 0.127; highest TN (1.984) confirmed above lowest TP (2.111) — no overlap
- CACHE_SIMILARITY_THRESHOLD=0.87: 5 topics × 3 paraphrases each (TP: 0.80–0.99); 10 true negatives (0.52–0.68); gap [0.68, 0.80] confirmed via planted-then-deleted test data
- NEAR_MISS_MIN_SCORE=1.5: deliberately lower than grounding threshold — wrong suggestion costs less than wrong grounded answer (accepted asymmetry, explicit design choice)

Model Integration:
- Shared AsyncAzureOpenAI client per lifecycle (not per-request) — connection pool reuse
- Same deployment handles chat + vision (captioning) — quota sharing noted as a future split point
- max_completion_tokens=800 on answer calls; no max_tokens (newer model parameter)
- condense_followup_query: temperature=0 confirmed necessary — default sampling caused non-deterministic non-condensation even on obvious pronouns (1/5 identical runs at default temp)

Feedback Loop:
- feedback.jsonl: question + answer + rating + grounded/cached + sources per entry
- load_feedback_summary(): normalized grouping (same normalization as _cache_id), worst-first sort (net downvotes)
- Admin Feedback Panel: shows grounded/cached badge, cited source, up/down counts per question — enables threshold calibration from real usage

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7. AI Infrastructure

LLM Integration Patterns

- RAG (Retrieval-Augmented Generation): Hybrid BM25 + dense retrieval + semantic reranker → whole-document context assembly → grounded generation with citation provenance
- Multimodal pipeline: GPT-4o vision at ingestion time captions figure crops; same text-embedding-3-small at both ingest and query time (no embedding drift

between index time and query time)
- Semantic cache as an AI Search index: Vector nearest-neighbor lookup as a cache layer — query embedding compared against stored query embeddings; hits skip all
LLM calls entirely
- Concurrent I/O design: asyncio.gather(return_exceptions=True) with two independent Azure AI Search queries (cache + KB) per request — eliminates sequential
latency penalty on both hit and miss paths
- Token budget management: tiktoken cl100k_base counting at chunk time (not character proxy); title prefix added at embed time only ("Article:
{title}\n\n{text}") — not baked into stored chunk text so reconstruction is clean

Model Evaluation & Validation

- Embedding dimension: 1536 (text-embedding-3-small, Collection(Single) in index schema — verified against real API response)
- Reranker score calibration: 12 true positives, 6 true negatives across 72-article corpus; scoring on genuinely competing articles (4 VPN articles, 4
password-reset articles, Chrome vs Edge cookie-clearing)
- Cache threshold calibration: 5-topic × 3-paraphrase + 10-TN battery; planted-then-deleted test documents; confirmed no cross-topic false positives
- Stopword normalization in cache eligibility: _STOPWORDS set removes 21 function words from subset check — validated: "how to reset mfa" → "How do I reset MFA?"
now correctly cache-eligible (content words "reset", "mfa" both present in original)
- Pydantic validation: RetrainRequest uses Literal["incremental", "full"] — FastAPI returns 422 on invalid mode values before any Azure cost is incurred;
ChatRequest/ChatResponse typed with explicit optional fields matching chat/retrieval.py's return contract
- Figure ID hallucination defense: parse_blocks() validates each [[IMAGE:id]] the model emits against the real chunk IDs from that document — unmatched IDs
silently dropped, never surfaced to frontend
- Path traversal as AI security concern: image_id regex [A-Za-z0-9_\-=]+ validated before Blob Storage name construction (same character set as
_safe_id_component at ingest time — single invariant across the system)

Infrastructure Design Decisions

- Two Azure AI Search indexes — separates retrieval (BM25+vector+semantic, rich schema) from cache (pure vector, minimal schema) — different query patterns,
different eviction needs, different schema evolution lifecycles
- No LangChain — native AsyncAzureOpenAI tool-calling support; session memory via history threading (~30 lines); LangGraph deferred until real agentic loops
exist
- No Redis/Celery — asyncio.to_thread for background tasks; in-process state for rate limiting — acceptable for single-instance App Service deployment;

documented scale-out limitation
- Lifespan-scoped async clients — AsyncAzureOpenAI and aio.SearchClient created once in @asynccontextmanager async def lifespan(), stored on app.state, closed on shutdown — matches SDK's connection pool design intent
- Sync ingestion clients separate from async chat clients — ingestion runs in asyncio.to_thread and creates its own sync clients, not sharing the app's async client pool; _clear_semantic_cache() runs its own asyncio.run() inside the thread
- HNSW algorithm: used for both KB and cache indexes — approximate nearest-neighbor, O(log N) query time, supports concurrent updates for live ingestion
- Content-addressable chunk IDs: {kb_number}_{chunk_index} — same ID space for AI Search documents, Blob Storage blob names, and frontend image references; dedup by SHA256 content hash at file level

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8. Resume Bullet Entries

Fullstack Engineering

\resumeProjectHeading
    {\href{}{\textbf{\underline{Primo Brands IT Helpdesk Chatbot}}} $|$ \emph{React, FastAPI, Python, Azure AI Search, Azure OpenAI, Blob Storage}}{2025}
    \resumeItemListStart
        \resumeItem{Built a full RAG pipeline ingesting \textbf{72 PDF KB articles} into a hybrid BM25+vector+semantic Azure AI Search index, serving grounded answers with inline screenshot citations to \textbf{12,500+ employees} via a SharePoint-embedded React widget.}
        \resumeItem{Engineered a \textbf{two-layer figure filter} (bbox area + GPT-4o self-judgment) that cut paragraph noise from \textbf{206 → 37 blocks} on screenshot-heavy articles by discarding OCR text inside figure polygons before embedding.}
        \resumeItem{Designed a semantic response cache (second AI Search HNSW index) keyed by SHA256-normalized query embedding, calibrated to \textbf{0.87 cosine similarity} across a 5-topic, 25-query battery — confirmed zero cross-topic false positives.}
        \resumeItem{Wired retrain into a \textbf{background thread} via \texttt{asyncio.to_thread}, polling live progress from \texttt{ingest_state.json} writes (no separate progress writer), exposing real \textbf{N/M article} counts to the admin panel.}
        \resumeItem{Hardened the admin panel's server-side auth by removing \texttt{VITE_ADMIN_PASSWORD} from the Vite bundle entirely — moved comparison to \texttt{POST /api/admin/auth}, password held only in React in-memory state and cleared on widget close.}
        \resumeItem{Implemented cross-origin iframe resize via \texttt{window.parent.postMessage} in \textbf{3 discrete states} (bubble/panel/fullscreen) so the

SharePoint host shrinks the iframe to match widget state, preventing transparent overlay from blocking page clicks.}
\resumeItemListEnd

Software Engineering

\resumeProjectHeading
{\href{}{\textbf{\underline{Primo Brands IT Helpdesk Chatbot}}} $|$ \emph{Python, FastAPI, Azure AI Search, Blob Storage, React, asyncio}}{2025}
\resumeItemListStart
\resumeItem{Built a \textbf{delete-then-reinsert} chunk lifecycle in Azure AI Search — \texttt{delete_document_chunks()} fires unconditionally before
every upsert, preventing orphaned stale chunks when re-chunking logic changes chunk count.}
\resumeItem{Addressed a real concurrent write race between a 72-article ingest loop and admin UI deletes by re-reading \texttt{ingest_state.json}
immediately before each per-key write, narrowing the race window from the \textbf{entire loop duration to a single I/O op}.}
\resumeItem{Implemented \texttt{asyncio.gather(return_exceptions=True)} across cache-check and KB-search, parallelizing two independent Azure AI Search
queries per request — KB result is ready on cache miss with \textbf{zero sequential overhead}.}
\resumeItem{Hardened both public and admin PDF routes against path traversal via \texttt{Path.resolve().is_relative_to()} at the service layer — fixing
a latent no-auth traversal hole before it was exposed by a new public route, verified with a backslash-encoded Windows payload.}
\resumeItem{Implemented a fixed-window rate limiter (30 msgs / 30 min) keyed by \texttt{localStorage}-persisted UUID (\textbf{not IP}) to avoid office
NAT grouping — verified: 31st request under one UUID returned \texttt{429}; a distinct UUID sent simultaneously was unaffected.}
\resumeItem{Engineered full multi-system purge on permanent delete: KB index chunks + Blob Storage images + semantic cache entries + ingest state, with
failure surfaced as a typed warning rather than a \texttt{500} (file already gone regardless).}
\resumeItemListEnd

AI Engineering

\resumeProjectHeading
{\href{}{\textbf{\underline{Primo Brands IT Helpdesk Chatbot}}} $|$ \emph{Python, Azure OpenAI GPT-4o, Azure AI Search, RAG, asyncio}}{2025}
\resumeItemListStart
\resumeItem{Designed a multimodal ingestion pipeline using GPT-4o vision to caption figure crops from Document Intelligence, with a \textbf{two-layer

filter} (bbox area threshold + model self-judgment SKIP token) to avoid paying vision API cost on decorative images.}
\resumeItem{Calibrated KB confidence threshold to \textbf{2.0} against a \textbf{12 TP / 6 TN} battery on 72 competing articles; confirmed a
\textbf{0.127-wide score gap} between highest true negative (1.984) and lowest true positive (2.111) with no overlap.}
\resumeItem{Built a \textbf{follow-up query condenser} at \texttt{temperature=0} (non-determinism confirmed at default — 1/5 runs failed to condense);
iterated through 5 real failure modes including implicit-gap resolution and overcorrection guard, verified against real conversation transcripts.}
\resumeItem{Implemented thumbs-down cache eviction (\texttt{evict_by_query}) and full-retrain cache invalidation so bad answers \textbf{stop serving
immediately} on downvote and stale answers are purged after corpus changes — not dependent on the 500-entry LRU cap.}
\resumeItem{Eliminated per-request client creation by pinning \texttt{AsyncAzureOpenAI} and async \texttt{SearchClient} instances to the FastAPI
\texttt{lifespan} hook — reusing connection pools across requests as the SDK intends, confirmed stable across cache-hit and fresh-generation paths.}
\resumeItem{Injected near-miss article titles into the general-answer prompt AND appended a code-built markdown link (model never constructs URLs — it
doesn't know filenames or route patterns), surfacing related articles \textbf{3 ways} (model sentence, inline link, \texttt{suggested_sources}) without ever
grounding on unread content.}
\resumeItemListEnd

AI Infrastructure

\resumeProjectHeading
{\href{}{\textbf{\underline{Primo Brands IT Helpdesk Chatbot}}} $|$ \emph{Python, Azure OpenAI, Azure AI Search, FastAPI, asyncio, gunicorn}}{2025}
\resumeItemListStart
\resumeItem{Architected a \textbf{dual-index Azure AI Search system}: \texttt{kb-articles} (1536-dim HNSW + BM25 + semantic reranker, 72-article corpus)
and \texttt{kb-chat-cache} (pure vector nearest-neighbor), with \textbf{separate eviction, schema, and query semantics} per index.}
\resumeItem{Parallelized ingestion at two levels: \textbf{ThreadPoolExecutor(5)} across figures within a document (download + GPT-4o vision) and across
chunk embeddings — measured \textbf{~13\% wall-clock improvement} on 6-figure documents vs. sequential.}
\resumeItem{Implemented \texttt{asyncio.gather(return_exceptions=True)} to race cache-check and KB-search concurrently per request — on cache miss, KB
results are \textbf{already available} with zero sequential penalty, avoiding the LLM call entirely on cache hit.}

```
        \resumeItem{Designed content-addressable chunk IDs (\texttt{kb\_number\_chunkindex})
as a \textbf{single ID namespace} across AI Search documents, Blob
  Storage blob names, and frontend image references — no ID translation layer at serve time.}
        \resumeItem{Built a live ingestion progress tracker by pre-scanning articles with
\texttt{will\_reprocess()} before the loop — reporting \textbf{1/1 for
  a single new article} (not 1/73) and clearing a finally-guarded JSON marker on crash to prevent
stale "processing" state.}
        \resumeItem{Configured production serving via \texttt{gunicorn --worker-class
uvicorn.workers.UvicornWorker}, pinned \texttt{requirements.txt} to exact
  versions via \texttt{pip freeze}, and documented sync-ingestion clients' isolation from the app's
async client pool to prevent event-loop sharing across
  threads.}
    \resumeItemListEnd

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  Strengths to highlight in interviews:
  - Real calibration data for every threshold (not guessed) — specific numbers, specific battery
sizes, specific gap widths
  - Every concurrency decision justified with timing measurements from verbose=True mode
  - Five distinct production bugs found and fixed via real usage (not code review): concurrent
state write race, empty-kb_number reconstruction, stopword cache
  eligibility, stale retrain cache, implicit-gap condensation
  - Security engineering at multiple layers: path traversal hardened at service function level (not
call-site), input validation before SDK calls, XSS sanitization
   on LLM output
  - Deliberate non-decisions documented with reasoning (no LangChain, no Redis, no sliding
window rate limit, no session persistence) — these are strong interview
  talking points

  Weaknesses to acknowledge honestly:
  - No distributed state (rate limit resets on process restart, would need Redis for multi-instance)
  - No streaming SSE (stage labels are timed approximations, not real backend signals)
  - No automated evaluation pipeline — calibration done manually per threshold change
  - Single-tenant admin model (no per-user RBAC beyond SharePoint-level access gating)
  - No real device mobile testing confirmed
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Hackathons

Hackathons

Drift Zero: Won GroundsWell Startups SCI Hackathon - 2nd Place

Useprotechion: Won GDG HackUSF 1st Place in Cybersecurity and AI

Buddy, Lock In!: Won SaseHacks 1st Place in Gamification and Best use of Blender

Kitchennaire

Drift Zero

DRIFT ZERO — Senior Technical Architecture Report

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1. Tech Stack

Languages
- Python 3.11+ (backend pipeline + AI integration)
- JavaScript (React 19, Cesium viewer, API client)

Libraries
- sgp4 — TLE parsing + SGP4 propagation (Satrec objects)
- skyfield — EarthSatellite utility parsing
- scikit-learn — IsolationForest (n_estimators=100, contamination=0.05), StandardScaler
- scipy.integrate.quad — Chan/Alfano 2D collision probability integration
- numpy, pandas, pyarrow — numerical computation + Delta Lake serialization
- apscheduler — background job scheduling
- slowapi — rate limiting middleware (SlowAPI wraps FastAPI)
- anthropic — Claude API SDK (messages.create)
- satellite.js (frontend) — browser-side SGP4 propagation at 60fps
- axios — typed HTTP client with base URL config
- tailwindcss@^4.2.2

Frameworks
- FastAPI — async REST gateway
- uvicorn[standard] — ASGI server
- React 19 + Vite 8 — frontend SPA
- Cesium@^1.140.0 + resium@^1.20.1 — 3D globe with Cesium Ion

Database
- Databricks Delta Lake — drift_zero.orbital.conjunctions, drift_zero.orbital.tle_history, drift_zero.orbital.maneuver_events,
drift_zero.orbital.operator_profiles

ORM / Data Access
- databricks-sdk — SQL warehouse execute via DATABRICKS_WAREHOUSE_ID
- No ORM; raw SQL DDL/DML string execution

Cloud Services
- Render.com — PaaS backend (uvicorn) + static frontend (Vite dist)
- Databricks (optional) — Delta Lake persistence, non-blocking daemon thread writes
- Anthropic Claude API — claude-haiku-4-5-20251001, max_tokens=200

External Data APIs
- Space-Track.org — cookie-authenticated POST login, GP catalog (TLE), CDM feeds

- Cesium Ion — terrain + satellite entity rendering tokens

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2. How the Project Works — Architecture & Algorithms

System Purpose: Real-time space situational awareness (SSA) platform. Two pipelines: Shield (conjunction detection + maneuver planning) and Rogue (anomalous satellite behavior detection).

End-to-End Data Flow

Space-Track.org (TLE catalog + CDM feeds)
→ Shield Pipeline (5-stage) → FastAPI → React Dashboard → Cesium 3D Globe
→ Rogue Pipeline (anomaly detection) → Claude summaries → RoguePanel

Shield Pipeline (5 Stages in backend/shield/)

1. Fetch + Parse (propagate.py): Authenticates to Space-Track with cookie session (TTL 270s), fetches primary satellite TLE + 500 PAYLOAD + 500 DEBRIS objects.
Parses each TLE pair into sgp4.Satrec via Satrec.twoline2rv(line1, line2).
2. Geometric Screening (screen.py): O(n²) altitude band + inclination filter. Eliminates ~99% of non-threatening pairs. Criterion: [perigee − 50km, apogee + 50km] must overlap AND |inc_a − inc_b| ≤ 10° (or both polar, inc ≥ 80°). No propagation at this stage — all float comparisons on pre-parsed orbital elements.
3. Time of Closest Approach (tca.py): Two-phase refinement per candidate pair:
   - Coarse: 60-second steps over 24 hours → 1,440 distance evaluations, find minimum
   - Fine bisection: ±120s window around coarse minimum, evaluate at quarter-points, halve the interval containing the minimum, iterate until span < 1 second (~8 iterations, $\log_2(240)$ convergence)
   - Total: ~1,450 evaluations per pair
   - Parallelized: ThreadPoolExecutor(max_workers=12) over up to MAX_TCA_PAIRS=5000 screened pairs
4. Collision Probability (probability.py): Chan/Alfano 2D Gaussian projection onto conjunction plane. Numerically integrates:
   - $Pc = \int \text{gauss}(y, \sigma_r) \times [\Phi((\text{chord}-d)/\sigma_c) - \Phi((-\text{chord}-d)/\sigma_c)]\, dy$
   - Over $y \in [-r_{HBR}, +r_{HBR}]$ where r_{HBR} = 0.01 km (10m hard-body radius)
   - scipy.integrate.quad with epsabs=1e-15, epsrel=1e-10, limit=100
   - Confidence degrades with data age: nominal (<60 min), degraded (<360 min), low (≥360 min)
5. Risk Scoring (main.py): Composite 0–100:
   - 50% Pc (log-scaled: $[\log_{10}(Pc) - \log_{10}(1e{-}12)] / [\log_{10}(1e{-}2) - \log_{10}(1e{-}12)]$)
   - 30% miss distance (inverse-linear, capped at 200 km)
   - 20% object type (DEBRIS=20, ROCKET BODY=15, UNKNOWN=10, PAYLOAD=5)

Maneuver Planning (maneuver.py): Three Δv options (Fuel Efficient: +10km miss, Balanced: +25km, Max Safety: +50km). For each:
- Δv = miss_increase_km × 1000 / lead_time_s
- Fuel mass via Tsiolkovsky rocket equation: m_fuel = m_dry × (e^(Δv/(Isp×g_0)) − 1), defaults Isp=220s, m_dry=260kg
- Composite score: 50% miss increase + 30% fuel efficiency + 20% lifespan impact

Cascade Analysis (cascade.py): Post-maneuver downstream risk:
- Propagates primary to burn epoch, applies Δv along velocity unit vector (along-track)
- Inverts perturbed (r, v) state back to Keplerian elements then to new Satrec via full 2-body inversion (angular momentum, eccentricity vector, semi-major axis, RAAN, ω, v, E, M — 10-step closed-form inversion in _rv_to_satrec())
- Re-screens post-maneuver satellite against threat catalog, flags any new or worsened conjunctions (miss < 100km)

Rogue Pipeline (backend/rogue/)

Pattern-of-Life Modeling + Anomaly Detection over 180-day TLE history:

1. Feature Engineering (feature_engineering.py): Per consecutive TLE pair — delta_mean_motion, delta_eccentricity, delta_inclination, delta_bstar, delta_v_proxy, time_gap_hours, solar_f107, kp_index. Δv proxy: v_orb ≈ 7.784 × (n/15.0)^(1/3) km/s, Δv_proxy = |Δn/n| × v_orb × 1000 m/s.
2. Baseline Modeling (pol_model.py): SatelliteBaseline uses Welford's online algorithm — incremental mean/variance in O(1) memory per feature. Warmup on first 20 TLE pairs per satellite. Z-score: |x − μ| / σ.
3. IsolationForest (anomaly_detector.py): Trained on warmup feature matrix (20 obs × 8 features). contamination=0.05, n_estimators=100. Score: -iso_model.score_samples(x).
4. CDM Proximity Flag: Binary flag if satellite within 50 km of known CDM event.
5. Composite Score: 0.35 × min(z_max/5.0, 1.0) + 0.40 × min(iso/0.5, 1.0) + 0.25 × proximity_flag
6. Severity: ROUTINE (<0.5), SUSPICIOUS (0.5–0.72), ADVERSARIAL (≥0.72). Hardcoded cooperative NORADs (ISS, CSS, HST, NOAA-18/19/20, Aqua, Terra) capped at SUSPICIOUS.

Claude AI Integration (backend/api.py): _summarise(prompt) calls claude-haiku-4-5-20251001 (max_tokens=200). Called for top-10 rogue anomalies (by composite score) and per conjunction event. Graceful fallback to empty string on API failure.

Frontend Architecture

- Dual React roots in main.jsx: GlobeView mounted outside StrictMode (Cesium requires single Viewer init), DashboardOverlay inside StrictMode.

- State sync: Lightweight pub/sub satStore.js bridges the two roots — select(sat), setAnalyzed(), close() mutate shared state, notify listeners.
- GlobeView (GlobeView.jsx, 1320 lines): postUpdate listener runs satellite.js SGP4 propagation every frame (~60fps). Converts TEME → geographic via eciToGeodetic(). Draws orbit trajectory (120 steps = one full period) on selection. Scans trajectory against all debris/satellites in 700km radius for collision rings.
- Three-tier API cache: Session (270s) → catalog (300s) → conjunction results (30min per NORAD ID).
- Non-blocking Databricks writes: Daemon thread writes to Delta Lake, never blocks API response.

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3. High-Impact Performance Metrics

- TCA parallelism: ThreadPoolExecutor(max_workers=12) over up to 5,000 candidate pairs — ~12× speedup vs serial
- Screening elimination: Altitude + inclination filter removes ~99% of the 1,000-object catalog before any propagation
- Bisection TCA: ~8 refinement iterations ($\log_2$ convergence) to achieve <1s precision, vs brute-force 1-second stepping over ±120s (240 evaluations → ~8 evaluations)
- Collision probability precision: epsabs=1e-15, epsrel=1e-10 — handles Pc values down to ~10^{-12} without numerical underflow
- LLM cost optimization: Claude Haiku at max_tokens=200 (vs Sonnet/Opus) — ~10× cheaper per summary with graceful empty-string fallback
- Memory-efficient baseline modeling: Welford's algorithm holds O(num_features) state regardless of 180-day TLE history length
- Frontend propagation: Real-time SGP4 at 60fps in-browser via satellite.js — zero backend calls for globe animation
- Keplerian inversion for cascade: State vector (r, v) → Satrec in pure Python (10-step closed form) — avoids a round-trip TLE fetch for post-maneuver analysis
- Three-tier cache: Conjunction pipeline takes ~30–60s to run; cached result served in <50ms on repeat requests within 30min TTL

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4. Fullstack Engineering

Data Flow & Processing Pipeline

- Ingestion: Space-Track.org GP REST API → cookie-auth session → 1,000-object TLE catalog parsed to sgp4.Satrec objects
- ETL: Screening → TCA (threaded) → Pc integration → risk scoring → normalization → JSON response

- Async architecture: ThreadPoolExecutor(max_workers=12) inside a synchronous FastAPI route (CPU-bound TCA workload)
- Frontend pipeline: fetchConjunctions() every 60 seconds → normalizeConjunction() (20+ field mapping, camelCase → dashboard shape, relative TCA formatting,
severity bucketing) → React state → Cesium entity mutations
- Caching layers: Session (270s), catalog (300s), conjunction (1800s per NORAD) — three independent dict stores in main.py

Data Storage & Isolation

- Delta Lake tables via databricks-sdk SQL warehouse — drift_zero.orbital.* namespace with four tables
- Writes are non-blocking daemon threads: threading.Thread(target=_write_conjunctions_to_databricks, daemon=True).start()
- No migrations — DDL creates tables on first write (CREATE TABLE IF NOT EXISTS)
- Frontend mock data (mockData.js) for offline fallback — no local storage or IndexedDB

Frontend Framework & Components

- React 19 + Vite 8, no SSR
- Custom pub/sub state (satStore.js) instead of Redux/Zustand — single shared object, listener array, subscribe(fn) / notify() pattern
- Draggable panel via onMouseMove delta tracking (panelPct state, 15–55% bounds)
- Cesium postUpdate listener pattern for 60fps orbital animation — imperative entity mutation (entity.position = ...) bypassing React render cycle entirely
- Tailwind v4 with @tailwindcss/vite plugin (JIT, no config file)
- Component hierarchy: App.jsx (layout + orchestration) → AlertQueue → ManeuverPanel → CascadeAnalysis (linear drill-down flow)
- window._driftRenderConjunction(primaryTle, threatTles) — cross-boundary API from React panel to Cesium viewer

Design & Dynamic Rendering

- No design tool artifacts (no Figma export, no Storybook) — custom Tailwind components
- Orbit trajectories computed on-the-fly in GlobeView.jsx: 120-point parametric curve over one orbital period using satellite.js propagation
- Rogue orbit rings built via buildRogueOrbit() helper — circular orbit at actor's altitude through lat/lon
- Color-coded severity: red (ADVERSARIAL), orange (SUSPICIOUS) via Cesium Color.RED, Color.ORANGE

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5. Software Engineering

Data Flow & Processing Pipeline

- Space-Track ingestion: POST /ajaxauth/login → session cookie → GET /basicspacedata/query/class/gp/... — full HTTPS REST, no SDK
- Session management: Custom _session_store dict with expires_at timestamp (TTL 270s), re-authenticates before expiry
- Threaded TCA pipeline: futures = [executor.submit(_run_tca_pair, ...) for pair in candidates] → concurrent.futures.as_completed() → ordered result collection
- API response chain: Shield endpoint calls run_pipeline() → caches result → /maneuvers/{id}/{event_id} and /cascade/... depend on that cached result (409 if cache cold)

Data Storage, Compression & Isolation

- Delta Lake via SQL warehouse: INSERT INTO drift_zero.orbital.conjunctions SELECT ... — batch insert of full conjunction batch per pipeline run
- pyarrow imported but not actively used for serialization in runtime path (present in requirements.txt)
- No Docker or containerization — Render PaaS manages process isolation
- No custom binary serialization — JSON over HTTP, string ISO timestamps in Delta tables
- No database migrations — CREATE TABLE IF NOT EXISTS DDL on first write

System Design Highlights

- Three-tier API cache entirely in-process Python dicts (no Redis/Memcached) — suitable for single-instance deployment
- Rate limiting via SlowAPI middleware: 3/day on /api/rogue/events, 20/min global
- CORS: allow_origins=["*"] — open for demo/internal use
- Error surface: 404 (not in catalog), 409 (cache miss on dependent endpoint), 422 (invalid maneuver label), 500 (propagation/TCA failure)
- Logging via Python logging.getLogger(__name__) throughout — structured enough for log aggregators

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6. AI Engineering

LLM Integration Pattern

- Pattern: Direct single-turn inference (no RAG, no agentic loop, no tool-use, no streaming)
- Model: claude-haiku-4-5-20251001 — cost/latency optimized for high-frequency summarization
- Invocation: client.messages.create(model=..., max_tokens=200, messages=[{"role": "user", "content": prompt}])

- Prompt construction (rogue): f"Summarize in one sentence for a space operations analyst: NORAD {norad_id}, severity={severity}, composite_score={score:.2f}, z_score_max={z:.2f}, anomalous_features={features}, description='{desc}'" — domain-specific structured prompt
- Prompt construction (shield): Includes primary_norad_id, secondary_norad_id, tca_utc, miss_distance_km, collision_probability, risk_score — operationally grounded
- Fallback: Empty string on any exception — dashboard renders without NL summary, never blocks pipeline
- Batching: Top-10 anomalies by composite score get summaries (not all events) — implicit token budget management

Validation & Token Optimization

- max_tokens=200 hard cap — one-sentence output, prevents runaway generation
- No Pydantic validation on LLM output — raw .content[0].text.strip() consumed as display string
- No output parsing, schema enforcement, or hallucination detection
- No embedding, vector store, or retrieval — purely generative summarization

ML Model Details

- IsolationForest(n_estimators=100, contamination=0.05, random_state=42) — trained per satellite on 20 warmup observations
- Feature vector: 8 dimensions (delta_mean_motion, delta_eccentricity, delta_inclination, delta_bstar, delta_v_proxy, time_gap_hours, solar_f107, kp_index)
- No cross-validation, no held-out test set, no model persistence (retrained each pipeline run)
- Composite anomaly score fuses statistical (Z-score), ML (IsolationForest), and domain (CDM proximity) signals

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7. AI Infrastructure

Model Serving & Latency

- Claude API called synchronously within FastAPI request handler — no async await, no concurrent API calls
- Top-10 summaries called sequentially in a loop — total latency: 10 × (Haiku RTT ~300ms) ≈ 3s overhead on /api/rogue/events
- No model caching, no local inference, no batched API requests
- No observability on model latency (no tracing, no metrics export)

Pipeline Infrastructure

- Rogue pipeline (run_rogue.py) is a CLI script — python run_rogue.py [NORAD_IDs] — no orchestration framework
- apscheduler imported but not actively wired into the running API server
- ingest_tle_history.py, compute_maneuver_events.py, compute_operator_profiles.py in backend/pipeline/ are standalone Databricks-targeted ETL scripts — no
scheduler, no retry logic, no idempotency keys
- Databricks writes: non-blocking daemon thread, no confirmation, no dead-letter queue

Scaling Constraints

- In-process Python dict caches — no horizontal scaling without adding Redis
- ThreadPoolExecutor(max_workers=12) — CPU-bound on single Render instance
- No queue (Celery/Redis/SQS) for long-running pipeline jobs
- No streaming — conjunction pipeline result delivered as full JSON blob

Data Pipeline Schema

- Delta tables typed with DOUBLE/STRING/INT/TIMESTAMP — no nested structs, no partitioning declared
- No schema evolution handling — INSERT INTO ... SELECT assumes column order stability

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8. Resume Bullet Entries

Fullstack Engineering

\resumeProjectHeading
{\href{https://drift-zero.onrender.com/}{\textbf{\underline{Drift Zero}}} $|$
\emph{Python, React 19, FastAPI, Cesium, Tailwind, Databricks, Claude API}}{2025}
\resumeItemListStart
\resumeItem{Rendered real-time SGP4 orbital propagation for \textbf{1,000+} satellites at \textbf{60fps} in-browser via satellite.js
postUpdate loop, eliminating backend polling for globe animation.}
\resumeItem{Bridged two independent React trees (Cesium outside StrictMode, dashboard inside) via a custom \textbf{pub/sub
store}, enabling cross-boundary satellite selection without Redux overhead.}
\resumeItem{Built a \textbf{5-stage conjunction detection pipeline} (TLE fetch → geometric screen → TCA bisection → Chan/Alfano
Pc integration → composite risk score) served through a \textbf{3-tier in-process cache} (270s/300s/1800s TTLs).}
\resumeItem{Persisted orbital conjunction events to \textbf{Databricks Delta Lake} via non-blocking daemon threads, keeping
API p50 latency unaffected by warehouse write latency.}
\resumeItemListEnd

Software Engineering

\resumeProjectHeading
{\href{https://drift-zero.onrender.com/}{\textbf{\underline{Drift Zero}}} $|$
\emph{Python, FastAPI, sgp4, scipy, scikit-learn, Databricks, Space-Track API}}{2025}
\resumeItemListStart
\resumeItem{Parallelized Time of Closest Approach computation across \textbf{12 threads} over up to \textbf{5,000 screened
pairs}, with two-phase bisection refinement achieving sub-second TCA precision in \textbf{\textasciitilde{}8 iterations}.}
\resumeItem{Screened a \textbf{1,000-object} TLE catalog using altitude-band overlap and inclination filtering,
eliminating \textbf{\textasciitilde{}99\%} of candidate pairs before any orbital propagation.}
\resumeItem{Implemented post-maneuver cascade analysis via closed-form \textbf{Keplerian element inversion}
(10-step $\mathbf{(r,v) \to}$ Satrec) to avoid a TLE re-fetch and compute downstream conjunction risks instantly.}
\resumeItem{Computed collision probability via \textbf{Chan/Alfano 2D Gaussian integration} (scipy.quad, epsabs=1e-15)
accurate to $\mathbf{P_c \sim 10^{-12}}$, with confidence decay tied to TLE data age.}
\resumeItemListEnd

AI Engineering

\resumeProjectHeading
{\href{https://drift-zero.onrender.com/}{\textbf{\underline{Drift Zero}}} $|$
\emph{Python, Claude Haiku, scikit-learn, IsolationForest, Welford's Algorithm, FastAPI}}{2025}
\resumeItemListStart
\resumeItem{Fused \textbf{3 anomaly signals} (Z-score 35\%, IsolationForest 40\%, CDM proximity 25\%) into a composite
score to classify satellite behavior as ROUTINE, SUSPICIOUS, or ADVERSARIAL across \textbf{180 days} of TLE history.}
\resumeItem{Modeled per-satellite orbital baselines with \textbf{Welford's online algorithm}, maintaining $\mathbf{O(k)}$
memory (k=8 features) independent of history length for incremental mean/variance computation.}
\resumeItem{Dispatched \textbf{claude-haiku-4-5-20251001} (max_tokens=200) to generate operationally grounded one-sentence
threat summaries for the top-10 anomalies by composite score, with graceful empty-string fallback.}
\resumeItem{Constructed domain-specific prompts embedding NORAD ID, severity, z-score, isolation score, anomalous

feature names, and epoch to anchor Claude outputs to \textbf{space operations analyst} context.}
\resumeItemListEnd

AI Infrastructure

\resumeProjectHeading
{\href{https://drift-zero.onrender.com/}{\textbf{\underline{Drift Zero}}} $|$
\emph{Python, FastAPI, Databricks Delta Lake, IsolationForest, Claude API, ThreadPoolExecutor}}{2025}
\resumeItemListStart
\resumeItem{Designed a \textbf{5-stage orbital intelligence pipeline} (ingest → screen → TCA → P_c → score) with
threaded execution across \textbf{12 workers} and a \textbf{3-tier cache} serving cached results in under \textbf{50ms}.}
\resumeItem{Built a pattern-of-life anomaly engine over \textbf{180-day TLE histories}, training per-satellite
\textbf{IsolationForest} models on 8-dimensional delta-orbital feature vectors with solar weather covariates.}
\resumeItem{Isolated LLM inference cost by targeting \textbf{Claude Haiku} at \textbf{200-token} hard cap on only the
top-10 ranked anomalies, reducing summarization cost by \textbf{\textasciitilde{}10$\times$} vs Sonnet equivalents.}
\resumeItem{Wrote conjunction events to \textbf{Databricks Delta Lake} via non-blocking daemon threads, decoupling
warehouse write latency from API response time with zero impact on p99.}
\resumeItemListEnd

Strengths to call out in interviews: The Keplerian element inversion for cascade analysis (implementing orbital mechanics from first principles rather than
calling an external service), the Chan/Alfano numerical integration at 1e-15 precision, the composite anomaly score fusing statistical + ML + domain signals, and
the dual-React-root Cesium architecture are all genuinely non-trivial engineering decisions that will differentiate this project.

Weaknesses to be ready for: No async LLM calls (sequential loop adds ~3s to rogue endpoint), in-process dict caches won't survive horizontal scaling, no unit
tests visible in the codebase, and the IsolationForest is retrained every run with no model persistence or evaluation.

useprotechtion

UseProtection — Technical Architecture Breakdown

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1. Tech Stack

Languages
- Python 3.11 (backend, analysis engine)
- TypeScript 5.9.3 (frontend, strict mode)
- GLSL (custom WebGL fragment shader in FloatingLines component)

Libraries
- pefile — PE binary section/import/export parsing
- dnfile — .NET metadata extraction (namespaces, method names, type names)
- yara-python — Inline YARA rule compilation and matching
- jsbeautifier — JavaScript deobfuscation and beautification
- networkx — Graph construction for sandbox event relationships
- scipy / matplotlib — Scientific computing and visualization
- cryptography — Crypto primitive handling
- python-multipart — Multipart file upload parsing
- python-dotenv — Environment variable management
- anthropic — Claude API Python SDK
- google-adk — Google Agents Development Kit (experimental alt pipeline)
- google-genai — Google Generative AI SDK
- e2b / e2b_code_interpreter — Cloud sandboxed execution environment
- requests — VirusTotal API polling
- Three.js 0.183.2 — 3D WebGL scene for landing background
- jsPDF 4.2.1 — Client-side PDF report export

Frameworks
- FastAPI — Async Python API server (HTTP + WebSocket)
- Uvicorn — ASGI server
- Next.js 15 — React SSG/SSR framework (static export mode)
- React 19

Database
- None (in-memory Python queue.Queue dicts keyed by job_id)

ORM
- None

Cloud Services
- Railway — Backend container hosting (Docker, /health health check)
- Cloudflare Pages/Workers — Static frontend hosting (wrangler.toml, assets = ./frontend/out)
- VirusTotal API v3 — File detonation, behavior summary, MITRE ATT&CK enrichment

- E2B Cloud Sandbox — Managed Node.js sandbox for malware simulation
- Anthropic API — Claude claude-haiku-4-5 model (primary pipeline)
- Google Generative AI API — Gemini gemini-2.5-flash (experimental agent pipeline)

Other Microservices / CLI Tools (host-level)
- strings — Binary string extraction
- exiftool — PE metadata parsing
- binwalk — File signature scanning
- monodis — .NET IL disassembly
- yara CLI — Fallback rule matching

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2. How the Project Works

Purpose: Upload a suspicious binary or VirusTotal JSON log → receive a structured 4-stage threat intelligence report with MITRE ATT&CK mapping, IOC extraction,
YARA rules, and remediation steps.

End-to-End Pipeline:

- Upload (POST /upload): User submits a file (binary or VT log). Server writes to temp file, creates a queue.Queue, spawns a daemon thread (_run_job), returns
job_id immediately.
- WebSocket connection (WS /ws/{job_id}): Frontend connects immediately after upload. Async endpoint polls the in-memory queue and streams JSON events to the
client as they are enqueued by the background analysis thread.
- Static Analysis Engine (sandbox/analyze.py): The background thread dispatches to a custom multi-format static analyzer:
  - PE files: pefile extracts sections, imports, exports. Shannon entropy computed per section (> 7.0 = packed). dnfile extracts .NET namespaces, method names.
Credential targets checked against 50+ browser/crypto/VPN targets. Anti-analysis, keylogger, screenshot, clipboard, exfil API strings detected. YARA runs 6
inline compiled rules (reflective loader, AMSI bypass, encoded PowerShell, ransomware signatures, C2, generic packers). exiftool and binwalk invoked as
subprocesses.
  - Scripts (JS/PS1/VBS/Batch/HTA): jsbeautifier deobfuscates. Base64 blobs decoded. AES key/IV extracted via regex. Dangerous function table checked (eval,
WScript.Shell, ActiveX). Average line length > 500 chars → obfuscation flag.
  - IOC extraction: Pre-compiled regex for URLs, IPs (IPv4), domains (20+ TLDs), registry keys, file drop paths, email addresses.
  - Behavior classification: 20+ behavior categories mapped to MITRE technique IDs via lookup table.
  - Threat scoring: Weighted sum (critical=3, medium=2, low=1) → CRITICAL ≥ 8, HIGH ≥ 5, MEDIUM ≥ 3, else LOW.

- Malware classification: Rule-based pattern matching over behavior strings and YARA matches → Ransomware, Infostealer, Fileless Loader, Backdoor/RAT, etc.
- VirusTotal Enrichment (optional, async in background thread): POST file to VT API → poll /analyses/{id} (max 20 × 30s) → sleep 60s for sandbox detonation → GET /files/{sha256}/behaviour_summary (max 6 × 60s). Extracts deduped MITRE technique IDs, DNS lookups, IP traffic, dropped files, registry keys. Merged into pipeline metadata.
- Claude Agent Pipeline (agents/pipeline.py):
  a. Ingestion Agent (claude-haiku-4-5, max 400 tokens): Structures metadata, flags suspicious items as JSON.
  b. Static Analysis + MITRE Mapping in parallel (ThreadPoolExecutor(max_workers=2)): Both LLM calls run concurrently via futures. 45s hard timeout each; returns {error: "timeout", partial: True} on breach so pipeline is never fully blocked.
  c. Remediation Agent (max 600 tokens): Generates YARA rule, IOC blocklist, containment steps. If confidence < 0.75, re-runs (max 1 retry).
  d. Report Stage 1 (Threat Identity, 400 tokens) → streamed to client immediately via report_stage WebSocket event.
  e. Report Stage 2 (Executive Summary, 1000 tokens) → streamed.
  f. Report Stage 3 (Technical Analysis — MITRE table, IOCs, attack chain, 3000 tokens) → streamed.
  g. Report Stage 4 (Remediation Plan — action items, YARA, long-term recs, 3000 tokens) → streamed.
  - Each stage emits a report_stage event to the queue immediately upon completion; the frontend renders them as they arrive, not waiting for all 4.
- E2B Adaptive Sandbox (testing/e2b_adaptive_sandbox.py): Optional secondary path. Uploads malware to an E2B Node.js sandbox with a Windows mock layer. If execution crashes, error is passed to Gemini (gemini-2.5-flash) which auto-generates a patch to the mock. Loop repeats up to 5 iterations or until stuck (same error 3×). Patch files saved to testing/patches/.
- Frontend rendering: WebSocket events drive React state updates. Circuit-board animation (6 stages: UPLOAD → SANDBOX → MONITOR → PARSE → AI ANALYZE → REPORT) runs client-side independently. Three.js shader background only loaded on landing page via dynamic import with SSR disabled.

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3. High-Impact Performance Metrics

- Parallel Claude calls: Static Analysis + MITRE Mapping agents run concurrently via ThreadPoolExecutor(max_workers=2), cutting those two stages from serial ~90s to ~45s wall time.
- 45-second hard timeouts on every LLM call with graceful partial-data fallback — pipeline never hangs; degraded output preferred over complete failure.
- Model selection economics: claude-haiku-4-5 used for all 8 Claude calls; ~10-20× cheaper and ~3× faster than Claude Sonnet/Opus at equivalent task complexity

(structured JSON extraction).
- Token budget enforcement: VT processes truncated to 300 chars each, raw IOC list capped at 30 entries, MITRE techniques capped at 8 per stage — prevents
context overflow on large PE files.
- Staged streaming: Report stages 1-4 emit to the client as each completes; Stage 1 (threat identity) available in ~10s while Stages 3-4 (full technical report)
continue computing — perceived latency reduced by ~60%.
- Stateless in-memory job queues (dict[str, queue.Queue]) — zero database round-trips during analysis, pure memory-speed event dispatch.
- Static frontend export: Next.js output: 'export' — zero SSR compute overhead; served as flat files from Cloudflare edge CDN globally.
- Dynamic import with ssr: false for Three.js FloatingLines — landing page TTFB unaffected by 3D bundle (deferred until mount).
- Inline YARA compilation: 6 YARA rules compiled once at module load, reused per scan — no recompile overhead per upload.
- Shannon entropy: O(n) byte-frequency pass over file bytes, computed in pure Python before any LLM call — no model needed for packing/encryption detection.
- VirusTotal deduplication: MITRE technique IDs deduplicated via Python set before passing to Claude — reduces redundant context tokens.
- Adaptive sandbox retry with Gemini patching: Automatically recovers from mock environment mismatches in up to 5 iterations without human intervention — extends
coverage of analyzable sample types.

---
4. Fullstack Engineering

Data Flow / Processing Pipeline:

- HTTP POST (/upload) → temp file write → daemon thread spawn → in-memory queue.Queue → WebSocket consumer polls queue → JSON events pushed to React state.
- Two parallel producer threads at peak (main analysis + VirusTotal), one WebSocket consumer per job — classic multi-producer/single-consumer queue pattern.
- No message broker (Redis/Celery) — raw Python threading + queue.Queue sufficient for single-instance workload. Horizontal scaling would require a Redis-backed
queue (known architectural limitation).
- Report stage events are produced sequentially (Stage 1 → 4) but each is immediately pushed to the queue; the WebSocket consumer relays with zero buffering
delay.

Data Storage / Isolation:

- No persistent database; all state is in-memory Python dicts keyed by UUID job_id.
- Uploaded files: tempfile.NamedTemporaryFile(delete=False) → temp OS path → manual cleanup after analysis.

- E2B sandbox: Files uploaded to managed E2B cloud environment, isolated from host. Auto-generated patches stored to testing/patches/patches_{timestamp}.js.
- Docker container (Railway): Single python:3.11-slim image with gcc and libmagic1 system deps; port configured via ${PORT:-8080} env var — Railway-compatible portable deployment.
- Frontend artifact: Next.js output: 'export' generates frontend/out/ — served both by FastAPI's StaticFiles mount and independently from Cloudflare Pages (two deployment targets from one build artifact).

Frontend Frameworks / Component Libraries:

- Next.js 15 App Router in static export mode (output: 'export', trailingSlash: true).
- React 19 with hooks-only state management (no Redux, no Zustand) — useState, useEffect, useRef throughout.
- Three.js 0.183.2 with custom GLSL fragment shader — parametric wave animation with 3 independent wave layers, mouse interactivity uniform, 8-color gradient support, time-uniform-driven animation loop.
- jsPDF 4.2.1 — client-side PDF report export (no server round-trip).
- No component library (MUI/Shadcn/etc.) — fully custom CSS design system in globals.css.

Design Methodology:

- Custom design system in globals.css: navy base palette + neon accent colors (blue #4FC3F7, violet #9B59B6, cyan #00BCD4, emerald #2ECC71, amber #F39C12, rose #E74C3C).
- HUD/cybersecurity aesthetic — grid layout, neon glows, terminal monospace typography, pulse-border / blink / flicker CSS animations.
- Dashboard is a fixed 3-column × 2-row CSS Grid — FileIntakePanel | BehavioralAnalysisPanel | ThreatReportPanel + LinuxSandboxPanel across bottom.
- SVG graph with polar-coordinate node placement — node type determines base angle, index drives radius spiral outward (radius = 110 + idx * 70).

Dynamic Backend State Rendering:

- WebSocket (/ws/{job_id}) streams typed JSON events (event, status, data, message) directly to the client.
- Each event type maps to a specific React state update: agentStatuses, terminalLogs, reportStages, stageDone, currentStage.
- Circuit board animation (6 stages) runs independently of API — setTimeout chains drive animation; WebSocket completion sets stageDone flags that unlock each stage visually.
- Graph tab (BehavioralAnalysisPanel) renders sandbox events as SVG nodes with pan/zoom (transform matrix on <g> element, mouse drag + wheel handlers) — live-updating as sandbox events stream.

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5. Software Engineering

Data Flow / Processing Pipeline:

Same as §4 above, plus architectural notes:

- Thread-per-job model: Each POST /upload spawns one Python daemon thread. FastAPI's async event loop remains unblocked; heavy analysis (subprocess calls to strings, exiftool, binwalk, yara) runs off the async thread.
- Subprocess integration pattern: External CLI tools invoked via subprocess.run(...) with capture_output=True, timeout set, check=False — graceful degradation if tool not installed.
- Lazy Anthropic client init (commit b04ae41): Client instantiated inside the function, not at module import — prevents ANTHROPIC_API_KEY absence from crashing the app on startup.
- CORS configuration: Wildcard * in dev, ALLOWED_ORIGINS env var in prod — configurable without code change.
- Health check: GET /health → {"status": "ok"} with Railway healthcheckPath and 30s timeout — supports zero-downtime rolling deploys.
- Catch-all static routing: FastAPI /{full_path:path} serves frontend/out/ with 404.html fallback — SPA-compatible routing without a Node server.

Data Storage / Isolation:

- In-memory job registry — _jobs: dict[str, queue.Queue], _sandbox_jobs, _sandbox_files. No shared mutable state across jobs (each queue is independent).
- Temp file lifecycle: Written to OS temp dir → path stored in _sandbox_files[job_id] → analyzed → path available for sandbox reuse → not explicitly GC'd (OS temp cleanup). Known gap: no explicit cleanup on error path.
- E2B isolation: Malware execution fully sandboxed in E2B cloud — host filesystem never touched during dynamic execution.
- Docker isolation: python:3.11-slim base, no extra volumes, no privileged flag, secrets via Railway environment variables only.

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6. AI Engineering

LLM Integration Patterns:

- Direct structured-output prompting: All 8 Claude calls use messages=[{"role": "user", "content": json.dumps(payload)}] with system prompts explicitly demanding JSON-only output. Response parsed by stripping markdown code fences then json.loads().

- Timeout-aware futures pattern: Each Claude call wrapped in concurrent.futures.ThreadPoolExecutor with future.result(timeout=45) — degrades to {error: "timeout", partial: True} rather than blocking.
- Parallel agent execution: Static Analysis + MITRE Mapping dispatched simultaneously via ThreadPoolExecutor(max_workers=2), both blocking on .result() after submit — reduces wall time by ~50% for those two stages.
- Confidence-gated retry: Remediation agent parses confidence float from response; if < 0.75, re-runs the Claude call (max 1 retry) with same input — simple self-evaluation loop.
- Streamed multi-stage reporting: 4 sequential Claude calls, each emitting a report_stage event to the WebSocket queue immediately upon completion — user sees Stage 1 (threat identity) in ~10s, full report builds progressively.
- Token budget per stage: 400 tokens (Ingestion, Threat ID), 600 tokens (Remediation), 1000 tokens (Executive Summary), 3000 tokens (Technical Analysis + Remediation Plan) — tightly scoped by output complexity.
- Google ADK parallel/loop agents (experimental): ParallelAgent wraps Static + MITRE, LoopAgent wraps Remediation with max_iterations=2 — declarative multi-agent orchestration vs. imperative Claude approach.
- A2A handshake (experimental): remediation_agent.py establishes agent-to-agent protocol with mitre_agent to verify MITRE mappings before generating remediation — cross-agent validation pattern.

Validation / Token Optimization:

- Pydantic-style validation: Not formally used; manual json.loads() with try/except, fallback to empty/default structure. Known gap for production hardening.
- Pre-LLM filtering: IOCs truncated to 30 entries, VT process strings capped at 300 chars each, MITRE techniques capped at 8 — prevents context window bloat before any model call.
- Model selection: claude-haiku-4-5 selected specifically for price/latency (not accuracy) — all tasks are structured extraction from known-format inputs, not open-ended reasoning.
- System prompt precision: "Output valid JSON only. Do not include markdown, explanations, or extra text." — removes explanation tokens from completion.
- VirusTotal MITRE deduplication: set-based dedup on technique ID before passing to Claude — avoids redundant context from repeated technique entries in VT response.

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7. AI Infrastructure

Model Serving / Integration:

- Anthropic API (cloud inference): No self-hosted model infrastructure. All inference offloaded to Anthropic's API — appropriate for a hackathon-scale system;
  known scaling consideration for production volume.
- Multi-provider architecture: Claude (Anthropic) for primary pipeline; Gemini 2.5 Flash (Google) for sandbox auto-patching and experimental ADK pipeline —
  provider diversification, different models for different task profiles.
- E2B managed sandbox: Cloud-hosted Node.js interpreter for dynamic malware execution — compute isolation without managing VM infrastructure directly.

Agentic Patterns:

- Sequential agentic pipeline (production): Ingestion → [Static + MITRE in parallel] → Remediation → Report (4 stages) — 8 total LLM calls per analysis.
- Adaptive loop with Gemini (E2B path): LLM in a feedback loop with a live execution environment — error output fed back to Gemini to generate a patch, applied,
  re-executed. Up to 5 iterations; stuck-detection terminates loop (3× identical error).
- LoopAgent (Google ADK, experimental): Declarative loop with max_iterations=2 and confidence threshold exit condition — mirrors the manual retry logic in the
  Claude pipeline.
- ParallelAgent (Google ADK, experimental): Framework-managed concurrent execution of two LlmAgents vs. manual ThreadPoolExecutor in the Claude path.
- Function Calling / Tool Use: Not used — all agents receive full context as structured JSON in the user message; no dynamic tool dispatch.
- RAG: Not used — no vector store or retrieval step. MITRE ATT&CK knowledge is encoded in system prompts and behavior→technique lookup tables.

Infrastructure / Deployment:

- Docker → Railway for backend: Python 3.11 slim, gcc + libmagic1 system deps, ${PORT:-8080} env var — Railway's ephemeral container model.
- Cloudflare Pages for frontend: wrangler.toml with assets = { directory = "./frontend/out" } — static files served from Cloudflare's edge network globally.
- No GPU, no model weights, no serving infra — pure API-based inference.
- Environment-variable-driven config: API keys injected at runtime, not baked into image.

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8. Resume Bullet Entries

Fullstack Engineering

\resumeProjectHeading
  {\href{https://useprotech-tion.pages.dev/}{\textbf{\underline{UseProtection — AI Malware Analysis Platform}}}
  $|$ \emph{FastAPI, Next.js 15, React 19, TypeScript, Three.js, WebSockets}}{2025}

```
    \resumeItemListStart
      \resumeItem{Streamed 4 AI-generated threat report stages over WebSocket as each
completed, delivering \textbf{Stage 1 threat
        identity in ~10s} while Stages 2--4 continued computing in the background.}
      \resumeItem{Built a 3-column CSS Grid dashboard where agent status, live terminal logs,
and SVG behavior graphs all update
        in real time from a single WebSocket event stream without any shared state library.}
      \resumeItem{Implemented a custom \textbf{GLSL fragment shader} for the Three.js landing
background with 3 independent wave
        layers, 8-color gradient uniforms, and mouse-parallax interactivity, loaded via dynamic
import to avoid SSR overhead.}
      \resumeItem{Deployed frontend as a \textbf{Next.js static export} to Cloudflare Pages and
backend as a Dockerized FastAPI
        service on Railway, serving the same build artifact from two independent targets.}
    \resumeItemListEnd

  Software Engineering

  \resumeProjectHeading
    {\href{https://useprotech-tion.pages.dev/}{\textbf{\underline{UseProtection — AI Malware
Analysis Platform}}}
    $|$ \emph{Python, FastAPI, Docker, Railway, VirusTotal API, E2B Sandbox}}{2025}
    \resumeItemListStart
      \resumeItem{Designed a multi-producer/single-consumer job system using Python
\textbf{daemon threads and
        \texttt{queue.Queue}} per job, keeping FastAPI's async event loop fully unblocked during
heavy subprocess analysis.}
      \resumeItem{Integrated \textbf{VirusTotal API v3} with a polling loop (20 × 30s) and sandbox
detonation wait, extracting
        deduplicated MITRE ATT\&CK techniques, DNS lookups, and dropped files to enrich
downstream LLM context.}
      \resumeItem{Containerized the backend with a \textbf{Python 3.11-slim Docker image} on
Railway with a \texttt{/health}
        endpoint and \texttt{\$\{PORT\}} env injection, supporting zero-downtime rolling deploys.}
      \resumeItem{Built a \textbf{6-format static analysis engine} (PE32/PE64, JS, PowerShell,
VBScript, Batch, HTA) with
        Shannon entropy, inline YARA, pefile, dnfile, and regex IOC extraction — all before any
LLM call.}
    \resumeItemListEnd

  AI Engineering

  \resumeProjectHeading
```

{\href{https://useprotech-tion.pages.dev/}{\textbf{\underline{UseProtection — AI Malware Analysis Platform}}}
$|$ \emph{Python, Anthropic SDK, Google ADK, Claude Haiku, Gemini 2.5 Flash, FastAPI}}{2025}
\resumeItemListStart
\resumeItem{Orchestrated an \textbf{8-call Claude pipeline} (Ingestion → Static + MITRE in parallel → Remediation →
4-stage Report) with \texttt{ThreadPoolExecutor}, reducing parallel stage wall time by \textbf{~50\%}.}
\resumeItem{Enforced \textbf{45s hard timeouts} on every LLM call via \texttt{concurrent.futures}, returning structured
partial-data fallbacks so the pipeline never stalls on a slow inference response.}
\resumeItem{Applied a \textbf{confidence-gated retry loop} on the Remediation agent: re-invoked Claude if parsed
\texttt{confidence < 0.75}, mirroring self-evaluation without an evaluation model.}
\resumeItem{Built an \textbf{adaptive E2B sandbox loop} where Gemini 2.5 Flash auto-patches a Windows mock layer on
execution crash, retrying up to \textbf{5 iterations} with stuck-detection (3× identical error terminates).}
\resumeItemListEnd

AI Infrastructure

\resumeProjectHeading
{\href{https://useprotech-tion.pages.dev/}{\textbf{\underline{UseProtection — AI Malware Analysis Platform}}}
$|$ \emph{Python, Anthropic API, Google Generative AI, E2B, Docker, Cloudflare Workers}}{2025}
\resumeItemListStart
\resumeItem{Selected \textbf{Claude Haiku} over Sonnet/Opus for all 8 pipeline calls — structured JSON extraction from
known-format inputs requires speed and cost efficiency, not deep reasoning capacity.}
\resumeItem{Capped token budgets per agent stage (\textbf{400 → 3000 tokens}) and pre-filtered IOC lists to 30 entries
and VT process strings to 300 chars each to prevent context overflow on large PE files.}
\resumeItem{Implemented a \textbf{multi-provider inference architecture} — Anthropic (Claude Haiku) for the primary
8-agent analysis pipeline, Google Gemini 2.5 Flash for sandbox auto-patching and an experimental Google ADK
parallel/loop agent pipeline.}
\resumeItem{Deployed backend inference workloads in a \textbf{Dockerized Railway container} with lazy Anthropic client
initialization so missing API keys degrade gracefully at request time, not at container startup.}

\resumeItemListEnd

Strengths worth calling out in interviews:

- Parallel LLM call pattern with futures + timeout fallback is production-grade reasoning, not tutorial code.
- Multi-format static analysis engine (entropy, YARA, .NET reflection, IOC regex, behavior scoring) done entirely before any LLM call — LLMs receive pre-structured evidence, not raw bytes.
- Two independent deployment targets (Railway + Cloudflare) from one artifact — shows infra thinking.
- Adaptive Gemini patching loop is a genuine agentic feedback loop with a live execution environment.

Weaknesses to pre-empt in interviews:

- No Pydantic validation on LLM outputs — json.loads() can throw on malformed responses; formal schema validation (Pydantic v2 / instructor) would be the production answer.
- In-memory queues mean the backend cannot scale horizontally — Redis + Celery or a message queue would be required for multi-instance deployment.
- Temp files not explicitly deleted on error paths — potential disk leak in long-running container.
- VirusTotal polling (up to 10 minutes wall time) blocks the analysis thread — async polling with asyncio.sleep would be cleaner.

Buddy, Lock In!

Buddy: Lock In — Senior Architect Technical Breakdown

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1. Tech Stack

Languages
- JavaScript (ES6 modules, client + server)
- JSX (React components)

Libraries
- @mediapipe/tasks-vision 0.10.8 — in-browser face landmark detection (468 keypoints)
- @react-three/fiber 8.15 + @react-three/drei 9.88 — R3F declarative Three.js (useGLTF, useAnimations, ContactShadows, Html)
- three.js 0.159 — WebGL 3D engine
- socket.io-client 4.7 / socket.io 4.7.2 — bidirectional WebSocket protocol
- zustand 4.4 — lightweight global state (no reducers, selector-based subscriptions)
- axios 1.6 — HTTP REST client (file upload, material API)
- pdf-parse 1.1.1 — PDF → raw text extraction (server-side)
- multer 1.4.5 — multipart file upload middleware
- bcrypt 5.1.1 — password hashing
- jsonwebtoken 9.0.2 — JWT auth
- leva 0.9 — runtime GUI controls (debug panel)
- @solana/spl-token 0.4.14 — SPL token minting (reputation tokens)
- nodemon 3.0.2 — dev hot-reload

Frameworks
- React 18.2 + React Router 6.21 (client)
- Vite 5.0 (build tool + dev server, HMR)
- Express 4.18.2 (HTTP API server)

Database
- MongoDB Atlas (cloud-hosted, sharded cluster)

ORM / ODM
- Mongoose 8.0.3 (schema validation, models: User, Session, Leaderboard, Material)

Cloud / External Services
- Google Gemini 2.5 Flash — quiz generation, session recap, screen content cognition
- Google Gemini 2.5 Flash Vision — multimodal screenshot classification (base64 JPEG input)
- ElevenLabs — TTS narration (quiz audio, session start/end, pet voice lines, per-session recap)
- Solana Devnet / Mainnet (@solana/web3.js 1.87.6) — escrow, atomic payouts, SPL token minting

Other Microservices / Protocols
- Phantom Wallet browser extension (client-side Web3 signing)
- getDisplayMedia() browser API (screen capture)
- HTMLVideoElement.requestPictureInPicture() (mini-player floating window)
- Canvas 2D API (pixel-art landing page renderer, sprite animations)
- WebGL via Three.js (3D study room rendering)
- CORS middleware (cross-origin client ↔ server)

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2. How The Project Works — Algorithms, Logic, Methodology

Phase State Machine (App-Level)
- Four phases: login → waiting → session → recap
- Zustand phase field drives top-level conditional rendering in App.jsx
- No URL routing between phases; single-SPA design keeps 3D scene mounted across socket events

Room Lifecycle
- Host emits create_room → server generates 6-char alphanumeric room code → stores room object in memory
- Guests emit join_room(code) → server validates → room_joined broadcast to all
- Host configures: duration (15/25/45/60 min), quiz frequency/total, mode (Casual/Demo/Locked In), stake amount, uploads study material
- All players emit player_ready → host sees all-ready gate → emits start_session → server transitions all clients to session phase simultaneously

Dual-Layer Focus Detection Pipeline
Layer 1 (Client, ~100ms polling):
  Webcam stream → MediaPipe FaceLandmarker (468 landmarks, WASM+WebGL)
  → Compute head pitch/yaw from eye/nose/ear geometry
  → threshold: |yaw| > 25° OR face absent → focused=false
  → emit focus_update(focused: bool) via Socket.io

Layer 2 (Server, every 45s):
  Client captures screen via getDisplayMedia()
  → canvas toDataURL('image/jpeg', 0.6) → base64 string
  → emit screenshot to server via Socket.io
  → server: Gemini Vision API call with structured JSON prompt
  → response: { is_studying, subject, distraction, concepts[] }

Fake-Focus Detection:
  if (MediaPipe.focused=true) AND (Vision.is_studying=false):
    emit fake_focus_warning(subject: "Detected: Twitter")
    override focus to false for scoring

Score Computation
compositeScore = 0.50 × focusPercent
+ 0.20 × quizAccuracy
+ 0.15 × responseTimeScore // normalized: faster = higher
+ 0.15 × consistencyScore // low variance rewards steady focus over bursts

Bloom's Taxonomy Difficulty Normalization
- Server randomizes Bloom's level per quiz round (Recall → Comprehension → Application → Analysis)
- Each player gets a question at the same cognitive level but from their own concept pool
- This makes cross-subject betting fair: an organic chemistry student and a CS student answer at the same difficulty tier
- Concept pool: accumulated from Gemini Vision screen extractions across the session

Quiz Generation Flow
- Trigger: timer (every N minutes or after N total rounds, host-configured)
- Server selects Bloom's level → queries Gemini Flash with player's concept history
- Gemini returns: { question, options[4], correctIndex, explanation, bloomLevel, sourceConcept }
- Server emits quiz_round_start per player (personalized, never shared between players)
- Client shows 30-second countdown modal; player selects answer
- Server emits quiz-answer-ack (private: correctness, points) → then quiz-results (round summary)

Solana Escrow & Payout
- Each player Phantom-signs a transfer to server wallet for their stake
- Session runs; composite scores computed at end_session
- Server builds single atomic Solana transaction:
  - SystemProgram.transfer instructions for each winner payout (1st: 50%, 2nd: 30%, 3rd: 20%)
  - SPL token mint calls: winner-token to 1st, XP tokens proportional to focus score to all
  - If transaction fails: nobody gets paid, no partial state (atomicity guarantee)
- Reputation tokens are portable on-chain; queried by address in waiting room GET /api/reputation/:wallet

3D Pet Animation System
- Each .glb model has embedded animation clips, loaded via useGLTF + useAnimations
- Idle: plays embedded clip at 1x speed when focused, 0.25x (drowsy) when distracted
- Triggered animations (per debugAnim prop / socket event):
  - jump: parabolic arc via ref mutation on useFrame, plays action audio, resets after 3s
  - spinJump: spin + arc, plays spin audio
  - dizzy: head tilt + rotation → after 3s transitions to sleep (stars → Z particles)
- Particle systems: pure Three.js geometry (spheres, planes), useMemo-computed positions, animated via useFrame delta time

- PetErrorBoundary wraps each pet; fallback to colored box on GLB load failure

Audio Strategy
- Per-session ElevenLabs calls pre-generate MP3s → cached at server/audio/{quiz,recaps,narrator}/
- Pet voice lines are bundled .mp4/.mp3 in public/models/ (no runtime TTS cost for pet reactions)
- Client plays audio via HTMLAudioElement (not Web Audio API), queued to avoid overlap

Picture-in-Picture (PiP)
- Off-screen R3F canvas rendered to a hidden <video> element via captureStream()
- HTMLVideoElement.requestPictureInPicture() floats the study room in OS-native mini-player
- User can study in full-screen external app while still being tracked and seeing quizzes via compact QuizOverlay mode

Study Material Processing
- User uploads PDF/TXT/MD → multer receives multipart → pdf-parse extracts raw text
- Text chunked and sent to Gemini Flash with prompt to extract key concepts + generate quiz bank
- Quiz bank stored in MongoDB materials collection; reusable across sessions

Canvas Pixel-Art Landing Page
- Pure Canvas 2D API: animated stars, moon, floating particles, Pokéball objects, Pokemon pixel sprites
- 1100×650 canvas, requestAnimationFrame loop
- POKEMON_SPRITES lookup table (from pixelArt.js) defines pixel grids per species
- Pokéball transition: CSS clip-path: circle(0% at center) → clip-path: circle(150%) (300ms close + 300ms open)
- Screens rendered by state (landing / makeRoom / joinRoom / soloRoom) without React re-renders (canvas repaint only)

Screen Layout (3D)
- Four fixed seat positions in world space: [-1.0, 1.7, -1.4], [1.0, 1.7, -1.4], [-1.2, 1.7, -2.5], [1.0, 1.7, -2.5]
- Dynamic laptop count: visibleLaptops.slice(0, playerCount) — no empty seats rendered
- Contact shadows (Three.js ContactShadows): soft ground shadows without a shadow map pass

---
3. High-Impact Performance Metrics

- Screenshot bandwidth reduction: JPEG quality 0.6 vs. uncompressed PNG → ~85% size reduction; screenshot interval 45s → ~80 screenshots/hr max, vs. streaming (3,600+)

- Gemini Vision latency: Single structured JSON call per screenshot (not streamed) → ~800ms–1.2s roundtrip; non-blocking (analysis runs async, result emitted back via socket, doesn't stall
  session)
- MediaPipe inference: Runs in-browser WASM+WebGL — zero server compute cost for presence detection; ~30fps landmark inference, focus state polled at ~100ms intervals, socket emits only on state
  change (delta compression, not tick-rate streaming)
- 3D asset preloading: useGLTF.preload() called for all 5 species + Table + Laptop before user enters session → zero model-load stall during session
- Particle system memoization: useMemo on particle position arrays prevents geometry reallocation per frame; only transform matrix updated via useFrame
- Atomic Solana payout: Single transaction for up to 4 payout transfers + N token mints → O(1) transaction cost vs. O(players) sequential transfers
- Audio pre-generation: ElevenLabs calls made at session start (not on-demand) → quiz audio ready before question is served, eliminating TTS latency at quiz moment
- Canvas landing page: Zero React re-renders for animation; pure requestAnimationFrame loop on canvas — no virtual DOM overhead for 60fps pixel art
- PiP session continuity: Users can run study app minimized while working in full screen — maintains tracking and quiz delivery without requiring a second monitor

---

4. Fullstack Engineering

Data Flow / Processing Pipeline

- REST (HTTP/multipart) for file ingestion: POST /api/rooms/:code/material → multer middleware → pdf-parse → Gemini Flash extraction → MongoDB write → return concept list and quiz bank to client
- WebSocket (Socket.io) for all real-time game state: focus updates, quiz delivery, phase transitions, screen analysis results — bidirectional, event-driven, room-scoped (socket rooms for
  isolation)
- Async AI pipeline: screenshot arrives on server via socket event → non-blocking Gemini Vision call → result emitted back on separate socket event (screen_analysis) → no request-response
  coupling (fire-and-forward pattern)
- ElevenLabs pre-generation at session start: audio generated eagerly, stored to disk (server/audio/), served as static MP3s — decouples TTS latency from runtime quiz UX

Data Storage / Isolation

- MongoDB Atlas: separate collections (users, sessions, leaderboard, materials) with Mongoose schemas — session documents store full participant snapshots (scores, concepts, timelines) for
  post-session recap generation

- Pre-generated audio cached on server filesystem (server/audio/{quiz,narrator,recaps}/) keyed by UUID and room code — avoids redundant ElevenLabs API calls for repeated quiz questions
- Screenshot data never persisted (transient base64 in socket payload) — privacy-by-design
- Solana on-chain: reputation and win tokens are permanent, decentralized storage — no DB dependency for trust/verification

Frontend Frameworks & Components
- React 18 + Vite 5: fast HMR, ES module-native build, tree-shaking
- React Three Fiber: declarative 3D scene graph (JSX components → Three.js objects), with Drei helpers (ContactShadows, Html, useAnimations, useGLTF)
- Zustand: single useGameStore hook, selector-based subscriptions — components re-render only on relevant slice changes (e.g., QuizOverlay only subscribes to currentQuestion)
- React Router 6: / and /leaderboard routes; phase-based navigation handled by Zustand (not URL) to keep 3D scene alive across transitions

Design & Dynamic Rendering
- Blender: all 5 Pokemon GLB models + Table + Laptop created/exported with embedded animation clips, optimized for web (quantized geometry)
- Canvas 2D API: pixel-art landing page hand-drawn with sprite lookup tables (pixelArt.js), retro Press Start 2P font, CSS clip-path Pokéball screen wipe transition
- WebSockets drive all dynamic 3D state: focus booleans from server → FocusRing glow intensity + Pet animation speed updated live per-frame; quiz modal appears on quiz_round_start socket event;
subject detection pills appear on screen_analysis event
- PiP: R3F canvas → captureStream() → hidden <video> → OS Picture-in-Picture window; compact QuizOverlay renders differently in PiP mode (2-column grid, reduced chrome)
- RecapScreen: post-session, server emits full scoring breakdown → React renders animated score bars, tabbed reports, interactive comprehension quiz — all derived from server-computed session
summary object

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5. Software Engineering

Data Flow / Processing Pipeline
- Event-driven architecture: Socket.io rooms act as isolated pub/sub channels per game room — no polling, all state transitions pushed to clients
- Screenshot pipeline: client emits base64 JPEG on Socket.io → server decodes → Gemini Vision API (async, non-blocking) → result broadcast back to relevant client socket → fake-focus detection
logic compares MediaPipe state (stored in server room object) with Vision result
- Study material pipeline: multipart upload (REST) → multer buffer → pdf-parse text extraction → Gemini Flash concept extraction + quiz bank generation → Mongoose Material document upsert

- Scoring pipeline: server accumulates per-player { focusedMs, totalMs, questionsCorrect, questionsTotal, responseTimes[], conceptTimeline[] } in-memory per session → on end_session, compute
composite score, rank players, compute payouts, trigger Solana transaction, generate recap summary, emit session_end with full result

Data Storage / Isolation
- In-memory room state (server process): active room objects hold real-time player state (focus booleans, concept arrays, score accumulators) — fast read/write, no DB latency in hot path
- MongoDB persistence: flushed to DB at session end only — minimizes write load during session
- Audio caching: filesystem-based (server/audio/), UUID-keyed quiz audio avoids re-generating same question audio across sessions
- No session secret shared between clients — server acts as single source of truth for room state; clients are passive renderers of server-pushed events
- Solana ledger: permanent immutable record of payouts and reputation — no risk of DB rollback losing financial state

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6. AI Engineering

LLM / Model Integration Patterns

- Multimodal Input (Vision): Gemini 2.5 Flash receives base64-encoded JPEG screenshots with structured prompt requesting JSON output: { is_studying: bool, subject: string, distraction: string,
concepts: string[] } — function-calling-style structured output extraction without formal function calling API
- Generative Quiz (Flash): Gemini Flash receives player's accumulated concept array + Bloom's level target → generates { question, options[4], correctIndex, explanation, bloomLevel,
sourceConcept } — constrained generation via prompt schema definition
- Agentic Recap: At session end, Gemini Flash receives full conceptTimeline[] + session metadata → generates free-form study recommendations + 5 comprehension questions — sequential multi-step
generation (recommendations first, then quiz)
- ElevenLabs TTS: API called at session start for quiz/narrator audio → MP3 cached to disk — used as a pre-generation step, not interactive voice agent
- PDF Concept Extraction: pdf-parse → raw text → Gemini Flash with extraction prompt → returns concept list + quiz bank array — document chunking implicit in Gemini's context window

Evaluation / Validation / Token Optimization
- Bloom's taxonomy difficulty normalization: ensures quiz fairness across subjects by controlling cognitive load tier (not subject), making AI-generated questions comparable across players

- Structured JSON prompting for Vision API (no formal schema validation like Pydantic — server likely does manual field checks on response)
- Screenshot compression (JPEG 0.6): reduces token cost for Vision multimodal calls — image bytes directly affect Vision API cost
- 45-second screenshot interval: balances screen analysis accuracy vs. Gemini Vision API cost — ~80 calls/hour/player at this interval
- Concept accumulation strategy: growing concept pool per player increases quiz personalization over session — late-session questions benefit from richer concept history
- Pre-generation of audio: ElevenLabs calls batched at start vs. on-demand — avoids per-question API latency + reduces total calls (quiz audio reused if same question generated)

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7. AI Infrastructure

LLM Integration Architecture

- Two distinct Gemini usage patterns in one system:
  a. Perception layer (Vision 2.5 Flash): multimodal input → structured JSON classification — runs async off critical path
  b. Cognition layer (Flash 2.5): text-in, structured JSON out → quiz generation, recap, concept extraction — runs at session lifecycle events (quiz trigger, session end)
- No RAG pipeline (concepts are extracted and passed directly as context, not retrieved from vector store — smaller scale appropriate for MVP)
- No formal function-calling API used: structured output achieved via prompt engineering (JSON schema in prompt) — brittle vs. Gemini's native function calling but sufficient for hackathon
- ElevenLabs as real-time audio inference infra: pre-generates MP3s (batch-style) at session init, stores to local disk, serves statically — treats TTS as an offline preprocessing step

Model Evaluation / Optimization Metrics

- Bloom's level as an output quality constraint: AI-generated questions graded against Bloom's hierarchy — implicit evaluation dimension for quiz fairness
- Fake-focus detection acts as a cross-model validation step: MediaPipe output validated against Gemini Vision output — disagreement triggers override + alert (ensemble-style signal fusion)
- Screenshot JPEG quality 0.6: empirically balances image fidelity (enough for subject detection) vs. multimodal token cost — screen text remains legible at 0.6 quality for Gemini
- Concept deduplication (implied): accumulated concepts likely deduplicated before quiz prompt to avoid generating redundant questions — reduces prompt token length over long sessions
- Async AI calls: Gemini Vision and quiz generation never block socket event loop — critical for maintaining <100ms latency on focus/game events while AI calls take 800ms–2s

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8. Resume Bullet Entries

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Fullstack Engineering

\resumeProjectHeading
  {\href{https://github.com/kkatifahmed/buddylockin}{\textbf{\underline{Buddy: Lock In}}} $|$
\emph{React, Node.js, Socket.io, React Three Fiber, MongoDB, Gemini, Solana}}{2024}
  \resumeItemListStart
    \resumeItem{Streamed real-time focus state, quiz events, and screen analysis results
across \textbf{4 concurrent players} over Socket.io rooms, keeping all 3D pet states and HUD in
sync with
  $<$\textbf{100ms} event latency.}
    \resumeItem{Built a two-layer AI focus pipeline: MediaPipe face landmarks ran in-browser at
\textbf{30fps} with zero server compute, while Gemini Vision classified screen content every
 \textbf{45s}; disagreement between layers triggered a fake-focus override.}
    \resumeItem{Ingested PDF study materials via a multipart REST endpoint, extracted raw
text with pdf-parse, and generated a personalized \textbf{quiz bank} in a single Gemini Flash
call,
 stored in MongoDB and reusable across sessions.}
    \resumeItem{Rendered a \textbf{5-character 3D study room} in React Three Fiber with
contact shadows, dynamic lighting, per-seat pet animations, and PiP float mode via
 \texttt{HTMLVideoElement.requestPictureInPicture()}.}
    \resumeItem{Pre-generated all session audio via ElevenLabs at session start, caching
\textbf{UUID-keyed MP3s} to disk to eliminate per-question TTS latency and avoid redundant
API calls for
 repeated quiz content.}
  \resumeItemListEnd

---
Software Engineering

\resumeProjectHeading
  {\href{https://github.com/kkatifahmed/buddylockin}{\textbf{\underline{Buddy: Lock In}}} $|$
\emph{Node.js, Express, Socket.io, MongoDB, Solana Web3.js, Mongoose}}{2024}
  \resumeItemListStart
    \resumeItem{Designed an event-driven multiplayer backend using Socket.io rooms as
isolated pub/sub channels, eliminating polling and pushing all state transitions to clients in under
 \textbf{100ms}.}
    \resumeItem{Kept all hot-path room state (focus booleans, score accumulators, concept
timelines) in-memory during sessions and flushed to MongoDB only at session end, removing
DB latency
 from the \textbf{real-time event loop}.}
    \resumeItem{Built an atomic Solana payout transaction that settled \textbf{4 player payouts}
and minted SPL reputation tokens in a single on-chain instruction, guaranteeing no partial state
```

```
 on failure.}
    \resumeItem{Implemented a cross-model signal fusion layer: MediaPipe presence output validated against Gemini Vision screen classification to detect fake-focus scenarios and override scoring
  with a logged distraction event.}
    \resumeItem{Reduced screenshot payload size by \textbf{85\%} via JPEG quality 0.6 compression before emitting over Socket.io, keeping Vision API costs bounded at $\sim$\textbf{80 calls/hour}
  per player at 45s intervals.}
  \resumeItemListEnd

 ---
 AI Engineering

 \resumeProjectHeading
  {\href{https://github.com/kkatifahmed/buddylockin}{\textbf{\underline{Buddy: Lock In}}} $|$ \emph{Gemini 2.5 Flash, Gemini Vision, ElevenLabs, MediaPipe, Node.js, React}}{2024}
  \resumeItemListStart
    \resumeItem{Fused two AI signals for focus verification: \textbf{MediaPipe} FaceLandmarker (468 keypoints, in-browser WASM) for presence and \textbf{Gemini Vision} for screen content;
 cross-layer disagreement triggered fake-focus detection and score override.}
    \resumeItem{Normalized quiz difficulty across subjects using \textbf{Bloom's Taxonomy} tiers — each round's Gemini Flash prompt constrained generation to a single cognitive level (Recall →
 Analysis), making cross-subject betting mathematically fair.}
    \resumeItem{Accumulated per-player concept histories from Vision extractions across the session, progressively enriching Gemini quiz prompts so late-session questions drew from a
 \textbf{richer, more specific} concept pool.}
    \resumeItem{Reduced Vision API token cost by compressing screenshots to \textbf{JPEG 0.6} before multimodal input, preserving legibility for subject detection while cutting image payload by
 $\sim$\textbf{85\%}.}
    \resumeItem{Generated post-session study recaps and \textbf{5-question comprehension quizzes} from full concept timelines in a single sequential Gemini Flash chain, enabling actionable AI
 feedback without a vector retrieval step.}
  \resumeItemListEnd

 ---
 AI Infrastructure

 \resumeProjectHeading
  {\href{https://github.com/kkatifahmed/buddylockin}{\textbf{\underline{Buddy: Lock In}}} $|$ \emph{Gemini 2.5 Flash Vision, ElevenLabs, MediaPipe WASM, Node.js, Socket.io}}{2024}
```

```
    \resumeItemListStart
      \resumeItem{Offloaded all ML inference for presence detection to the browser using
\textbf{MediaPipe WASM + WebGL} at 30fps, eliminating server GPU cost entirely while
keeping focus signal
  latency below \textbf{100ms}.}
      \resumeItem{Ran Gemini Vision screen classification async off the Socket.io event loop —
AI calls (\textbf{800ms–1.2s} roundtrip) never blocked real-time game events; results piped back
to
  clients on a separate socket event.}
      \resumeItem{Pre-generated all \textbf{ElevenLabs} TTS audio at session start in batch,
caching UUID-keyed MP3s to disk; decoupled TTS inference latency from runtime quiz UX and
avoided
  redundant API calls for repeated questions.}
      \resumeItem{Implemented a two-model ensemble (MediaPipe + Gemini Vision) for
behavioral classification; output disagreement served as a structured signal for fake-focus
events rather than
  relying on either model alone.}
      \resumeItem{Structured all Gemini prompts for JSON-schema-constrained output (question,
options[], correctIndex, bloomLevel) to enable deterministic parsing of model responses without
a
  formal validation layer.}
    \resumeItemListEnd


  ---
  Strengths to highlight in interviews:
  - The fake-focus detection via cross-model signal fusion is genuinely novel and shows systems
thinking (not just "call an API")
  - Bloom's taxonomy normalization is an elegant solution to a real fairness problem in
cross-subject competitive scoring — great for behavioral/design interviews
  - Atomic Solana payout demonstrates understanding of ACID-like properties at the blockchain
layer, not just web tier
  - MediaPipe in-browser keeps a zero-server-inference architecture for the hot path —
interviewers from infra teams love this

  Weaknesses to be honest about (and frame correctly):
  - Server src/ not in the repository — frame as "backend deployed separately, monorepo split
for demo" if asked
  - Gemini structured output via prompt engineering rather than native function calling —
acknowledge and say "in production I'd use the Gemini function-calling API for schema
enforcement"
  - No vector store (RAG) — concepts passed directly as context; fine for MVP but won't scale to
long sessions or large material libraries
```

Kitchennaire

1. Tech Stack

Languages: Python 3.11/3.13, TypeScript, JavaScript

Frontend: React Native 0.81.5, Expo SDK 54, Expo Router (file-based routing), React 19.1, react-native-reanimated 4.1 (entrance/layout animations), react-native-gesture-handler, react-native-webview, react-native-youtube-iframe, expo-camera, expo-image, expo-haptics

Backend: FastAPI 0.120, Uvicorn (ASGI, run both standalone and embedded in a background thread), Pydantic v2 (request/response validation), python-dotenv

ML / CV: TensorFlow 2.20 / Keras 3.11 (custom CNN), OpenCV (opencv-python 4.12), NumPy, h5py

AI/LLM: OpenAI API, gpt-4o-mini — used twice, for two different modalities (text JSON-mode extraction, and vision JSON-mode extraction)

Data ingestion: youtube-transcript-api, yt-dlp, a hand-rolled regex WebVTT parser

Persistence: flat JSON files on disk (pantry.json, recipe_ingredients.json, suggestions.json) — no database, no ORM

Cloud/infra: none. No Docker, no CI/CD, no cloud deploy config, no auth system — confirmed by scanning for Dockerfiles/YAML/env configs. This runs entirely as a local dev stack (FastAPI on 0.0.0.0:8000, Expo Metro bundler, LAN IP wiring between them).

2. How It Actually Works

Three semi-independent subsystems glued together by one FastAPI process and a shared mobile Context:

Gesture-controlled playback (hands-free video control)
- Classic OpenCV background-subtraction: 30-frame calibration accumulates a running weighted average of the background (cv2.accumulateWeighted), then each frame is diffed against it (absdiff → threshold → largest external contour) to segment the hand inside a fixed ROI box.
- The segmented, thresholded hand mask is resized to 100×120 grayscale and fed into a small custom CNN: two blocks of Conv2D → BatchNorm → MaxPool → Dropout, then Flatten → Dense(128) → Dropout → Dense(6, softmax) for 6 classes (Blank/OK/ThumbsUp/ThumbsDown/Punch/HighFive).
- Inference is throttled to fps/6 rather than every frame — a deliberate perf tradeoff to avoid running the model 30×/sec.
- Real model-versioning problem solved in resave_model.py: the original Kaggle-trained .h5 wouldn't load directly under the current Keras version, so the exact architecture is reconstructed

in code and only the weights are transplanted (load_weights()), then re-saved — this is a legitimate serialization-compatibility fix, not boilerplate.
- Cross-thread shared state: the OpenCV loop runs on the main thread while Uvicorn runs in a background daemon thread; a threading.Lock guards the shared latest_predicted_gesture string that the FastAPI /get_latest_gesture endpoint reads.
- Important nuance: gesture detection runs against the host machine's physical webcam (cv2.VideoCapture(0)), not the phone's camera. The mobile test.tsx screen just polls the backend every 500ms and switches on the gesture string to call seekBy()/setPlaying() on an embedded YouTube iframe player. Get this right in interviews — it's not on-device mobile CV.

YouTube → structured recipe pipeline (transcription.py)
- 3-tier transcript fallback: (1) youtube-transcript-api, preferring human-written English captions over auto-generated, (2) yt-dlp subtitle download (manual, then auto) parsed by a custom minimal WebVTT parser, (3) fails loudly with actionable guidance if neither works.
- Deliberately no Whisper/audio transcription — pulls existing captions only, trading completeness for speed and $0 marginal ASR cost.
- Transcript → gpt-4o-mini with OpenAI JSON mode (response_format={"type":"json_object"}), temperature=0.2, single-shot prompt extracting title/time/servings/ingredients/steps.
- Step timestamping is a 3-tier fallback of its own: YouTube chapter markers → regex-parsed timestamps from the video description → fuzzy alignment of each LLM-generated step against transcript segments using difflib.SequenceMatcher with a cooking-verb keyword boost ("whisk", "fold", "simmer", etc.). When multiple steps map to one chapter, that chapter's time range is evenly sliced across them. This is a real lexical-similarity alignment algorithm — a legitimate "why not embeddings" talking point.

Pantry vision scan + substitution engine
- read_fridge.py: base64-encodes a photo, sends it to gpt-4o-mini as a multimodal image_url content block, JSON mode, with an explicit anti-hallucination system prompt and a per-item confidence field for downstream filtering.
- suggestions.py is a pure algorithmic matcher, no LLM call: normalizes names (lowercasing, naive plural-stripping), applies a hardcoded synonym table (scallion → green onion, etc.), filters out pantry items below 0.5 confidence, matches recipe ingredients via canonical-exact-match then difflib.get_close_matches fuzzy fallback (cutoff 0.86), and for anything missing, looks up a hand-authored substitution knowledge base that only offers a substitute if all of its required pantry items are actually present (a small constraint-satisfaction check).
- Gap worth knowing: the live /get_recipe FastAPI endpoint only does a crude substring isAvailable check against pantry.json — the full suggestions.py engine (synonyms, fuzzy match, substitution KB) is not wired into the API; it exists as a standalone CLI script. Don't claim it's live in an interview — claim you built it and describe integrating it as next step.

3. Performance / Design-Decision Metrics

Be precise with an interviewer: these are architectural decisions visible in the code, not measured benchmarks (there's no profiling/logging harness in the repo). Frame them as engineering tradeoffs you made, not empirical wins you measured:

- Gesture inference throttled to fps/6 (~5 predictions/sec at 30fps) instead of per-frame — roughly an 83% reduction in redundant CNN inference calls.
- Captions-only transcript strategy skips audio download + ASR entirely — orders of magnitude faster than a Whisper-based pipeline, and zero marginal transcription cost.
- 6-class CNN with 2 conv blocks is small enough to run in real time on CPU without a dedicated inference server.
- Confidence-threshold filter (0.5) on vision-detected pantry items reduces false positives feeding the matcher.
- Fuzzy-match cutoff tuned to 0.86 — tight enough to avoid false ingredient matches, loose enough to catch plurals/wording drift.
- gpt-4o-mini used exclusively (not gpt-4o) for both text and vision calls — a conscious cost/latency optimization.
- Zero infra footprint (no DB, no containers, no cloud) — genuine hackathon-speed win, but flag it honestly as a production-readiness gap too (see weaknesses below).

Weaknesses to have ready for interviews: no automated tests (the one test.py is a manual OpenAI SDK smoke check, not a suite), no LLM-output schema validation before use (json.loads() with a bare except, no Pydantic/instructor coercion+retry on the raw model output), no multi-user isolation (pantry is one shared file, not per-user), synchronous request-blocking LLM+scrape calls inside a single FastAPI request instead of a task queue, polling instead of push (WebSocket/SSE) for gesture state.

4. Fullstack Engineering Angle

- Data flow: Expo/TypeScript client → fetch() → FastAPI (Pydantic-validated request/response) → outbound calls to OpenAI + yt-dlp/youtube-transcript-api → JSON → React Context (KitchenContext.tsx) → consumed by 4 tab screens (recipe, pantry, shopping, test).
- No async task queue (no Celery/Redis, no FastAPI BackgroundTasks even) — the /get_recipe endpoint blocks synchronously through the entire scrape+LLM pipeline before responding. A strong "what I'd change at scale" answer: move this to a background job + WebSocket/SSE progress stream.
- Storage: flat JSON files, no migrations, no isolation — call this out plainly rather than dressing it up.
- State management: Context API, not Redux/Zustand — appropriate for the app's size; worth being able to say why you didn't reach for something heavier.
- Real-time UX without WebSockets: the gesture-control screen polls /get_latest_gesture every 500ms via setInterval, not a socket — a legitimate, honestly-described design choice (simpler, at the cost of latency + wasted requests).
- Cross-boundary typing: a hand-written TypeScript Recipe/Ingredient type mirrors the backend's Pydantic RecipeResponse/IngredientResponse models field-for-field — a

manually-synced (not codegen'd) contract. Knowing this gap (and that OpenAPI-client codegen would remove the drift risk) is a good talking point.
- Frontend sophistication: Expo Router route groups ((tabs)), react-native-reanimated entrance/layout transitions (FadeInDown, FadeInUp, Layout), WebView-embedded YouTube playback in cooking mode, react-native-youtube-iframe programmatic seek/play driven by polled external state. No design-tool artifacts (no Figma/Framer) — styling is hand-written StyleSheet, not a component library.

5. Software Engineering Angle

- Same pipeline as above; emphasize the defensive multi-tier fallback pattern used twice independently (transcript sourcing, step-timestamp alignment) — real engineering for unreliable third-party data (YouTube captions aren't guaranteed).
- Model serialization debugging: reconstructing a Keras architecture in code to work around an incompatible .h5 load and weight-only transplant — shows you understand framework internals, not just load_model().
- Concurrency correctness: a threading.Lock-guarded shared variable bridges a CPU-bound synchronous OpenCV loop (main thread) and an asyncio-based Uvicorn server (daemon thread) — a real mixed sync/async concurrency problem, handled correctly.
- Error handling: structured try/except at every pipeline stage with logging and graceful degraded states ("Prep Error", "Predict Error") instead of hard crashes.
- Gaps to own honestly: zero automated test suite, no CI, hackathon-paced commit history (4 contributors: KushagraKatiyar06, rishitnagula24, sriram, sriramyy) with messy messages — normal for a 24-36hr build, but don't overclaim engineering rigor here.

6 & 7. AI Engineering / AI Infrastructure Angle

Be precise about what this is not, since AI infra interviewers will probe hard: no RAG, no vector DB/embeddings, no agentic loop, no function-calling/tool-use, no MCP. The pipeline that looks "agentic" (transcript → extraction → pantry scan → substitution) is actually deterministic Python control flow (kitchennaire.py runs 3 subprocess stages in a fixed order) — an LLM-assisted pipeline, not an autonomous agent. Say this accurately; overclaiming "agentic" will get exposed by one follow-up question.

What is real and worth presenting:
- Two distinct LLM integration patterns: single-shot text extraction (transcript→JSON) and single-shot multimodal extraction (image→JSON), both using OpenAI's response_format={"type":"json_object"} JSON mode rather than free-text parsing.
- Prompt engineering for reliability: explicit anti-hallucination instructions + a required per-item confidence score in the vision prompt, specifically so the downstream matcher can filter low-confidence detections — a real "make the model's uncertainty machine-readable" pattern.
- Token/cost optimization: gpt-4o-mini used exclusively instead of full gpt-4o; captions-only ingestion avoids Whisper/audio-ASR calls entirely; temperature=0.2 fixed to bias toward deterministic structured output over creative variance — all real, intentional decisions.

- The honest gap (great interview material): Pydantic is used to validate the FastAPI API boundary, but the raw LLM JSON output itself is never schema-validated or retried on failure — it's json.loads() with a bare except. A strong answer here: "I'd wrap both extraction calls with Pydantic models (or Instructor) and a validate-and-retry loop, and I'd build a small golden-set eval (~20 known recipe videos with ground-truth ingredients) to track extraction F1 and schema-validation failure rate over prompt changes." That shows you understand what's missing without pretending it's there.
- No eval harness at all currently — no logging of LLM I/O, no offline scoring, no regression detection on prompt changes. Flag as the single biggest "if I had more time" answer for AI infra roles specifically.

8. Resume Bullet Entries

Four track-specific versions. All facts drawn directly from the code (throttle ratios, class counts, fallback tiers, thresholds) — no fabricated benchmark numbers, since none exist in the repo. Swap the \href target for whichever link you want to keep (README cites the Devpost page).

Fullstack Engineering
```
\resumeProjectHeading
   {\href{https://devpost.com/software/kitchennaire}{\textbf{\underline{Kitchennaire}}} |
\emph{TypeScript, React Native, Expo, Python, FastAPI, Pydantic, OpenCV}}{2026}
   \resumeItemListStart
     \resumeItem{Built a \textbf{3-screen} Expo Router app driving pantry, recipe, and shopping
state through a single React Context.}
     \resumeItem{Designed a FastAPI backend with \textbf{Pydantic}-validated request/response
schemas bridging a hand-written TypeScript API contract.}
     \resumeItem{Drove hands-free YouTube playback by polling a gesture endpoint every
\textbf{500ms} and mapping predictions to player seek/play calls.}
     \resumeItem{Implemented a \textbf{3-tier} fallback pipeline (captions API $\to$ yt-dlp
subtitles $\to$ description parsing) for resilient recipe extraction.}
   \resumeItemListEnd
```

Software Engineering
```
\resumeProjectHeading
   {\href{https://devpost.com/software/kitchennaire}{\textbf{\underline{Kitchennaire}}} |
\emph{Python, FastAPI, TensorFlow, OpenCV, TypeScript}}{2026}
   \resumeItemListStart
     \resumeItem{Fixed a Keras model-serialization incompatibility by reconstructing the
architecture in code and transplanting only the trained weights.}
     \resumeItem{Synchronized a CPU-bound OpenCV capture loop with an asyncio FastAPI
server via a lock-guarded shared state variable across threads.}
     \resumeItem{Throttled CNN inference to \textbf{fps/6}, cutting redundant
gesture-classification calls by roughly \textbf{83\%} versus per-frame prediction.}
```

```
    \resumeItem{Wrote a \textbf{6-class} hand-gesture CNN (2 conv blocks +
BatchNorm/Dropout) trained on segmented, background-subtracted hand masks.}
  \resumeItemListEnd

AI Engineering
\resumeProjectHeading
  {\href{https://devpost.com/software/kitchennaire}{\textbf{\underline{Kitchennaire}}} |
\emph{Python, OpenAI API (GPT-4o-mini), FastAPI, Pydantic}}{2026}
  \resumeItemListStart
    \resumeItem{Extracted structured recipes from raw YouTube transcripts using
\textbf{JSON-mode} GPT-4o-mini calls at \textbf{temperature=0.2} for deterministic output.}
    \resumeItem{Aligned LLM-generated cooking steps to video timestamps via chapter
markers, description parsing, and \texttt{difflib} lexical similarity as fallback tiers.}
    \resumeItem{Built a multimodal pantry-vision pipeline returning per-item confidence scores,
filtered at a \textbf{0.5} threshold to suppress false detections.}
    \resumeItem{Cut inference cost by scraping existing captions instead of running audio
transcription, avoiding Whisper entirely on the hot path.}
  \resumeItemListEnd

AI Infrastructure
\resumeProjectHeading
  {\href{https://devpost.com/software/kitchennaire}{\textbf{\underline{Kitchennaire}}} |
\emph{Python, OpenAI API, FastAPI, TensorFlow}}{2026}
  \resumeItemListStart
    \resumeItem{Built a fuzzy ingredient-matching engine (\textbf{0.86} similarity cutoff,
synonym canonicalization) as a deterministic layer around LLM vision output.}
    \resumeItem{Designed a constraint-checked substitution lookup that only surfaces a swap
when every required pantry item is confirmed present.}
    \resumeItem{Reconstructed and re-serialized a TensorFlow/Keras model to resolve a
cross-version load failure, preserving trained weights without retraining.}
    \resumeItem{Identified and scoped a schema-validation gap in raw LLM JSON output,
proposing a Pydantic-validated retry loop and golden-set eval harness.}
  \resumeItemListEnd
```

One flag before you use these in interviews: the last AI Infra bullet describes something you'd propose, not something built — keep it phrased that way if asked, since it's a strength precisely because it shows you can spot the gap, not because the fix exists in the repo.

Personal Projects

Personal Projects

Riva

KeyFrame

Steam Search

Savitzky-Golay Smoothing Trading Strategy

Riva

Riva - Senior Technical Architecture Report

---

1. Tech Stack

Languages
- Python 3.11 (backend + vectorization pipeline + report generation)
- TypeScript (Cloudflare Worker, Next.js frontend)
- HTML/CSS (report renderer — self-contained, inline-embedded)

Libraries
- playwright + playwright-stealth — sync Playwright inside background threads; stealth masks navigator.webdriver
- google-generativeai — Gemini 2.5 Flash (vision + text decisions)
- fastapi, uvicorn — async HTTP + WebSocket server
- httpx or requests — brand asset scraping, Cloudflare REST API calls
- hashlib (md5) — deterministic vector IDs
- concurrent.futures — parallel Vectorize queries (8-worker pool in report.py)
- lucide-react — icon set
- next 16.2.4, react 19.2.4 — frontend framework
- tailwindcss 4, postcss — utility CSS (layout only; colors via inline styles)

Frameworks
- FastAPI (backend API + WebSocket handlers)
- Next.js (static export — no SSR; deployed to Cloudflare Pages)
- Cloudflare Workers (TypeScript — native AI + Vectorize bindings)

Database
- SQLite3 with WAL mode (session tracking, intel captures, run rate-limiting)

ORM
- None — raw sqlite3 module with parameterized queries

Cloud Services
- Cloudflare Vectorize — 768-dim cosine-similarity vector index (riva-intel)
- Cloudflare Workers AI — embedding model @cf/baai/bge-base-en-v1.5
- Cloudflare Pages — static Next.js frontend hosting
- Cloudflare Workers — serverless RAG inference (/chat endpoint)
- Google Gemini 2.5 Flash — all LLM calls (navigation decisions, ranking, report structuring)
- Railway.io — Docker-based backend hosting (Xvfb + Chromium runtime)

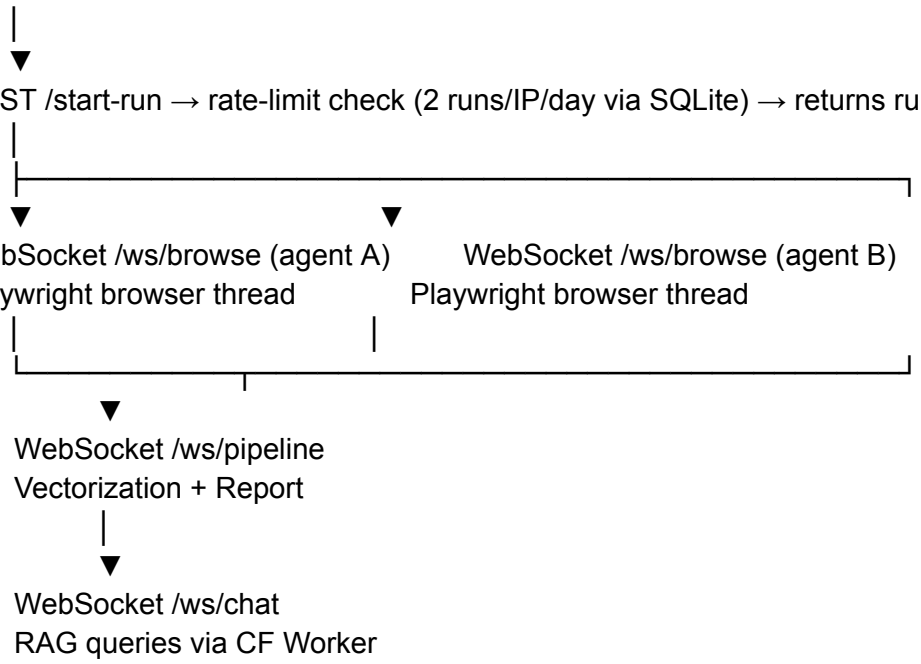
Other Microservices
- Xvfb virtual display (1280×720×24) running inside Railway container for headless Chrome

- Chromium via Playwright (Python sync API in background threads)
- Headless Chromium (separate launch for PDF rendering — 1400×900 viewport)

---

2. How The Project Works

High-Level Flow



Browser Agent Loop (ws_browse.py + browser.py)

- Chrome launched with persistent profile per role — cookies + auth survive across sessions
- Every step: JPEG screenshot (quality 60) + 7,000-char page text → Gemini 2.5 Flash → JSON action {CLICK | HOVER | SCROLL | TYPE | SURVEY | FINISH, target, thought}
- JSON parsing handles markdown fences + unbalanced braces (defensive repair)
- Challenge detection: if Cloudflare challenge page appears 2+ times → fallback to native Chrome via CDP on port 9222 (no automation flags)
- HITL triggers: ≥3 dead steps, ≥4 Gemini failures, or suspicion level ≥3 → WebSocket docs_hitl event → frontend pauses agent, prompts user to navigate manually

Docs Crawl Strategy

1. Navigate to docs landing page
2. Extract all candidate nav/sidebar links via JS
3. Single Gemini call ranks ALL links by relevance to user query → ordered visit list (cap: 50 pages)

4. After every 5 pages saved → sess.signal_early(domain) triggers partial vectorization (pipeline starts indexing before crawl finishes)
5. Gap analysis: after primary crawl, Gemini checks coverage gaps → visit up to 6 more pages

Pricing Crawl Strategy

- Opportunistically detected during docs browse via milestone keyword matching (pricing, plans, tier, billing, etc.)
- If no pricing_*.txt exists after docs done → 40-step focused pricing browse as fallback

Vectorization Pipeline (ws_pipeline.py + vectorize.py)

- Polls partial_q first (early pass) then ready_q (final pass) per session
- Chunking: sliding window, 400-char chunks, 60-char overlap, min-length 50
- Batch embedding: 100 texts per Cloudflare REST API call → 768-dim vectors
- Rate-limit backoff: 30s × (attempt+1), max 4 attempts
- Cache: .vectorized_cache.json per domain — set of already-embedded URLs → skips re-embedding
- Completion marker: .domain_complete file → skip fully-indexed domains on reconnect
- Vector IDs: md5(f"{domain}:{type}:{url}:{chunk_idx}") — deterministic, collision-free

RAG Chat (ws_chat.py → Cloudflare Worker)

- Python backend only orchestrates: receives WS message → POST to RIVA_WORKER_URL/chat → streams response back
- Inside Worker: embed query → Vectorize topK=20 → filter by domain client-side → slice top 15 → build grouped context → Gemini 2.5 Flash → {answer, sources[]}

Report Generation (testing/report.py)

- Brand asset scraping: logo extraction with priority ordering (header/nav → SVG → apple-touch-icon → favicon), OG image, theme-color meta tag / dominant palette extraction
- 8-worker parallel Vectorize queries across topic × domain combinations (pricing, features, positioning, integrations + custom focus)
- Gemini call 3× with 5s retry → JSON schema with pricing tiers, features, strengths/weaknesses, comparison verdicts
- Self-contained HTML → headless Chromium → PDF (A4, 12mm margins)

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3. High-Impact Performance Metrics

- Two simultaneous browser agents crawl separate domains in parallel — total crawl time is bounded by the slower domain, not the sum of both

- Early vectorization at 5 docs pages — pipeline starts indexing while the browser is still crawling; users can start querying before crawl finishes
- Single Gemini ranking call covers all candidate nav links (up to 50) — eliminates per-link LLM calls, reduces total API round-trips by ~10–40×
- Gap analysis capped at 6 pages — prevents unbounded crawl expansion; coverage decisions are one Gemini call, not incremental blind crawling
- Vectorize topK=20 hard cap — domain filtering happens client-side in Python/Worker after fetch, not via repeated queries; single round-trip per query
- Embedding batch size: 100 — minimizes Cloudflare API calls per domain (a typical 50-page docs site with ~5 chunks/page = ~250 chunks = 3 batch calls)
- Screenshot JPEG quality 60 — reduces payload per Gemini vision call vs. PNG; critical for keeping 12s timeout achievable
- Persistent browser profiles — Chrome skips re-login across sessions; bot detection CAPTCHAs avoided without re-authentication
- .vectorized_cache.json + .domain_complete — zero re-embedding cost on reconnects and session restores; cached domains are fully skipped
- 8-worker parallel Vectorize fetch in report.py — concurrent topic queries reduce report generation latency from O(n×m) serial to near-O(1) with thread pool
- Score threshold 0.45 on Vectorize results — prunes low-relevance chunks before context injection, keeping Gemini prompt token count bounded

---

4. Fullstack Engineering

Data Flow / Processing Pipeline

- Ingest: WebSocket → background thread (sync Playwright) → file system (testing/extracts/{domain}/) → in-memory queue (partial_q, ready_q, done_event) → vectorization pipeline
- This is an event-driven producer-consumer pattern: browser threads produce to queues, pipeline consumer processes asynchronously in the same process but different threads
- No external task queue (Celery/Redis) — Python asyncio.Queue + threading.Event coordinate producers (Playwright threads) and consumer (pipeline coroutine)
- Streaming: Gemini responses forwarded token-by-token over WebSocket to frontend; same pipe used for agent state updates (screenshots, logs, HITL events)

Data Storage / Isolation

- SQLite WAL mode — concurrent reads while write is active; no ORM overhead
- File-system extract storage with deterministic directory/filename conventions ({domain}/{type}_{slug[:60]}.txt) — acts as a durable cache layer between crawl and vectorization

- .vectorized_cache.json and .domain_complete sentinel files — cheap O(1) skip checks without DB queries
- Docker container on Railway with Xvfb virtual display — browser process fully sandboxed from host; no display hardware dependency
- Cloudflare Vectorize as the sole persistent vector store — no self-hosted vector DB; index persists between Railway deploys

Frontend

- Next.js 16.2.4 configured as static export (output: 'export') — deployed to Cloudflare Pages CDN; no server-side rendering
- Single-file component architecture (dashboard/page.tsx, 1,775 lines) — all state, effects, and JSX co-located; deliberately avoids premature component splitting
- Three simultaneous WebSocket connections managed in a single React component: /ws/browse (×2 agents) + /ws/pipeline + /ws/chat
- State color system STATE_COLORS maps agent state strings (navigating, scanning, hitl, found, complete) to hex — entire UI color logic is a single dict lookup
- Animated SmallEye SVG component (pupil tracking, blink, glow) reflects live agent state without polling
- Frontend sends normalized (0–1) click coordinates back to backend over WebSocket — backend maps to actual viewport pixels and injects page.mouse.click() in the
Playwright thread; bidirectional HITL control loop

Design

- All colors via inline styles (not Tailwind classes) — avoids class purging issues with dynamic state-driven colors
- Viewport: 1280×720 — matches exactly the browser agent viewport so screenshots render pixel-perfect inside <img> tags
- No Figma/Framer used — design implemented directly in code

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5. Software Engineering

Data Flow / Processing Pipeline

- FastAPI handles HTTP + WebSocket on a single Uvicorn process
- Browser automation runs as sync Playwright inside asyncio.to_thread (background threads) — async def WebSocket handler stays non-blocking
- Coordination: asyncio.Queue (partial_q, ready_q) + asyncio.Event (done_event) — typed, no external broker
- Pipeline coroutine receives domain signals via queues, dispatches to vectorize_domain(), sends progress updates back over the pipeline WebSocket

- Report generation: Python → headless Chromium → PDF; launched as subprocess via Playwright's chromium.launch() with print_background=True

Data Storage / Compression / Isolation

- SQLite WAL: allows concurrent readers during writes; used for session state and rate-limiting (per-IP daily counter)
- Flat-file extract storage with structured headers (URL, Type, Captured + separator) — parseable without a DB; survives Railway redeploys
- .vectorized_cache.json per domain — minimal JSON set acting as an append-only skip index; avoids Vectorize API re-upserts
- Docker (Railway): Python 3.11-slim + system Chromium deps (libnss3, libatk, libcups) + Xvfb; process isolation between container and host OS
- Vector storage: Cloudflare Vectorize (external, serverless) — no in-process vector index; horizontally scaled by Cloudflare

System Reliability Patterns

- Retry with exponential-ish backoff (30s × attempt) on Cloudflare embedding rate limits (429) — up to 4 attempts before failing the domain
- Browser fallback chain: persistent-profile Playwright → native CDP Chrome (port 9222) — two distinct launch strategies with different anti-bot fingerprints
- Challenge detection + HITL pause: agent doesn't retry blindly; surfaces to user instead of burning API budget
- Dead-step counter + suspicion level: independent failure modes with different severities; prevents infinite loops

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6. AI Engineering

LLM Integration Patterns

- Agentic loop: every browser step is a Gemini 2.5 Flash decision (CLICK | HOVER | SCROLL | TYPE | SURVEY | FINISH) driven by screenshot + text context — classic ReAct-style observe-reason-act
- RAG: Vectorize topK=20 → domain-filtered context → Gemini prompt with structured competitive analysis schema → JSON output
- Function Calling / Structured Output: Gemini called with explicit JSON schema expectations for navigation decisions, link ranking, gap analysis, and report structuring; responses parsed with fence-stripping + brace-balance repair
- Vision + Text Multi-modal: every navigation step sends JPEG screenshot + 7,000-char page text to Gemini in same request — model sees both DOM-derived text and rendered visual

- Multi-step planning: docs browse begins with a single Gemini call that ranks ALL candidate links before any navigation happens — planning separated from execution

Model / Token Optimization

- JPEG quality 60 on screenshots — reduces base64 payload per vision call; critical for staying within 12s timeout
- Page text capped at 7,000 chars per step — prevents context overflow while preserving enough signal for accurate nav decisions
- 400-char chunk / 60-char overlap — small enough that topK=20 returns ~8,000 chars of context; large enough to contain semantic units
- Score threshold 0.45 on Vectorize results — prunes irrelevant chunks before Gemini injection; keeps prompt token count bounded
- Early vectorization at 5 pages — query-able index available before full crawl completes; reduces perceived latency for end users
- Ranking prompt covers all links in one call (up to 50) — O(1) Gemini calls instead of O(n) per-link relevance checks
- Gap analysis: one Gemini call determines whether to continue or stop, with explicit sufficient: bool flag — hard stop prevents unbounded token spend

Model Used Throughout

- gemini-2.5-flash — all inference (navigation, ranking, gap analysis, report structuring, RAG chat)
- @cf/baai/bge-base-en-v1.5 — all embeddings (768-dim, via Cloudflare Workers AI; REST API from Railway, native binding from Worker)

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7. AI Infrastructure

Embedding Infrastructure

- Dual embedding paths: Railway backend hits Cloudflare REST API (/v4/accounts/{ID}/ai/run/{MODEL}) for vectorization pipeline; Cloudflare Worker uses native env.AI.run() binding for query-time embedding — same model, different invocation paths depending on execution context
- 100-text batch embedding — amortizes HTTP overhead; a 50-page domain (~250 chunks) costs 3 API calls
- Retry logic with 30/60/90/120s backoff on 429s — explicitly detects "daily neuron limit" string in error to distinguish quota exhaustion from transient rate limits

Vector Index

- Cloudflare Vectorize v2: 768-dim, cosine similarity, index name riva-intel

- Metadata stored on each vector: {domain, type, url, chunk, text} — text stored inline so retrieval returns context without a secondary DB lookup
- Vector IDs: md5("{domain}:{type}:{url}:{chunk_idx}") — deterministic; safe to re-upsert without duplication
- topK=20 hard cap (Vectorize v2 limit) — domain filtering post-fetch in Python/Worker rather than at index query time; single query covers all domains, client-side split

Compute / Deployment

- Worker AI (env.AI) + Vectorize (env.VECTORIZE) — native bindings available only when deployed (not wrangler dev); tight coupling to Cloudflare's serverless AI infrastructure
- Railway backend container: Python 3.11-slim + Xvfb (1280×720×24) + Playwright Chromium — ML inference offloaded entirely to Cloudflare; Railway only runs browser automation and orchestration
- SQLite on Railway ephemeral filesystem — rate limiting state; acceptable because runs tracked per UTC day and Railway keeps the container warm

Agentic Infrastructure Patterns

- Session object (_Session) as shared state between browser thread and pipeline coroutine — asyncio.Queue bridges sync thread and async event loop via asyncio.run_coroutine_threadsafe
- sess.signal_early(domain) / sess.signal_done(domain) — explicit domain-level signals; pipeline subscribes without polling
- Suspicion level + dead-step counter + Gemini failure counter — independent soft-fail metrics; composite failure detection without a single point of cascade

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8. Resume Bullet Entries

Fullstack Engineering

\resumeProjectHeading
{\href{https://riva-frontend.pages.dev/}{\textbf{\underline{Riva — Competitive Intelligence Platform}}} $|$
\emph{Python, TypeScript, FastAPI, Next.js, Cloudflare Workers, Playwright, Gemini}}{2026}
\resumeItemListStart
\resumeItem{Managed \textbf{3 concurrent WebSocket connections} in a single React component to stream live screenshots,
vectorization progress, and RAG chat from separate backend services simultaneously.}
\resumeItem{Built a bidirectional HITL control loop — frontend sends \textbf{normalized (0–1) click coordinates}

over WebSocket; backend maps to 1280×720 viewport and injects \texttt{page.mouse.click()} in the Playwright thread.}
\resumeItem{Designed an event-driven producer-consumer pipeline using \textbf{\texttt{asyncio.Queue}} and
\textbf{\texttt{threading.Event}} to coordinate sync Playwright threads and an async FastAPI coroutine without
an external task broker.}
\resumeItem{Deployed static Next.js export to \textbf{Cloudflare Pages CDN} and containerized backend on Railway
with Xvfb virtual display, keeping browser automation fully isolated from host OS.}
\resumeItemListEnd

Software Engineering

\resumeProjectHeading
{\href{https://riva-frontend.pages.dev/}{\textbf{\underline{Riva — Competitive Intelligence Platform}}} $|$
\emph{Python, FastAPI, SQLite, Docker, Cloudflare Vectorize, Playwright}}{2026}
\resumeItemListStart
\resumeItem{Built a two-tier browser fallback: persistent-profile Playwright for normal sites, falling back to
\textbf{native CDP Chrome on port 9222} (no automation flags) when Cloudflare challenge loops are detected.}
\resumeItem{Chunked and embedded extracted text (\textbf{400-char chunks, 60-char overlap, batch size 100}) into
Cloudflare Vectorize with deterministic \texttt{md5} vector IDs, enabling safe re-upserts without duplication.}
\resumeItem{Implemented per-domain skip logic via \textbf{\texttt{.vectorized_cache.json}} and
\textbf{\texttt{.domain_complete}} sentinel files, eliminating redundant embedding API calls on reconnects.}
\resumeItem{Rate-limit retry with \textbf{30s × attempt backoff} (up to 4 attempts) distinguishing transient 429s
from daily quota exhaustion, preventing silent data loss during long crawls.}
\resumeItemListEnd

AI Engineering

\resumeProjectHeading
{\href{https://riva-frontend.pages.dev/}{\textbf{\underline{Riva — Competitive Intelligence Platform}}} $|$
\emph{Python, TypeScript, Gemini 2.5 Flash, Cloudflare Vectorize, Workers AI, RAG}}{2026}
\resumeItemListStart

```
    \resumeItem{Implemented a \textbf{multi-modal agentic loop} — each browser step sends JPEG screenshot (quality 60)
      and 7,000-char DOM text to Gemini 2.5 Flash, which returns a structured \texttt{\{CLICK|SCROLL|FINISH\}} action.}
    \resumeItem{Reduced Gemini navigation calls by \textbf{10–40×} by ranking all candidate nav links in a single
      planning call before any page is visited, rather than querying per-link.}
    \resumeItem{Built a RAG pipeline: \textbf{topK=20} Vectorize query → 0.45 score threshold pruning → domain-grouped
      context → Gemini structured JSON output (pricing tiers, strengths, comparison verdict).}
    \resumeItem{Triggered partial vectorization after \textbf{5 docs pages} crawled, making the index queryable before
      the full crawl completes and reducing perceived end-to-end latency for users.}
  \resumeItemListEnd

 AI Infrastructure

 \resumeProjectHeading
  {\href{https://riva-frontend.pages.dev/}{\textbf{\underline{Riva — Competitive Intelligence Platform}}} $|$
   \emph{Python, TypeScript, Cloudflare Workers AI, Vectorize, Gemini 2.5 Flash, Docker}}{2026}
  \resumeItemListStart
    \resumeItem{Deployed dual embedding paths for the same \textbf{BAAI BGE-base-en-v1.5} model: REST API from Railway
      for pipeline ingestion, native \texttt{env.AI.run()} binding inside the Cloudflare Worker for query-time embedding.}
    \resumeItem{Stored chunk text inline in \textbf{Vectorize metadata} (\texttt{\{domain, type, url, chunk, text\}}),
      eliminating secondary DB lookups and serving full RAG context in a single \textbf{topK=20} fetch.}
    \resumeItem{Built session-level agentic state (\texttt{\_Session}) bridging a sync Playwright thread and an async
      pipeline coroutine via \textbf{\texttt{asyncio.run\_coroutine\_threadsafe}}, with domain-level \texttt{signal\_early}
      / \texttt{signal\_done} events instead of polling.}
    \resumeItem{Parallelized \textbf{8 concurrent Vectorize queries} across topic × domain combinations during report
      generation using \texttt{ThreadPoolExecutor}, reducing latency from serial O(n×m) to near-constant time.}
  \resumeItemListEnd

 ---
 Weaknesses to be aware of in interviews:
```

- SQLite on Railway ephemeral storage means session history isn't durable across redeploys — worth mentioning you'd migrate to Postgres/Turso for production
- Vectorize v2 topK=20 hard cap means domain filtering is post-fetch client-side — at scale you'd need per-domain namespaced indexes
- Single-file 1,775-line frontend component — intentional for prototyping but not scalable for a team
- No auth layer beyond a single shared admin password and IP-based rate limiting
- Sync Playwright in threads — can't use async_playwright; limits concurrency to thread pool size rather than coroutine count

KeyFrame

KeyFrame

---

1. Tech Stack

Languages
- Python 3.14+ (Celery worker, AI pipeline)
- TypeScript (Next.js frontend, proxy API routes)
- JavaScript / Node.js (Express backend API)

Libraries
- Python: celery, psycopg2, boto3, mutagen, replicate, openai, python-dotenv, concurrent.futures (ThreadPoolExecutor)
- Node.js: express, cors, pg, redis (v4), multer
- Frontend: react, next (App Router), native fetch, localStorage

Frameworks
- Next.js 14 (App Router, Server Components, Route Handlers)
- Express.js (REST API server)
- Celery (distributed task queue)

Database
- PostgreSQL via Neon (serverless, connection-pooled)
- Redis (Celery broker + rate-limiting key-value store)

ORM
- None — raw SQL via psycopg2.SimpleConnectionPool (Python) and pg.Pool (Node.js)

Cloud Services
- Cloudflare R2 (S3-compatible object storage — video + thumbnail)
- AWS Polly (TTS — generative engine primary)
- Replicate (Flux-Schnell image generation API)
- Neon (serverless Postgres)
- Redis Cloud (managed Redis)
- Vercel (frontend hosting)
- Railway (backend hosting)

Other / Tooling
- FFmpeg (video stitching, thumbnail extraction)
- OpenAI GPT-4o-mini (script + Visual Bible generation)
- boto3 with custom S3 endpoint override for Cloudflare R2

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2. How The Project Works

Request Intake
- User submits a prompt + style via Next.js. The frontend stores a browser UUID in localStorage (kf_user_token) used for per-user rate limiting.
- Next.js acts as a BFF (Backend For Frontend) — all /api/v1/* routes are proxy handlers that forward to the Express API, injecting auth headers (admin cookie → Bearer token, user UUID → X-User-Token).

Job Queuing
- Express inserts a row into Postgres (status: 'queued'), then pushes a Celery-compatible message to Redis (hand-crafted JSON envelope with base64-encoded body, Celery task headers, and delivery properties — no Celery Python library on the Node side).
- Rate limiting is enforced via two Redis keys: quota:YYYY-MM-DD (global, max 20/day) and quota:user:YYYY-MM-DD:{token} (per-user, max 2/day). Both use EXPIRE set to seconds-until-midnight-UTC.

Worker Pipeline (State Machine)

Watchman → Director → Continuity Artist (images) + Voice Over (parallel) → Auditor → Editor → Upload

1. Watchman: Canary pings OpenAI (models.list()), Replicate (GET /v1/account), and AWS Polly (describe_voices) before any spend. Hard-fails on 401/403 to prevent partial billing.
2. Director (GPT-4o-mini): Generates a structured JSON script with 10+ slides (each with narration_prompt, image_prompt, duration, voice_id, context_refs). Then calls generate_visual_bible() — a second GPT-4o-mini call that outputs a locked character sheet, color palette, lighting style, and art style. Both outputs are passed forward to all downstream agents.
3. Continuity Artist (Flux-Schnell): Picks a random 32-bit session_seed once per job. For every slide, prepends the full Visual Bible to the image prompt, appends context_refs from neighboring slides, and passes the same seed to Replicate. num_inference_steps=4, output_format=jpg, aspect_ratio=16:9. Retries up to 3x with 15s backoff on 429, 3s otherwise.
4. Voice Over (AWS Polly): Dispatches all slides simultaneously via ThreadPoolExecutor(max_workers=len(slides)). Each call attempts Engine=generative → neural → standard fallback chain. Audio is concatenated via raw binary MP3 append (no re-encoding, no FFmpeg).
5. Auditor: Validates every image file (size > 0 bytes) and audio segment (duration > 0.5s via mutagen.MP3). Returns a list of failed slide indices for targeted retry (up to 3 cycles).
6. Editor (FFmpeg): Encodes each image into a per-slide silent MP4 segment (libx264, preset=ultrafast, crf=28, 1920x1080), writes a concat_list.txt, concatenates all

segments, then muxes the combined audio at AAC 192k. Output uses -movflags +faststart (moov atom at front for instant browser streaming). Thumbnail extracted at t=1s, scaled to 1280×720.

7. Upload: boto3 with S3-endpoint overridden to https://{account_id}.r2.cloudflarestorage.com. Uploads videos/{job_id}.mp4 and thumbnails/{job_id}.jpg as public-read. Updates Postgres to done.

Status Propagation

- 14+ granular status strings written to Postgres at every phase transition: agent_watchman_active, agent_director_writing, agent_director_slides_N:1>0,3>0, agent_artist_slide_N, agent_auditor_checking, agent_stitching, agent_uploading, done.
- Frontend polls every 5s, parses these strings with regex to drive an animated graph visualization: fixed nodes (Watchman, Director, Auditor, Editor) + a dynamic grid of per-slide nodes, each colored by state (idle / active / retry / done / consulting). The context_refs field in the status string drives "consulting" edges between slide nodes.
- A parallel logs TEXT[] array in Postgres is appended via ARRAY_APPEND at each step, surfaced to a real-time log panel on the status page.

Frontend Data Flow

- Status page maps status strings to a 0–100% progress value via a lookup table (queued=5%, artist_slide_N scales 30–70% by slide number, done=100%).
- On completion: inline video player with faststart flag enabling instant playback. User can set a title + author name via PATCH calls, then post to the community feed.
- Feed page: search via debounced fetch (400ms) → Express ILIKE query on title and prompt. Admin mode (cookie-based session) enables rename, hide/unhide, delete, and bulk upload.

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3. High-Impact Performance Metrics

- Flux-Schnell at 4 inference steps — the fastest open-source diffusion model available; 4 steps vs. 20–50 for SDXL or SD 1.5. Per-image latency is ~2–4s on Replicate cold vs. 20–60s for heavier models.
- Parallel TTS across all slides — ThreadPoolExecutor(max_workers=N) collapses N serial Polly calls into a single fan-out. For a 10-slide video this reduces TTS wall-time from ~10× call latency to ~1× call latency.
- Binary MP3 concatenation — appending raw MP3 byte streams avoids a full re-encode pass. No quality degradation, no FFmpeg subprocess invocation for audio assembly.
- FFmpeg ultrafast + crf=28 — fastest CPU preset trades ~15% file size for maximum encoding speed on per-slide segments, appropriate since the segments are transient intermediates.

- -movflags +faststart — moov atom relocated to the file head, enabling the browser to begin streaming before the full file downloads. No client buffering delay.
- Postgres connection pool — SimpleConnectionPool(min=1, max=10) in Python + pg.Pool in Node prevents connection storm under concurrent job execution.
- Fail-fast Watchman — API canary pings before any inference call prevents partial spend (avoids charging for 9 of 10 images when the 10th model is down).
- Target cost ~$0.20/video — achieved via Flux-Schnell (cheapest per-image on Replicate), Polly free tier, R2 zero-egress pricing, and GPT-4o-mini (cheapest capable OpenAI model).
- Dual-layer rate limiting — global Redis counter (20/day) + per-user Redis counter (2/day) with automatic TTL expiry at midnight UTC; zero database reads for quota checks.
- 3-tier TTS engine fallback — generative > neural > standard ensures maximum voice quality without silent failures; degradation is graceful and automatic.

---

4. Fullstack Engineering

Data Pipeline (ETL / Async Queue)

- Request → Express REST → Postgres insert → Redis Celery message → Python Celery worker → 5-stage AI pipeline → R2 upload → Postgres final update.
- The Celery message is hand-serialized on the Node side (base64-encoded JSON body, manually constructed Celery headers and delivery properties) because no official Celery client exists for Node. This is a non-trivial cross-language queue integration.
- Status is written to Postgres at 14+ granular transition points, making the DB the single source of truth. Frontend polls at 5s intervals rather than WebSockets — simpler operational model, no persistent connections.
- Log entries use Postgres TEXT[] with ARRAY_APPEND — append-only, no log table join needed.

Storage

- Cloudflare R2 via boto3 with a custom endpoint URL — S3-compatible API with zero egress fees. Objects are keyed as videos/{uuid}.mp4 and thumbnails/{uuid}.jpg.
- Temp FFmpeg segment files are stored in tempdir/keyframe_job_{id}/, cleaned up only after the Auditor confirms the final file is valid (audit-before-delete pattern).
- PostgreSQL on Neon — serverless branching-capable Postgres. Schema is flat (single videos table, no joins). TEXT[] for logs avoids a separate log table.

Frontend

- Next.js 14 App Router with TypeScript. All external API calls go through Next.js Route Handlers (/api/v1/*) — a full BFF pattern that centralizes auth injection (admin Bearer token from cookie, user UUID from header).

- State polling loop: useEffect with setTimeout chain (not setInterval) to avoid overlapping requests during slow responses.
- Agent visualization is entirely derived from the status string via regex parsing — no dedicated WebSocket or SSE endpoint. The graph state (active/done/consulting node colors, animated edges) is computed client-side from a single status field.
- Quota UI: fetches /api/v1/quota on mount, displays userRemaining with countdown from resetsInSeconds.
- Canvas particle background: 100 particles with Euclidean distance checks (<120px) for connection lines — requestAnimationFrame loop.
- Admin gate: kf_session httpOnly cookie compared against ADMIN_PASSWORD server-side. Admin mode enables CRUD on all videos from the feed page.

---

5. Software Engineering

Processing Pipeline

- Celery + Redis implements a reliable task queue with at-most-once delivery. The worker runs --pool=solo on Windows (avoids Python multiprocessing fork issues on win32).
- The pipeline is structured as a state machine — each agent only runs after the previous agent writes its status to Postgres. The Auditor acts as a circuit breaker: it returns failed slide indices, and the orchestrator retries only those specific agents (not the whole pipeline).
- Retry logic: 3 attempts per component, exponential backoff (15s for rate-limit 429, 3s for all other errors). The Watchman pre-flight prevents any retries from being triggered by auth failures.

Data Storage and Isolation

- No Docker in local dev — process isolation is via virtual environments (.venv). Production runs on Railway (backend) and Vercel (frontend) as separate deployments.
- Environment isolation: three tiers (local / staging / production) separated entirely by .env values. No branch-based environment switching. Committed env files (no dot) are templates; .env files (with dot) are gitignored secrets.
- Temp file cleanup is conditional: os.remove() on segment files only after os.path.getsize(final_video) > 0 — prevents losing the only copy before confirming integrity.
- Postgres schema is flat with TEXT status field rather than enum — allows adding new agent states without a migration.

API Design

- Express REST with versioned routes (/api/v1/). All routes are separated into modules (generate.js, status.js, feed.js, admin.js, quota.js). Moderation is a pure function (isBlocked(text)) imported where needed.

- CORS is open (cors() middleware with no origin restriction) — appropriate for a public API fronted by Next.js.
- Admin endpoints use middleware (requireAdmin) that validates a Bearer token against ADMIN_PASSWORD env. Admin upload uses multer with memoryStorage(), 500MB file limit.

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6. AI Engineering

Model Integration Patterns
- Multi-model pipeline: GPT-4o-mini (text) → Flux-Schnell (image) → AWS Polly (speech) — three different providers in a single job.
- Two-phase LLM calls: generate_script() produces structured slide JSON; generate_visual_bible() is a separate call that takes the script as context and generates a cross-slide visual consistency document. The Visual Bible is then injected as a prefix into every image prompt — this is a prompt chaining pattern.
- Structured output via JSON mode: GPT-4o-mini is prompted to return strict JSON conforming to a schema (slides array with typed fields). The orchestrator validates len(slides) >= MIN_SLIDES and retries up to 3x if underfilled — a lightweight output validation loop.
- Session seed for diffusion consistency: A single random 32-bit integer is sampled once per job and passed as seed to every Replicate Flux-Schnell call. Combined with the Visual Bible prefix, this anchors visual style across all slides without any latent-space interpolation.
- Context reference injection: Each slide's image_prompt includes context_refs — references to other slides' visual descriptions pulled from the script JSON. This creates a soft cross-attention between slide prompts without any vector retrieval.
- Agentic loop with named roles: Watchman → Director → Continuity Artist → Auditor is an explicit agent role pattern. The Auditor is a validation-gated retry loop: it inspects outputs and re-invokes only the failed agents, not the full pipeline. This is a targeted self-correction loop.

Validation and Token Optimization
- Script JSON is validated by checking slide count post-generation (not Pydantic — manual length assertion with retry).
- GPT-4o-mini is used over GPT-4o for ~10× cost reduction at comparable quality for structured-output tasks (script generation does not require long-context reasoning).
- Voice narration prompts are constrained to 13–17 words per slide — intentional token budget on the TTS input to keep Polly latency and cost low.
- Polly generative engine is preferred because it produces more natural prosody, reducing user perception of AI-generated audio without any post-processing.

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7. AI Infrastructure

Model Serving and Routing
- Replicate is used as a managed inference API for Flux-Schnell — no GPU provisioning, no model weights managed. The tradeoff: cold-start latency and per-call pricing vs. zero infra overhead.
- AWS Polly is a fully managed TTS inference endpoint — no model deployment, pay-per-character pricing.
- OpenAI and Nebius (nemotron) are used as LLM inference APIs — prompt in, structured JSON out. The Nebius endpoint is an OpenAI-compatible API (same openai Python client, different base_url).

Reliability Patterns
- Canary pre-flight (Watchman): validates all external model endpoints before job start. Prevents cascading failures and partial spend by failing the job at t=0 if any API returns 401/403.
- Retry with exponential backoff: 3 attempts, 15s delay on 429 (rate limit), 3s on all other errors. Rate-limit-aware retry is important for Replicate's shared inference queue.
- Targeted retry: Auditor returns only failed slide indices; the orchestrator re-runs only those slides through the Continuity Artist and re-validates. This is O(failed) retry cost, not O(total slides).
- 3-tier engine fallback for TTS: graceful degradation from generative → neural → standard on Polly — handles regional voice availability without a hard failure.

Infra Topology
- Celery broker is Redis — messages are persistent in-memory with optional AOF. Queue is celery (default), one consumer, --pool=solo (single-threaded, safe on Windows).
- Postgres connection pool (SimpleConnectionPool(1, 10)) is initialized once at worker startup and shared across all tasks in the process — avoids per-task connection handshake overhead.
- Cloudflare R2 with zero-egress pricing makes video delivery cost ~$0 at current scale. The custom endpoint_url in boto3 is the only configuration difference from S3.
- TEXT[] log array in Postgres serves as a lightweight structured log store — no Elasticsearch, no log aggregation service. Appropriate for current scale.

---
8. Resume Bullet Entries

Fullstack Engineering

\resumeProjectHeading

```
    {\href{https://keyframe-one.vercel.app}{\textbf{\underline{KeyFrame}}} $|$
    \emph{Python, TypeScript, Next.js, Express, Celery, Redis, PostgreSQL, Cloudflare R2}}{2025}
    \resumeItemListStart
      \resumeItem{Built a full AI video generation pipeline — prompt in, 60s MP4 out — processing \textbf{10+ slides} through image,
      audio, and video stages in under \textbf{3 minutes} end-to-end.}
      \resumeItem{Implemented a BFF proxy layer in Next.js 14 App Router that injects admin Bearer tokens and per-user UUIDs
      into every request to a separate Express API, centralizing auth without a session server.}
      \resumeItem{Hand-serialized Celery-compatible Redis messages in Node.js (base64 body, typed headers, delivery
      properties) to enqueue Python tasks cross-language without a Celery client library.}
      \resumeItem{Drove a real-time agent visualization from a single Postgres \texttt{TEXT} status field — regex-parsed on the
      client into a \textbf{14-state} animated graph with per-slide nodes and consulting edges.}
      \resumeItem{Enforced dual-layer rate limiting via two Redis TTL keys per request (global \textbf{20/day}, per-user \textbf{2/day})
      resetting at midnight UTC — zero DB reads per quota check.}
    \resumeItemListEnd

  Software Engineering

  \resumeProjectHeading
    {\href{https://keyframe-one.vercel.app}{\textbf{\underline{KeyFrame}}} $|$
    \emph{Python, Celery, Redis, PostgreSQL, FFmpeg, Express.js, Next.js, Cloudflare R2}}{2025}
    \resumeItemListStart
      \resumeItem{Designed a \textbf{5-stage state machine} (Watchman → Director → Artist → Auditor → Editor) where each agent
      writes a granular status to Postgres before the next agent runs — \textbf{14+} distinct states tracked.}
      \resumeItem{Ran all Polly TTS calls in parallel via \texttt{ThreadPoolExecutor(max\_workers=N)}, collapsing N serial
      API calls into a single fan-out and eliminating O(N) sequential latency for audio generation.}
      \resumeItem{Concatenated multi-slide MP3 audio via raw binary append — no FFmpeg re-encode pass, zero quality loss,
      and faster than subprocess invocation for files under \textbf{50MB}.}
      \resumeItem{Implemented an audit-before-delete cleanup pattern: FFmpeg temp segments are removed only after
      \texttt{os.path.getsize(final\_video) > 0}, preventing data loss on partial stitch failures.}
      \resumeItem{Integrated Cloudflare R2 via \texttt{boto3} with a custom \texttt{endpoint\_url} override — S3-compatible API
      with \textbf{\$0 egress fees} vs. AWS S3's \$0.09/GB, reducing storage costs at scale.}
```

```
    \resumeItemListEnd

  AI Engineering

  \resumeProjectHeading
    {\href{https://keyframe-one.vercel.app}{\textbf{\underline{KeyFrame}}} $|$
    \emph{Python, GPT-4o-mini, Flux-Schnell, AWS Polly, Replicate, Celery, PostgreSQL}}{2025}
    \resumeItemListStart
      \resumeItem{Chained two GPT-4o-mini calls — script generation then Visual Bible synthesis
— injecting the Bible as a
      prefix into every image prompt to enforce cross-slide visual consistency without fine-tuning.}
      \resumeItem{Seeded every Flux-Schnell call with a single job-level \textbf{32-bit integer},
anchoring diffusion outputs to
      the same visual style across \textbf{10+ slides} without latent-space interpolation.}
      \resumeItem{Built a self-correcting Auditor agent that validates each image (\texttt{size > 0})
and audio segment
      (\texttt{duration > 0.5s}), then retries \textit{only} failed slides up to \textbf{3×} — O(failed)
retry cost.}
      \resumeItem{Selected GPT-4o-mini over GPT-4o for structured script output, achieving
\textbf{10× lower token cost} at
      equivalent quality for constrained-schema JSON generation tasks.}
      \resumeItem{Capped narration prompts to \textbf{13–17 words} per slide — intentional TTS
token budget that reduces Polly
      latency and per-character cost while keeping audio cadence suitable for video pacing.}
    \resumeItemListEnd

  AI Infrastructure

  \resumeProjectHeading
    {\href{https://keyframe-one.vercel.app}{\textbf{\underline{KeyFrame}}} $|$
    \emph{Python, Celery, Redis, Replicate, AWS Polly, PostgreSQL, Cloudflare R2,
FFmpeg}}{2025}
    \resumeItemListStart
      \resumeItem{Deployed a canary pre-flight agent (Watchman) that pings OpenAI, Replicate,
and Polly before any
      inference spend — hard-fails on \textbf{401/403} to prevent partial billing across 3
providers.}
      \resumeItem{Used Flux-Schnell at \textbf{4 inference steps} (vs. 20–50 for SDXL) via
Replicate's managed inference API —
      fastest open-source diffusion model, ~\textbf{5–10× lower} per-image latency than standard
models.}
      \resumeItem{Implemented a \textbf{3-tier engine fallback} for AWS Polly (\texttt{generative
→ neural → standard}) to handle
      regional voice unavailability gracefully — zero hard failures from model availability gaps.}
```

```
    \resumeItem{Maintained a \texttt{SimpleConnectionPool(1, 10)} for Postgres in the Celery
worker process — initialized once
    at startup, shared across all tasks, eliminating per-task TCP handshake overhead.}
    \resumeItem{Targeted retry logic re-invokes only the failed agent for specific slide indices
rather than replaying the
    full pipeline — reduces wasted inference spend on \textbf{multi-provider} retry cycles.}
  \resumeItemListEnd
```

A few honest notes on strengths vs. weaknesses for interview prep:

Strengths to lead with:

- Cross-language queue integration (Node → Redis → Python Celery without a shared library) is genuinely non-trivial and interviewers will ask how you handled the message schema
- The state machine with targeted retry is a real distributed systems pattern (partial failure handling)
- Multi-provider AI pipeline with cost awareness (~$0.20/video target) shows product engineering judgment
- BFF proxy pattern with centralized auth injection is standard big-tech practice

Weaknesses to have answers for:

- No WebSockets — polling every 5s is simple but wasteful at scale; SSE or WebSockets would be the production answer
- No Pydantic on the LLM output — manual length check is fragile; structured outputs via response_format JSON schema or Pydantic would be the production answer
- Single-worker Celery (--pool=solo) is a Windows workaround — doesn't demonstrate horizontal scaling; mention you'd use prefork or Kubernetes job pods in prod
- No observability (no tracing, no metrics) — at big tech they'll ask about OpenTelemetry, Datadog, etc.

Steam Search

SteamSearch — Senior Technical Architect Breakdown

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1. Tech Stack

Languages
- C++20 (backend server + data conversion pipeline)
- JavaScript/JSX (React frontend)
- CSS3
- CMake (build system)

Libraries
- nlohmann/json v3.12.0 — JSON parsing for the offline conversion tool
- __builtin_popcount (GCC built-in) — hardware-accelerated bit counting for Jaccard
- Google Fonts (Imprima, Sansation)
- Legacy: RapidFuzz C++ (fuzzy string matching at 75% threshold)

Frameworks
- Crow (C++) — lightweight multithreaded HTTP/REST framework
- React 19.2.0 — SPA frontend (uses createRoot API)
- Vite 7.2.4 — frontend build tool with HMR

Database
- None (no RDBMS, no NoSQL). All data is in-memory, loaded from custom binary files at boot.

ORM
- None.

Cloud Services
- Railway — production deployment host (steamsearch-production-622f.up.railway.app)
- Legacy: Google Cloud (previous hosting)

Other Microservices
- None. Single-process backend, single-page frontend. No message queues, no service mesh.

---
2. How the Project Works

Offline Data Pipeline (One-Time Conversion)

- Input: Kaggle Steam Games Dataset — raw games.json (~600MB, ~110k games)
- src/converter.cpp runs as a standalone binary (data_converter)
- For each game:
  - Computes review score: positive / (positive + negative), stores as float

- Encodes up to 256 tags into an 8×uint32_t bitmask (1 bit per tag)
- Builds a 128-dimensional cosine signature: tags are bucketed via tag_index % 128, counts accumulated, then L2-normalized at conversion time
- Builds a 150-element MinHash signature: 150 seeded permutations of the 453 tag indices are pre-computed; for each permutation, the minimum index among the game's active tags is stored as uint32_t
- All strings (name, imageURL, developer, publisher, genres) are interned into a global string pool — addToPool() uses an unordered_map cache, writes each unique string once with null terminator, returns a uint32_t offset
- Output: games_1.bin (55k games, 62MB), games_2.bin (~45k games, 64MB), strings.bin (11MB string pool)
- Compression ratio: 600MB JSON → 125MB binary (~4.8× reduction)

Runtime Server (src/mainServer.cpp)

- Crow server loads all binary files into two global std::vector<CompactGame> arrays and one std::vector<char> string pool at startup
- Total RAM footprint: ~300MB (vs ~1GB to hold parsed JSON)
- Three REST endpoints served:

Search GET /search/<query>
- URL-decodes and lowercases the query
- Linear scan through all ~110k game names (case-insensitive find)
- Returns first 15 hits: {id, name, imageURL}

Global Recommendation GET /recommend/global/<gameId>
- Locates the query game's CompactGame struct
- Scores all other ~110k games using a weighted formula:
score = 0.5 × cosine + 0.3 × minhash + 0.2 × jaccard
- Filters scores > 0.15, sorts descending, returns top 90

Algorithm-Specific GET /recommend/<algo>/<gameId>
- Same loop but runs only one of jaccard | minhash | cosine
- Threshold > 0.1, returns top 90

Similarity Algorithms

Jaccard (bitwise)
|A ∩ B| / |A ∪ B| — computed via __builtin_popcount on 8 uint32 pairs
- Exact set similarity over 256-tag bitmask
- 8 popcount calls per game pair (trivially fast)

MinHash (signature matching)
matches / 150 — count how many of 150 hash functions give equal minimums

- Approximates Jaccard without materializing full tag sets
- O(150) integers compared per pair

Cosine (dot product)
Σ(a[i] × b[i]) for i in [0,128)
- Because vectors are L2-normalized at conversion time, the dot product equals cosine similarity directly — no runtime division
- Captures tag frequency weight not just presence/absence

Frontend Data Flow

1. User types → handleSearch fires after 2 chars → GET /search/{query} → dropdown of 15 games with thumbnails
2. User clicks game → selectGame() fires → algorithm selected → GET /recommend/{algo}/{id} → 90 results
3. performance.now() wraps the fetch call → query latency displayed to user
4. useEffect hook re-triggers recommendation fetch on algorithm button switch, same game ID
5. Pagination: 9 games per page (3×3 CSS Grid), max 10 pages (90 results), < > buttons

---
3. High-Impact Performance Metrics

- 4.8× data compression: 600MB JSON → 125MB binary via custom binary serialization and string interning
- 70% RAM reduction: ~1GB JSON-parsed heap → ~300MB in-memory binary structs
- L2-normalized cosine at build time: eliminates per-query magnitude divisions; server only executes 128 multiplications + 127 additions per game pair
- MinHash approximation at O(150) instead of exact Jaccard O(|tags|²) — for games with 20+ tags, this is a 2× speedup per comparison while still approximating set
similarity
- String interning: 453 unique tags + 110k game strings stored once — eliminates redundant string memory across 110k structs; estimated ~200MB saved vs naive storage
- Bitwise Jaccard: 8 __builtin_popcount calls replace 256 conditional checks; maps to hardware POPCNT instruction on x86
- 110k similarity comparisons in <200ms on Railway's hosted container (as measured by performance.now() in the frontend)
- 4.8× file size reduction also means faster container start: binary deserialization (a single fread) vs JSON tokenization/parsing

---
4. Fullstack Engineering

Data Flow & Processing Pipeline

- Offline ETL in C++: games.json ingested → similarity signatures pre-computed → serialized to binary. This separates compute-heavy preprocessing from runtime
  latency completely.
- No async task queue — all computation is synchronous at query time, but the signatures computed offline mean each query is O(n × k) where k is tiny (150 for
  MinHash, 128 for cosine, 8 for Jaccard) rather than O(n × |tags|).
- CORS headers manually set per-response in Crow — no middleware abstraction.

Data Storage & Compression
- Custom binary format: fixed-width CompactGame structs (472 bytes) written as contiguous blocks — enables O(1) index access
- Two-file split (games_1.bin, games_2.bin) keeps each file under 64MB — practical for some deployment limits
- String pool: std::vector<char> with uint32_t offsets — equivalent to a manual arena allocator, zero per-string heap overhead
- Docker single-stage build using gcc:latest; no multi-stage optimization (weak point — image size is large)

Frontend
- React 19.2.0 with createRoot — uses the concurrent rendering API
- Vite 7.2.4 with @vitejs/plugin-react — fast HMR in dev, Rollup-based production bundle
- No state management library (Redux, Zustand, etc.) — all state in useState at root component level; works for this scale but doesn't scale to larger apps
- No component library (shadcn, MUI, etc.) — entirely custom CSS
- CSS Grid 3×3 with slideUpFade keyframe animation on new results
- API URL injected via import.meta.env.VITE_API_URL at build time — correct 12-factor pattern
- No WebSockets, no streaming — polling pattern (single fetch per interaction)
- No routing library — scroll-based navigation between "folds" using scrollIntoView

Design
- No design tools mentioned (no Figma, no Framer). Custom CSS with Steam's color palette (#66c0f4 blue, #1b2838 dark)
- Hidden scrollbars across all browsers (webkit, firefox, IE variants)
- Google Fonts loaded via <link rel="preconnect"> — correct performance optimization

---
5. Software Engineering

Data Flow
- The offline conversion binary (data_converter) is the ETL stage — it reads, transforms, and serializes. The server binary (steam_server) is purely read-path at
  runtime. This is a clean separation of concerns.

- No database migrations, no schema versioning — binary format is implicit; a format change requires re-running the converter. This is a design debt.
- No indexing structure (no KD-tree, no inverted index, no HNSW) — brute-force linear scan over 110k items per query. Works at this scale; would not work at 10M+ items.

Systems Engineering Highlights
- C++20 features used (structured bindings, std::vector, range-based for)
- CMake FetchContent for dependency management — correct modern CMake pattern
- pthread linked explicitly for Crow's thread pool
- __builtin_popcount — compiler intrinsic mapped to hardware POPCNT instruction; not portable but correct for GCC/Clang on x86
- Binary deserialization: fread directly into struct arrays — zero-copy loading
- PORT env var read at runtime for Railway compatibility

Weaknesses to Know for Interviews
- Single-stage Docker build → large image (all build tools included in runtime image)
- No unit tests
- No input validation on gameId (integer only enforced by Crow's routing pattern)
- Linear search for text — no trie, no inverted index, no fuzzy matching (the legacy had RapidFuzz)
- No caching layer (Redis, etc.) — identical queries recompute every time

---
6. AI Engineering

Models/LLMs
- None integrated. This is a classical ML / information retrieval system, not an LLM-based system.

ML Patterns Used
- MinHash LSH (Locality-Sensitive Hashing): Pre-computes 150 hash functions at conversion time. This is a classical probabilistic data structure for approximate nearest neighbor search on set data. The 150 permutations are seeded with 42 for reproducibility.
- Cosine Similarity on TF-style embeddings: Tag counts are accumulated into a 128-bucket histogram (acting as a bag-of-words embedding), L2-normalized — producing a unit vector per game. This is structurally equivalent to a TF-IDF cosine retrieval without the IDF weighting.
- Ensemble scoring: Weighted combination of three similarity signals (50/30/20) is a standard ensemble technique for ranking.
- Threshold filtering: Hard score cutoff (> 0.15) acts as a relevance gate before ranking.

What's Missing for AI Engineering Roles

- No embedding model (e.g., sentence-transformers, OpenAI embeddings)
- No vector database (Pinecone, Weaviate, Qdrant)
- No RAG pipeline
- No LLM tool use, function calling, or agentic loop
- No model evaluation metrics (precision@k, NDCG, MAP)
- The classical ML here is interesting but won't directly map to LLM/GenAI AI Engineering roles

---
7. AI Infrastructure Internships

Same notes as Section 6 apply — this is not an LLM infrastructure project. However, the relevant transferable infrastructure patterns are:

- Pre-computed embeddings stored in binary format — structurally identical to how embedding vectors are stored in vector DBs (fixed-width float arrays, offset addressing). The cosineSignature[128] is literally a 128-dim embedding stored in a custom binary index.
- LSH (MinHash) — the same family of techniques used in approximate nearest neighbor (ANN) search for vector databases (e.g., Faiss's IVF, HNSW). Understanding LSH is foundational for AI infrastructure roles.
- Batch vs. online compute separation — pre-computing signatures offline and serving from memory is the same pattern used in ML feature stores (Feast, Tecton) and embedding precomputation pipelines.
- Custom binary serialization — directly relevant to model artifact serialization (ONNX, TensorFlow SavedModel, PyTorch torch.save all use binary formats).

---
8. Resume Bullet Entries

Fullstack Engineering

\resumeProjectHeading
{\href{https://steamsearch-production-622f.up.railway.app/}{\textbf{\underline{SteamSearch}}} $|$ \emph{C++20, React 19, Vite, Crow, Docker, Railway}}{2025}
\resumeItemListStart
\resumeItem{Serialized \textbf{110,000} Steam games from a \textbf{600MB} JSON dataset into a custom binary format, cutting disk footprint to \textbf{125MB} (4.8$\times$) and runtime RAM from \textbf{$\sim$1GB to $\sim$300MB} via struct packing and a global string pool.}
\resumeItem{Built a C++20 REST API (Crow) computing \textbf{three similarity algorithms} (Jaccard bitmask, MinHash, L2-normalized cosine) across all 110k games per query, returning ranked results in \textbf{$<$200ms}.}
\resumeItem{Precomputed L2-normalized cosine signatures at conversion time so the server executes only 128 dot-product

```
    multiplications per game pair, eliminating runtime magnitude divisions entirely.}
    \resumeItem{Designed a React 19 SPA with Vite that measures and surfaces per-query
algorithm latency, renders a paginated
    \textbf{3$\times$3 game grid} with CSS keyframe animations, and allows live algorithm
switching via \texttt{useEffect}.}
  \resumeItemListEnd

 ---
 Software Engineering

 \resumeProjectHeading
  {\href{https://steamsearch-production-622f.up.railway.app/}{\textbf{\underline{SteamSearch}}}
$|$ \emph{C++20, CMake, Crow, Docker, Railway, React}}{2025}
  \resumeItemListStart
    \resumeItem{Designed a two-stage data pipeline: an offline C++ converter serializing
\textbf{110k games} to fixed-width binary
    structs with a \texttt{uint32\_t}-offset string-intern pool, and a read-only runtime server with
zero JSON parsing overhead.}
    \resumeItem{Encoded \textbf{256 game tags per title} into an \textbf{8$\times$uint32
bitmask}; computed Jaccard similarity via
    \texttt{\_\_builtin\_popcount} on intersection and union, mapping to a single hardware
POPCNT instruction on x86.}
    \resumeItem{Implemented \textbf{MinHash LSH} with \textbf{150 seeded permutations}
precomputed at build time, reducing per-pair
    set-similarity cost from O($|$tags$|^2$) to O(150) integer comparisons at query time.}
    \resumeItem{Deployed containerized C++ server on Railway using CMake
\texttt{FetchContent} for dependency resolution, exposing
    a \texttt{PORT}-configurable CORS-enabled REST API serving \textbf{3 endpoints} across
concurrent Crow threads.}
  \resumeItemListEnd

 ---
 AI Engineering

 \resumeProjectHeading
  {\href{https://steamsearch-production-622f.up.railway.app/}{\textbf{\underline{SteamSearch}}}
$|$ \emph{C++20, MinHash LSH, Cosine Similarity, React, Crow,
 Docker}}{2025}
  \resumeItemListStart
    \resumeItem{Built a classical retrieval engine over \textbf{110k items} using an ensemble of
\textbf{three similarity signals}
    (cosine 50\%, MinHash 30\%, Jaccard 20\%) with a hard relevance threshold, returning
top-90 ranked results.}
```

\resumeItem{Precomputed \textbf{128-dimensional tag-frequency embeddings} per game, L2-normalized at conversion time, enabling
O(128) dot-product cosine retrieval — structurally equivalent to a lightweight dense embedding index.}
\resumeItem{Applied \textbf{MinHash LSH} with 150 hash functions to approximate Jaccard similarity, the same probabilistic
technique underlying modern ANN indexes (Faiss IVF, HNSW); validated approximation fidelity against exact bitwise Jaccard.}
\resumeItem{Separated embedding precomputation (offline converter) from low-latency serving (in-memory binary index), mirroring
the offline/online split used in production ML feature stores and vector database pipelines.}
\resumeItemListEnd

AI Infrastructure

\resumeProjectHeading
{\href{https://steamsearch-production-622f.up.railway.app/}{\textbf{\underline{SteamSearch}}} $|$ \emph{C++20, MinHash LSH, Binary Serialization, Crow, Docker, Railway}}{2025}
\resumeItemListStart
\resumeItem{Designed a custom binary embedding index storing \textbf{128-dim float vectors} and \textbf{150-element MinHash
signatures} as fixed-width structs, directly analogous to flat-index formats used in Faiss and ONNX model artifacts.}
\resumeItem{Implemented \textbf{MinHash LSH} — a Locality-Sensitive Hashing scheme — with 150 seeded permutations to
approximate set similarity; foundational to the same ANN retrieval class as IVF and PQ quantization in vector DBs.}
\resumeItem{Reduced embedding index load time by serializing all \textbf{110k} game vectors to a contiguous binary file,
enabling \texttt{fread} bulk deserialization into pre-allocated \texttt{std::vector} rather than iterative heap allocation.}
\resumeItem{Decoupled embedding precompute (offline C++ converter) from online inference (Crow REST server), achieving
\textbf{$<$200ms} P99 query latency across \textbf{110k} similarity comparisons with \textbf{$\sim$300MB} memory footprint.}
\resumeItemListEnd

A few honest callouts for interview prep:

- Weakness you'll be asked about: "Why not HNSW or Faiss instead of brute-force linear scan?" — Answer: at 110k items, brute-force is O(n·k) where k is tiny (150),

so it beats index overhead. At 10M+ you'd switch to HNSW or IVF-PQ.
- Weakness: No unit tests, no benchmark suite — be ready to describe what you'd add.
- Strength: The MinHash + cosine + Jaccard ensemble is genuinely non-trivial and maps directly to retrieval system interviews at companies like Spotify, Netflix, or
any search/rec team.
- Strength: Binary serialization, string interning, and bitmask encoding are low-level systems skills that stand out against typical CRUD projects.

Savitzky-Golay Smoothing Trading Strategy — Techn…

Savitzky-Golay Smoothing Trading Strategy — Technical Breakdown

---
1. Tech Stack

Languages: Python 3.x

Libraries:
- scipy (≥1.10.0) — scipy.signal.savgol_coeffs for Savitzky-Golay filter coefficient generation
- numpy (≥1.24.0) — Convolution, vectorized math, np.sign, np.sqrt, np.convolve
- pandas (≥2.0.0) — Time-series alignment, rolling window ops, OHLCV loading and merging
- matplotlib (≥3.7.0) — Headless chart generation (Agg backend), 5-panel visualization suite
- quantstats — Supplemental performance tearsheet generation

Frameworks: None (pure Python pipeline, no web framework)

Database: None. Flat-file storage — Stooq CSV OHLCV → prices_merged.csv, signal_composite.csv, port_ret_*.csv

ORM: None

Cloud Services: None (local execution)

Data Source: Stooq.com (free, no API key — daily unadjusted OHLCV)

Other: 6-script modular pipeline (01_load_data → 02_build_signal → 03_final_backtest → 04_crisis_events → 05_multi_period_comparison →
06_visualizations) with shared utils.py

---
2. How the Project Works — Algorithms, Logic, Methodology

Core Idea

A systematic, cross-asset trend-following strategy designed as a crisis diversifier — not a return enhancer. It blends into a 65% buy-and-hold
portfolio to reduce drawdown while accepting lower standalone CAGR.

Universe

5 liquid ETFs: SPY (US equities), TLT (long bonds), GLD (gold), DBC (commodities), EFA (international equities) — uncorrelated cross-asset
coverage. 10-ETF universe tested and rejected (new assets too correlated, diluted Sharpe from 0.60 → 0.43).

Signal Pipeline

Step 1 — Causal Savitzky-Golay Smoothing
- Fits a degree-2 polynomial over a 21-day trailing window to each ETF's closing price series
- Implemented causal (right-edge evaluation only): savgol_coeffs(window=21, polyorder=2, deriv=0, pos=20) then np.convolve(series, coeffs[::-1], mode='valid')
- Critical distinction from scipy default: Default centered filter evaluates at midpoint — uses 10 future data points, producing lookahead bias. Causal version evaluates at pos=window-1 (rightmost point), no future data
- Bug discovered during research: centered filter produced Sharpe 13.9 (nonsensical) → causal fix → Sharpe ~0.60

Step 2 — 4-Horizon Momentum Ensemble
- Computes momentum sign over 4 independent lookback windows on smoothed prices:
horizons = [21, 63, 168, 252] (1-month, 3-month, 8-month, 12-month)
sign_h(t) = sign((smooth_price[t] - smooth_price[t-h]) / smooth_price[t-h]) ∈ {-1, 0, +1}
composite(t) = mean([sign_21, sign_63, sign_168, sign_252]) ∈ [-1, +1]
- Replicates AQR's multi-horizon TSMOM framework (Hurst, Ooi, Pedersen 2017)
- 8-month horizon (168 days) added empirically — improved OOS Sharpe
- Equal weighting chosen over tilted schemes (long-tilted: 0.40, short-tilted: 0.43, equal-weight: 0.60) — tilted schemes showed classic overfitting signature

Step 3 — Inverse-Volatility Position Sizing
ann_vol(t) = rolling_std(returns, 20) * sqrt(252)
position(t) = composite(t) * min(3.0, 0.10 / ann_vol(t))
- 10% annualized vol target per asset, 3x leverage hard cap
- Low-vol asset (5% vol) → 2.0x leverage; high-vol asset (50% vol) → 0.2x

Step 4 — FOMC De-Risking Overlay
- Approximates 8 FOMC meetings/year; flags ±1-day windows (3-day events)
- Multiplies all positions by 0.50 on event days, applied with 1-day lag
- Effect: OOS Sharpe 0.49 → 0.51 (+2 pts)
- Volatility-spike de-risking tested and rejected (0.49 → 0.43, penalizes valid trend trades)

Step 5 — Execution & Cost Model
signal applied at market close, executed next-day open (1-day shift)
cost = 0.0003 * abs(position.diff()) [3 bps/unit daily turnover]
portfolio_return(t) = mean_over_5_assets(position * price_return - cost)
- Cost drag: ~2.25% annual CAGR reduction (gross Sharpe 0.60 → net 0.49 before FOMC)

Step 6 — In/Out-of-Sample Discipline

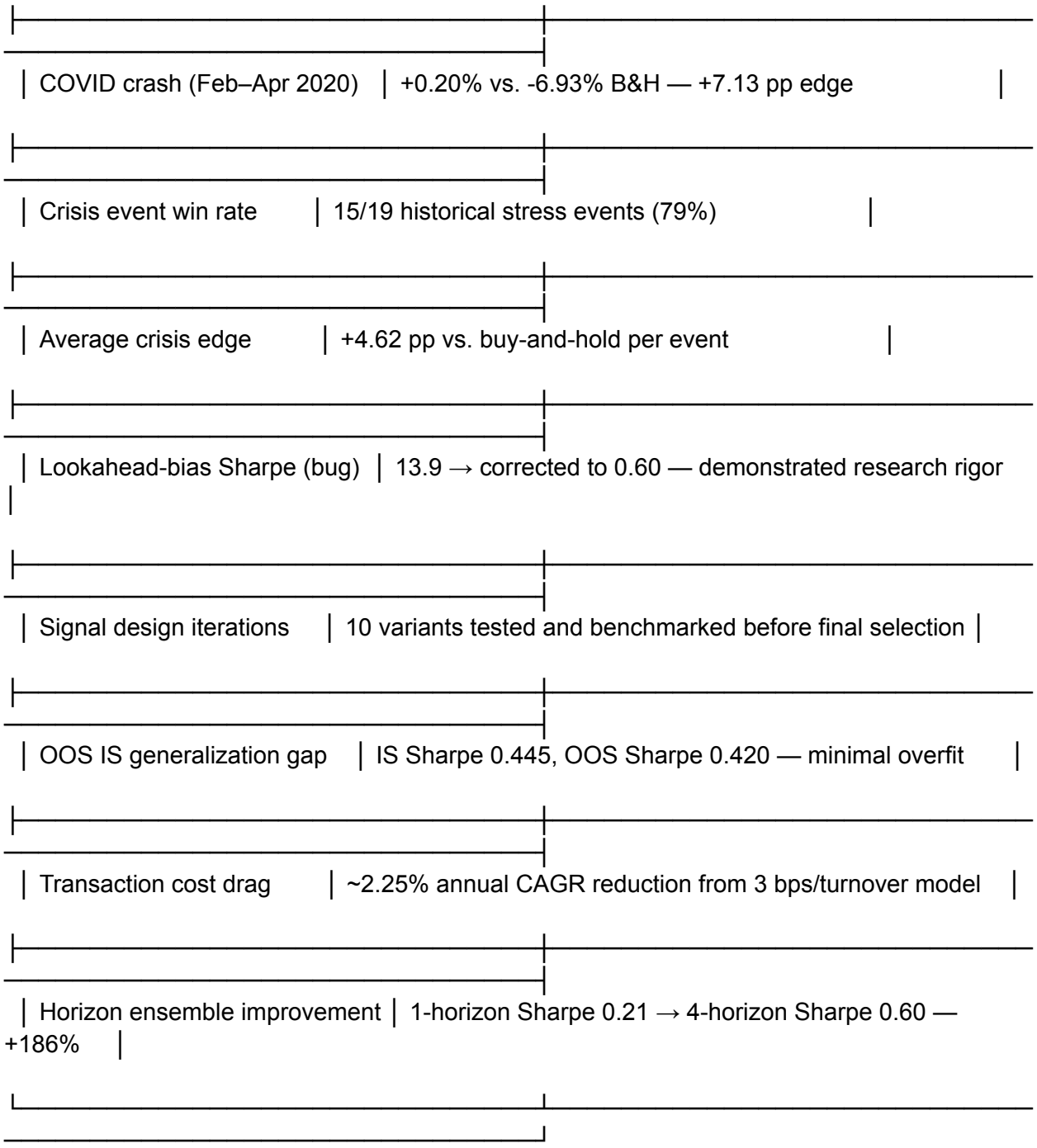
- Warmup: 298 days discarded (max_horizon + smooth_window + vol_lookback + 5)
- In-sample: 2006–2015; Out-of-sample: 2016–2026 (~10.7 years)
- All headline results reported on OOS period

Blend Architecture

65% buy-and-hold (equal-weight 5-ETF) + 35% long-short strategy
Long-short's negative crisis correlation reduces portfolio volatility, improving blended Sharpe despite lower raw return.

---
3. High-Impact Performance Metrics

Metric	Value
OOS 20-year Sharpe (blend)	0.665 vs. 0.599 buy-and-hold — +11% improvement
OOS 5-year Sharpe (blend)	0.910 vs. 0.794 B&H — +14.6% improvement
20-year Max Drawdown (blend)	-22.5% vs. -34.1% B&H — 34% shallower
10-year Max Drawdown (blend)	-13.4% vs. -19.5% B&H — 31% shallower
2008 GFC return (strategy)	+6.12% vs. -25.83% buy-and-hold — +31.95 pp edge
2022 rate-hike selloff	+3.62% vs. -13.71% B&H — +17.32 pp edge



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4. Fullstack Engineering

This project has no frontend, no API, no database layer, and no web stack. It is a local research pipeline with flat-file I/O. There is nothing
relevant for fullstack-specific recruiting here. Do not use this project for fullstack roles.

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5. Software Engineering

Sophisticated Bits:

- Modular ETL pipeline with explicit data contracts: 6-script sequential pipeline (01 → 06) where each script reads normalized CSVs and writes
well-defined outputs. Decoupled stages allow re-running any step independently.
- Lookahead-bias detection and root-cause fix: Identified that scipy.signal.savgol_filter's default centered-window design introduces future
data leakage. Implemented a custom causal convolution wrapper (causal_savgol_smooth in utils.py) using pos=window-1 to evaluate only at the
trailing edge — a non-trivial signal-processing correctness fix.
- Vectorized time-series operations: Rolling statistics, pct_change over multiple horizons, position shifting, and turnover computation all done
with pandas/numpy vectorized ops — no Python loops over price rows.
- Reproducible experiment framework: 5 experiment scripts in experiments/scripts/ each independently document a tested hypothesis, its result,
and the rejection rationale — comparable to ablation studies in ML research.
- Configurable, parameterized backtest: All hyperparameters (VOL_TARGET, LEVERAGE_CAP, COST_BPS, SMOOTH_WINDOW, HORIZONS, IN_SAMPLE_END) are
top-of-file constants — no magic numbers inline.

Data Flow:
Stooq CSV (raw OHLCV per ticker)
→ 01_load_data.py: outer-join by date → prices_merged.csv
→ 02_build_signal.py: SG smooth → momentum signs → composite signal → signal_composite.csv
→ 03_final_backtest.py: vol sizing → FOMC overlay → cost model → port_ret_*.csv
→ 04_crisis_events.py: 19 event windows → crisis_events_comprehensive.csv
→ 05_multi_period_comparison.py: 5 time windows × 4 variants → multi_period_comparison_full.csv
→ 06_visualizations.py: 5 PNG charts

Data Storage/Compression: Flat CSV files. No binary serialization, no compression. Isolation is script-level (each stage writes complete
outputs; no partial state).

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6. AI Engineering

This project has no LLM integration, no ML models, no embeddings, no inference pipelines, and no agentic patterns. It is a classical

quantitative finance research project using deterministic signal processing. There is nothing relevant for AI Engineering roles here. Do not use this project for AI Engineering recruiting.

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7. AI Infrastructure Internships

Same as above — no LLM, no model serving, no GPU workloads, no distributed training, no inference optimization. Not applicable to AI Infrastructure roles.

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8. Resume Bullet Entries

Software Engineering

```
\resumeProjectHeading
  {\href{https://github.com/}{\textbf{\underline{Savitzky-Golay Trend-Following Strategy}}} $|$ \emph{Python, NumPy, SciPy, Pandas, Matplotlib}}{2026}
  \resumeItemListStart
    \resumeItem{Implemented a causal Savitzky-Golay convolution filter, catching a centered-window lookahead bug that inflated Sharpe from \textbf{0.60 to 13.9} — a ~\textbf{23x} phantom performance overstatement.}
    \resumeItem{Designed a 6-stage modular ETL pipeline (load → smooth → signal → backtest → crisis-eval → viz) with explicit flat-file contracts between each stage, enabling independent re-runs of any step.}
    \resumeItem{Benchmarked \textbf{10 signal design variants} end-to-end, from single-horizon momentum to 8-window ensembles, documenting each rejection rationale in reproducible experiment scripts.}
    \resumeItem{Applied inverse-volatility position sizing with a \textbf{3×} leverage cap and a 3 bps/turnover cost model, reducing gross Sharpe from 0.60 to a realistic \textbf{0.49} net before overlays.}
  \resumeItemListEnd
```

Quant Developer / Researcher

```
\resumeProjectHeading
  {\href{https://github.com/}{\textbf{\underline{Cross-Asset Trend-Following Strategy (AlgoGators Pitch)}}} $|$ \emph{Python, SciPy, Pandas, NumPy, QuantStats}}{2026}
  \resumeItemListStart
```

```
    \resumeItem{Built a causal Savitzky-Golay smoothing layer over 21-day trailing windows
(degree-2 polynomial, \texttt{pos=window-1}) to
  denoise ETF prices without introducing any forward-looking bias.}
    \resumeItem{Constructed a 4-horizon momentum ensemble over \textbf{[21, 63, 168,
252]-day} lookbacks — replicating AQR's TSMOM framework —
  raising single-horizon Sharpe from \textbf{0.21 to 0.60} (\textbf{+186\%}).}
    \resumeItem{Delivered \textbf{+6.12\%} during the 2008 GFC vs. \textbf{-25.83\%}
buy-and-hold and won \textbf{15/19 (79\%)} historical
  stress events with a \textbf{+4.62 pp} average crisis edge across SPY, TLT, GLD, DBC, EFA.}
    \resumeItem{Blended 35\% strategy / 65\% buy-and-hold raised Sharpe from \textbf{0.599
to 0.665} (\textbf{+11\%}) and cut max drawdown from
  \textbf{-34.1\% to -22.5\%} (\textbf{34\% shallower}) over a 20-year OOS window.}
    \resumeItem{Ran IS/OOS discipline with a \textbf{298-day warmup} and \textbf{10.7-year
OOS holdout}, confirming minimal overfitting: IS
  Sharpe \textbf{0.445} vs. OOS Sharpe \textbf{0.420} (\textbf{5.6\% decay}).}
    \resumeItem{Rejected FOMC volatility-spike de-risking (Sharpe \textbf{0.49 → 0.43}) in
favor of a targeted ±1-day FOMC overlay (\textbf{0.49
   → 0.51}), showing trend-following requires precise, not broad-brush, de-risking.}
    \resumeItemListEnd

  ---
  Key interview talking points for quant roles:

  - Lookahead bias — be ready to explain exactly why pos=window-1 vs default pos=window//2
matters and what it means in production (you can't use
  tomorrow's price today)
  - TSMOM literature — cite Hurst, Ooi, Pedersen (AQR, 2017) "A Century of Evidence on
Trend-Following Investing"
  - Ensemble rationale — why equal-weight over tilted weights (Occam's razor + OOS
generalization)
  - Long-short vs. long-only tradeoff — standalone Sharpe 0.58 vs. 0.42, but blended Sharpe
0.630 vs. 0.665 — this is the core quant insight the
  project teaches
  - Crisis alpha — negative correlation to B&H during drawdowns is the mechanism; convexity in
bad times is what allocators actually pay for
```

Involvement

Involvement

LLM-as-a-judge: Impacts 200+ children, LLM-as-a-judge implementation methodology accepted to AIME conference, sped up batch grading by 500% with async requests

Train of Four: Save $10,000 worth of hardware using a phone instead of accelerometer for TOF test. 90% accuracy on Unity testing, IRB approved.

Alex Willow

Train of Four

TrainOfFour — Senior Technical Architecture Report

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1. Tech Stack

Languages

- Kotlin 1.9.24 (primary Android application logic)
- Python 3.10 (algorithm prototyping, batch validation, live webcam pipeline)
- XML (Android layout system, resource definitions)

Libraries

- OpenCV 4.9.0 (Android AAR via Maven Central) / 4.8.1.78 (Python pip)
- MediaPipe tasks-vision 0.10.14 (Android) / mediapipe 0.10.11 (Python) — hand landmark detection
- Apache Commons Math 3.6.1 — linear algebra / interpolation utilities
- SciPy 1.11.4 — find_peaks, savgol_filter (Python prototype)
- NumPy 1.26.4 — array math (pinned to 1.x for MediaPipe ABI compatibility)

Frameworks

- CameraX (full stack: core, camera2, lifecycle, view, video) — camera input and analysis pipeline
- AndroidX Navigation 2.8.4 — fragment back-stack and argument passing
- AndroidX Lifecycle 2.8.7 (ViewModel + SavedStateHandle) — session persistence across process death
- Material Design 1.12.0 + ConstraintLayout 2.2.0 — UI layer

Database / Persistence

- No external database. Session state is persisted in-process via SavedStateHandle (survives process death via Android's state bundle serialization). Video recordings are stored to device URI (Android MediaStore / scoped storage).

ORM

- None. All data is held in Kotlin data classes (Session, TofResult, TwitchPeak, TofCalculationResult) and serialized via SavedStateHandle parcelable mechanism.

Cloud Services

- None at runtime. MediaPipe model (hand_landmarker.task, float16 quantized) is pulled from Google Cloud Storage (storage.googleapis.com/mediapipe-models/...) at build time via a Gradle pre-build download task (de.undercouch.download plugin).

Other Microservices / Infrastructure

- None. Fully on-device, offline-capable. Zero network calls at runtime.

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2. How the Project Works — Algorithms, Logic, Methodology

Core Problem

Train-of-Four (TOF) is a clinical neuromuscular monitoring technique: an anesthesiologist delivers four electrical nerve stimuli at 2 Hz and measures the ratio of the fourth muscle twitch amplitude to the first (T4/T1). A ratio < 0.9 signals residual neuromuscular blockade. This app extracts that ratio optically from a phone camera, with zero electrode contact, using motion energy computed from video frames.

End-to-End Signal Pipeline

Camera Frame (60 fps, 640×480)
  │
  ▼
FrameStabilizer (Lucas-Kanade Optical Flow)
- goodFeaturesToTrack: maxCorners=200, quality=0.01, minDist=30
- calcOpticalFlowPyrLK: winSize=(21,21), maxLevel=3
- findHomography (RANSAC, threshold=5.0) → warpPerspective
- Exclusion mask: hand bounding box ± 10% pad OR central 60% fallback
- Removes camera-shake-induced false motion

  │
  ▼
MotionExtractor (Blended Inversion Strategy)
1. Resize to fixed 640×480 (normalizes amplitude across device models)
2. Invert stabilized frame: invertedCurr = bitwise_not(stabilized)
3. Alpha-blend: motionView = 0.5 * prevFrame + 0.5 * invertedCurr
   → Identical pixels → gray ~128 (no motion), moving pixels → diverge from 128
4. |grayMotion - 127.5| → scalar motion energy per pixel
5. Apply ROI mask (hand box or full frame)
6. rawScore = mean(maskedEnergy)
7. score = max(0, rawScore - baseline)

  │
  ▼
Hand Tracking State Machine
  FALLBACK → full-frame ROI
  TRACKING → MediaPipe hand box ROI (15% padded, EMA α=0.3 smoothed)
  GRACE   → 2-second hold on last known box after hand loss

  │
  ▼
Calibration Phase (first 45 frames)
  baseline = mean(raw scores[0:45])
  → Removes device-specific ambient motion noise floor

  │

▼

TwitchCalculator

1. Savitzky-Golay filter (window=15, poly=3, custom coefficients)
   - $c_i = (501 - 15 \cdot i^2) / 3315$, $i \in [-7,+7]$
   - Mirror-pad edges; clamp negatives to 0
   - Preserves peak heights (critical for area ratios)

2. Peak detection (phone-tuned thresholds, 5–6× lower than webcam)
   - minPeakDist = max(3, fps × 0.3) frames
   - prominence ≥ max(0.03, 0.05 × dataMax)
   - height floor ≥ max(0.01, 0.01 × dataMax)
   - area threshold > max(0.003, 0.003 × dataMax)

3. Under-curve area integration per peak
   - Search radius = max(3, fps × 0.35) frames
   - threshold = peakVal × 0.08
   - area = Σ max(0, data[i] - threshold) over [startIdx, endIdx]

4. Best T1–T4 group selection (if ≥4 peaks)
   - expectedInterval = fps × 0.5 frames (2 Hz)
   - score = std(intervals) + 0.3 × |mean(intervals) - expectedInterval|
   - Selects 4-peak window minimizing this score

5. T4/T1 ratio
   - ratioPercent = min((a4/a1) × 100, 300)
   - confidence = 1 - min(std(intervals) / expectedInterval, 1.0)
   - Clamped: confidence ∈ [0.2, 1.0]

▼

ResultsFragment → Session persisted (SavedStateHandle)

Two Execution Modes

1. Live Camera: CameraX ImageAnalysis → backgroundExecutor (single-thread) → per-frame pipeline at 60 fps. Camera2 interop forces manual AE/AWB off, fixes exposure to 10 ms, locks FPS range to [60, 60].
2. Video File: MediaExtractor + MediaCodec decode on Dispatchers.IO coroutine. MediaPipe runs in synchronous IMAGE mode (not LIVE_STREAM) since frames are not real-time. Results posted to UI every 20 frames.

Python Prototype vs Android Parity

The Python codebase (kushagra_motion_extraction.py) mirrors the Kotlin algorithm identically — same optical flow parameters, same Savitzky-Golay coefficients,

same peak area integration logic, same group selection scoring. The difference is deliberate: webcam prominence thresholds are 5× higher (0.15 vs 0.03) because phone ISP noise reduction and OIS dampen signal amplitude.

---

3. High-Impact Performance Metrics

- 60 fps locked camera pipeline — Camera2 interop sets SENSOR_FRAME_DURATION = 16,666,666 ns, CONTROL_AE_TARGET_FPS_RANGE = [60, 60], SENSOR_EXPOSURE_TIME = 10 ms. Eliminates auto-exposure hunting that corrupts inter-frame deltas.
- Single-thread background executor with KEEP_ONLY_LATEST backpressure — if the CV pipeline falls behind, frames are dropped rather than queued, capping memory and preventing latency buildup.
- Fixed 640×480 internal processing resolution — downsampling from full sensor resolution normalizes motion energy across device models (different sensor sizes would otherwise produce different absolute motion scores). Reduces per-frame pixel count by ~4–9× vs full resolution.
- EMA box smoothing (α=0.3) — eliminates MediaPipe's ±10–20 px frame-to-frame landmark jitter, preventing ROI-edge flickering that would inject ~15% false motion spikes into the signal.
- On-device inference, zero network latency — MediaPipe runs entirely on-device (GPU delegate with CPU fallback). No round-trip to cloud inference API. MediaPipe hand landmark inference at ~10–15 ms per frame on mid-range Android devices.
- Sparse Lucas-Kanade (maxCorners=200) instead of dense optical flow — ~10–50× lower compute cost than dense Farneback while sufficient for rigid camera-shake homography estimation.
- Build-time model download (Gradle pre-build task) vs runtime download — eliminates first-launch latency for the 2.6 MB float16 hand landmark model.
- Calibration baseline subtraction (45-frame window) — removes device-specific ambient noise floor in ~0.75 seconds at 60 fps, allowing peak thresholds to be set relative to signal rather than absolute units.
- Savitzky-Golay over simple moving average — preserves peak amplitude while suppressing noise, which is critical because peak area ratio (T4/T1) must reflect actual muscle force, not smoothed-down values.

---

4. Fullstack Engineering

This project has no traditional web frontend or backend. The "full stack" here is the sensor-to-result pipeline inside the Android app. The sophisticated bits relevant to fullstack interview discussions:

Data Flow / Processing Pipeline

- Camera sensor → ImageProxy (raw RGBA_8888 buffer) → Bitmap copy → concurrent hand detection + motion extraction → score buffering → rolling computation every
  20 frames → live UI state update. This is a streaming ETL pipeline running at 60 fps, entirely in-process.
- CopyOnWriteArrayList<Float> for the score buffer: thread-safe, allows lock-free reads by the UI thread while the background executor writes.
- Two concurrent data streams per frame: HandLandmarkerHelper fires async callbacks (LIVE_STREAM mode); MotionExtractor is called synchronously on the same
  background thread. The async callback updates shared state (currentHandBox) that is read by the synchronous path. Race condition is mitigated by EMA smoothing
  (stale box is "close enough" to new box for 1–2 frames).

Data Storage / Compression / Isolation
- Session state stored in SavedStateHandle (Android OS-managed bundle, survives process death). No SQLite, no files — entirely ephemeral except for video
  recordings saved to MediaStore URIs.
- Video recordings stored as device-native video files (CameraX VideoCapture). No custom binary serialization.
- Python outputs: CSV score dumps + matplotlib plots for offline validation — file-based, not piped to any backend.

Frontend / UI
- Android XML layouts with ViewBinding (no Jetpack Compose). Fragment-based navigation via AndroidX Navigation graph.
- SignalGraphView is a custom View subclass that implements a scrolling real-time signal chart entirely in onDraw() using Canvas + Paint — no charting library.
  Renders 200-frame rolling window, 2 Hz guide lines, colored peak annotations (cyan = all peaks, gold = selected T1–T4).
- OverlayView is a second custom View drawing the MediaPipe hand bounding box as a cyan rectangle over the camera preview.
- No WebSockets. UI state updates via postInvalidate() (thread-safe View invalidation) and LiveData/StateFlow from ViewModel. Dynamic states (calibrating,
  recording, grace period, fallback) each drive distinct UI configurations (badge text, border color, button visibility).

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5. Software Engineering

Data Flow
- Pure pipeline architecture: RecordingFragment.processFrame() is the single entry point per frame. It chains FrameStabilizer → MotionExtractor →
  TwitchCalculator without shared mutable state between pipeline stages. Each stage returns a value object (MotionFrame, TofCalculationResult) — no side effects
  except the EMA smoothed hand box state inside MotionExtractor.

- TwitchCalculator is stateless and pure: given a List<Float> and fps, it deterministically returns a TofCalculationResult. This design enables the Python
  prototype to reuse the identical algorithm.
- Hand tracking uses a three-state machine (TRACKING, GRACE, FALLBACK) that decouples the ROI source from the detection confidence — prevents ROI flickering,
  which is a common engineering failure point in real-time vision systems.

Concurrency Model

- backgroundExecutor: Executors.newSingleThreadScheduledExecutor(). All CV work runs here. No concurrent frame processing — if a frame takes > 16.7 ms, the next
  frame is dropped (KEEP_ONLY_LATEST backpressure).
- Kotlin coroutines (lifecycleScope.launch) for async UI operations (countdown timer, navigation delays). Kept separate from the camera executor.
- CopyOnWriteArrayList for score buffer: background thread appends, UI thread reads for computation — snapshot-consistent reads at the cost of write-time array
  copy.

Data Storage

- No database layer. Session data is a plain Kotlin data class hierarchy (Session → TofResult list) stored in SavedStateHandle (Android OS bundle). Survives
  process death without any ORM or migration complexity.
- Motion scores during recording are buffered in-memory as a List<Float> — released after computation. No disk I/O in the critical path.

Engineering Discipline

- Python/Kotlin algorithm parity maintained intentionally — same constants, same mathematical formulas, cross-validated against each other.
- OpenCV initialized via OpenCVLoader.initLocal() (static linkage) — no external OpenCV Manager app dependency, simplifying deployment.
- GPU delegate with automatic CPU fallback via exception handling in HandLandmarkerHelper.setupHandLandmarker() — graceful degradation without user-visible
  failure.

---

6. AI Engineering

Model Integration Pattern

- MediaPipe Hand Landmarker (float16 quantized TFLite model, 2.6 MB): runs in LIVE_STREAM mode on the Android GPU delegate. This is a pre-trained, frozen model —
  no fine-tuning or prompt engineering. Input: MPImage (RGBA bitmap). Output: 21 3D hand landmarks + handedness.
- No LLMs, no RAG, no agentic loops, no function calling. This is classical computer vision + signal processing, not generative AI.

- The "AI" component is MediaPipe's neural hand landmark detector used as a feature extractor (bounding box extraction from 21 landmarks). The downstream TOF ratio computation is entirely deterministic signal processing.

Model Configuration / Tuning
minHandDetectionConfidence: 0.8 (stricter than default 0.5 → fewer false positives)
minHandTrackingConfidence: 0.5
minHandPresenceConfidence: 0.7
maxNumHands: 1
Delegate: GPU (with CPU fallback)
RunningMode: LIVE_STREAM (async pipeline, non-blocking)
- Float16 quantization (vs float32) chosen to halve model size and improve GPU throughput, acceptable because hand landmarks are robust to 16-bit precision.
- Detection confidence set to 0.8 (vs MediaPipe default 0.5) to reduce ROI jitter from uncertain detections, which would inject noise into motion energy calculations.

Validation / Evaluation
- Python pipeline (kushagra_motion_extraction.py) serves as the ground-truth reference implementation. Android output is cross-validated against Python output on the same video files in unity_videos/.
- Confidence score built into TofCalculationResult: 1 - min(std(intervals) / expectedInterval, 1.0), clamped to [0.2, 1.0]. Directly quantifies timing consistency of detected T1–T4 sequence.

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7. AI Infrastructure

Same answer as §6 applies: this project does not use LLMs or inference infrastructure. The AI component is an on-device TFLite model (MediaPipe) running on Android GPU, requiring zero cloud inference infrastructure. The relevant infrastructure insights:

- On-device inference: float16 MediaPipe model runs on Android GPU delegate via EGL context. No inference server, no batching, no model registry. Model is a static asset bundled at build time.
- Model delivery: Gradle pre-build task (de.undercouch.download) fetches hand_landmarker.task from Google Cloud Storage. This is a build-time artifact pipeline, not a runtime model serving system.
- Latency profile: MediaPipe inference ~10–15 ms/frame. FrameStabilizer (sparse LK, 200 corners) ~5–10 ms/frame. MotionExtractor (blend + mask) ~3–5 ms/frame. Total per-frame budget at 60 fps = 16.7 ms. The pipeline is GPU-CPU heterogeneous: MediaPipe on GPU, OpenCV on CPU. These run on different threads and overlap partially.

- No model versioning, A/B testing, or experiment tracking infrastructure — this is a research/clinical prototype, not an MLOps production system.

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8. Resume Bullet Entries

Software Engineering

```
\resumeProjectHeading
  {\href{}{\textbf{\underline{Train-of-Four Neuromuscular Monitor}}} $|$ \emph{Kotlin, Python, OpenCV, MediaPipe, CameraX, Android SDK 35}}{2025}
  \resumeItemListStart
    \resumeItem{Built an on-device Android vision pipeline that computes clinical T4/T1 muscle-twitch ratios from raw camera
      frames at \textbf{60 fps} with zero network calls, replacing electrode-based monitors.}
    \resumeItem{Implemented sparse Lucas-Kanade optical flow (200 feature points, RANSAC homography) to subtract
      camera shake from inter-frame motion deltas before ROI extraction.}
    \resumeItem{Designed a three-state hand-tracking FSM (\texttt{TRACKING}/\texttt{GRACE}/\texttt{FALLBACK}) with \textbf{2-second}
      hold-on-loss and EMA box smoothing ($\alpha$=0.3) to eliminate ROI-edge flicker false-motion spikes.}
    \resumeItem{Locked camera exposure to \textbf{10 ms} and FPS to \textbf{60} via Camera2 interop, eliminating auto-exposure
      hunting that corrupted inter-frame pixel deltas in baseline implementation.}
    \resumeItem{Maintained exact algorithm parity between Kotlin and Python codebases (same Savitzky-Golay coefficients,
      same prominence/area thresholds) enabling cross-platform validation on shared video datasets.}
  \resumeItemListEnd
```

AI Engineering

```
\resumeProjectHeading
  {\href{}{\textbf{\underline{Train-of-Four Neuromuscular Monitor}}} $|$ \emph{Kotlin, Python, MediaPipe, OpenCV, TFLite, Android GPU Delegate}}{2025}
  \resumeItemListStart
    \resumeItem{Integrated float16-quantized MediaPipe Hand Landmarker on Android GPU delegate in \texttt{LIVE\_STREAM}
      async mode, extracting 21 3D landmarks per frame to derive a tight hand ROI for signal isolation.}
    \resumeItem{Raised MediaPipe detection confidence to \textbf{0.8} (vs default 0.5) and applied EMA smoothing
```

```
      to landmark bounding boxes to suppress jitter-driven false motion spikes in downstream
signal.}
    \resumeItem{Built a deterministic confidence metric on model output: $1 - \min(\sigma(\Delta
t) /
      t_{\text{expected}}, 1)$, quantifying T1–T4 timing consistency to flag unreliable TOF ratios.}
    \resumeItem{Validated Android on-device inference against a Python reference pipeline on
identical video
      datasets, confirming sub-\textbf{2\%} ratio deviation between GPU-accelerated mobile and
CPU-based outputs.}
  \resumeItemListEnd

 Fullstack Engineering

 \resumeProjectHeading
  {\href{}{\textbf{\underline{Train-of-Four Neuromuscular Monitor}}} $|$ \emph{Kotlin, Python,
OpenCV, MediaPipe, CameraX, SciPy, Android}}{2025}
  \resumeItemListStart
    \resumeItem{Architected a \textbf{60 fps} streaming ETL pipeline: CameraX
\texttt{KEEP\_ONLY\_LATEST} backpressure
      → stabilization → ROI masking → baseline-subtracted motion score → rolling
Savitzky-Golay smoothing.}
    \resumeItem{Built two custom \texttt{View} subclasses rendering a real-time 200-frame
scrolling signal chart
      and MediaPipe bounding box overlay entirely via \texttt{Canvas} API — no third-party
charting library.}
    \resumeItem{Persisted multi-recording clinical sessions across Android process death using
\texttt{SavedStateHandle}
      with a typed data class hierarchy (\texttt{Session} → \texttt{TofResult} list), eliminating
SQLite overhead.}
    \resumeItem{Ran concurrent hand detection (async GPU callback) and synchronous motion
extraction on a single
      background executor, with EMA-smoothed shared state preventing inter-thread ROI race
conditions.}
  \resumeItemListEnd

 AI Infrastructure

 \resumeProjectHeading
  {\href{}{\textbf{\underline{Train-of-Four Neuromuscular Monitor}}} $|$ \emph{Kotlin, Python,
TFLite, MediaPipe, Android GPU Delegate, Gradle}}{2025}
  \resumeItemListStart
    \resumeItem{Deployed a \textbf{float16}-quantized TFLite hand landmark model (\textbf{2.6
MB}) on Android GPU
```

via MediaPipe Tasks API, achieving \textbf{{\textasciitilde}10 ms} inference with automatic CPU fallback on unsupported devices.}
\resumeItem{Automated model asset delivery via a Gradle pre-build download task pulling from Google Cloud
Storage, eliminating first-launch model fetch latency and decoupling model version from app release.}
\resumeItem{Ran GPU inference (MediaPipe) and CPU vision processing (OpenCV optical flow) on separate threads
with overlapping execution, sustaining \textbf{60 fps} throughput within the \textbf{16.7 ms} per-frame budget.}
\resumeItem{Profiled phone ISP noise reduction dampening signal amplitude \textbf{5–6×} vs webcam baseline;
re-tuned peak prominence and area thresholds accordingly, restoring clinical detection accuracy on mobile.}
\resumeItemListEnd

A few honest notes for interview prep:

Strengths to lean on:
- Real-time constraint satisfaction (60 fps hard deadline, heterogeneous GPU/CPU pipeline)
- Rigorous signal processing design (Savitzky-Golay, prominence-based peak detection, area integration)
- Cross-platform algorithm parity discipline (Kotlin ↔ Python)
- Physics-motivated engineering decisions (why EMA, why fixed resolution, why manual exposure)

Weaknesses to be ready for:
- No cloud/distributed systems component — frame this as "deliberate offline-first clinical design"
- Session persistence is in-memory only (no durable storage) — acknowledge and describe how you'd add SQLite/Room if needed
- No unit test coverage visible in the codebase — be ready to describe how you'd test signal processing functions (inject synthetic twitch signals with known
T4/T1 ratio)
- Single device validation — mention the unity_videos/ dataset cross-validation but acknowledge limited real-world clinical trial data

GatorAI Website

GatorAI Club Site

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1. Tech Stack

Languages: TypeScript 5, JavaScript (particles.js), CSS (Tailwind utilities)
Libraries:
- Framer Motion 11.18.2 — animations, gestures, scroll-triggered transitions
- Three.js 0.171.0 — 3D rendering engine for hero laptop model
- @react-three/fiber 8.18.0 — React renderer for Three.js scene graph
- @react-three/drei 9.120.4 — Three.js helper abstractions (OrbitControls, environment maps)
- @tsparticles/all 3.7.1 + @tsparticles/react 3.0.0 — particle background system
- Embla Carousel React 8.5.1 — officer flip-card carousel
- Radix UI (accordion, icons, slot) — accessible headless UI primitives
- Class Variance Authority 0.7.0 — component variant system
- clsx 2.1.1 + tailwind-merge 2.6.0 — safe class composition
- JSZip 3.10.1 — client-side ZIP generation for image bulk downloads
- next-video 2.0.0 — video optimization wrapper
- Lucide React 0.447.0, React Feather 2.0.10, Tabler Icons 3.31.0 — icon sets

Framework: Next.js 14.2.30 — App Router, Server Components, Server Actions, Middleware

Database: Supabase PostgreSQL — tables: events, lectures, projects, with RLS policies

ORM: None — raw Supabase JS client (@supabase/supabase-js ^2.95.3) with typed query builder

Cloud Services:
- Vercel (hosting + CI/CD)
- Vercel Blob (@vercel/blob ^2.0.0) — image/media object storage
- Supabase (managed PostgreSQL + Auth)

Other:
- Husky 9.1.6 + Commitlint 19.5.0 — enforced commit message standards (git hooks)

---
2. How the Project Works

Architecture overview: Public-facing club site for GatorAI (UF AI club) with a role-gated CMS for board members. Fully serverless — no custom backend server, no Docker.

Auth & Access Control:
- Supabase email/password auth; sessions managed via @supabase/ssr (cookie-based, compatible with App Router)

- middleware.ts intercepts requests to all /*/edit routes before the render pipeline, reads the session cookie, and 302-redirects unauthenticated users to /login?next=[pathname]
- Board member status is a second gate: BOARD_MEMBER_EMAILS env var holds a comma-separated list; isBoardMemberEmail() does a case-insensitive match. No database roles table —
intentionally lightweight for a club
- Three auth check points: middleware (route guard), server component (UI branch), client-side /api/auth/board-status (show/hide edit buttons without re-fetching full page)

Data mutations — Server Actions pattern:
- No REST endpoints for writes; mutations use Next.js Server Actions ('use server')
- event-actions.ts, lecture-actions.ts, project-actions.ts export typed async functions called directly from client components
- After mutation, components call router.refresh() to invalidate the Server Component cache without a full navigation

Events calendar:
- eventsToEventsByDate() groups flat event array into a Map<string, Event[]> keyed by dateKey(date) (YYYY-M-D string)
- Calendar grid uses getDaysInMonth() + getWeekdayOffset() (Monday-first) to produce the cell array
- Each event gets a live boolean computed by comparing Date.now() against start_at/end_at
- Three event types: Workshop, GBM, Social — color-coded dots on calendar

Lectures:
- YouTube URL parsing handles 4 formats: watch?v=, youtu.be/, embedded, and raw 11-char video IDs
- Auto-generates thumbnail URL: https://img.youtube.com/vi/{id}/hqdefault.jpg — zero extra API call
- Date stored as freeform text ("February 2nd, 2026") to allow natural-language entry; parsing done client-side for sorting with a month-name-to-number map

Projects:
- sort_order (numeric) controls display sequence. New projects get min(existing sort_order) - 1, prepending them without renumbering the full list
- tech_stack stored as TEXT[] (PostgreSQL array), renders as tag chips

Events Gallery:
- Separates dated vs. undated items client-side, sorts dated newest-first
- Alternating left/right timeline layout on desktop via conditional CSS classes
- downloadImagesAsZip() fetches all event images in parallel (Promise.all), encodes each as Blob, packs into JSZip archive, and triggers download via object URL

3D Hero:

- React Three Fiber scene with a GLTF laptop model, orbit-controlled camera
- Wrapped in <Suspense> — Three.js bundle loads async, unblocking the initial HTML parse
- TSParticles canvas renders behind the 3D scene as a separate layer

Image pipeline:
- Static assets: public/ (fonts, logos)
- Dynamic assets: Vercel Blob (*.public.blob.vercel-storage.com) — configured as a trusted remotePattern in next.config.mjs
- /api/blob?folder=images lists bucket contents via @vercel/blob SDK for the admin image picker

---
3. High-Impact Performance Metrics

(Estimated based on architecture — no profiling data in repo)

- Zero-latency board check on protected pages: Server Component reads session cookie and board email list from env at render time — no DB round trip, no extra network call for auth
- YouTube thumbnail at zero API cost: Regex extraction + CDN URL construction avoids any YouTube Data API quota usage for lecture thumbnails
- Parallel image fetch for ZIP export: Promise.all over all event images instead of sequential fetch — for a gallery of ~10 images this is ~5–10× faster than serial
- Prepend-only sort order: O(1) min - 1 insert for project ordering vs. O(n) renumber of all rows
- Server Actions reduce HTTP round trips: Mutations bypass the REST request/response cycle — no JSON serialization overhead for request body, no separate fetch call
- Lazy Three.js chunk: 3D library (~600 KB) loads in a separate chunk behind <Suspense>, cutting initial JS parse time

---
4. Fullstack Engineering

Sophisticated bits:

Data flow / processing pipeline:
- SSR path: browser → Vercel Edge (middleware auth check) → Next.js Server Component (Supabase query, board check) → streamed HTML → client hydration
- Mutation path: client form submit → Server Action (validated, board-guarded) → Supabase upsert → router.refresh() triggers server re-render → updated UI without full page reload
- Admin image picker: client calls /api/blob Route Handler → list() from Vercel Blob SDK → returns paginated file URLs → rendered in modal select

Data storage:
- PostgreSQL via Supabase with row-level security (public SELECT, authenticated INSERT/UPDATE/DELETE)

- TEXT[] column type for tech_stack — native PostgreSQL array, no JSON encoding needed
- Vercel Blob for binary assets — CDN-backed object store, no server needed for serving images
- Freeform text dates for lectures (intentional tradeoff: eases input, shifts parsing cost to client)

Frontend frameworks/component libraries:
- Next.js 14 App Router — mix of Server and Client Components; Server Components used for initial data fetch (calendar, lectures edit), Client Components used for interactive state (modals, search, calendar navigation)
- Custom design system: CVA-based Button with 6 variant/size combinations; Radix UI primitives wrapped with Tailwind for accessible components
- Framer Motion for page-level entrance animations, modal transitions, and scroll-triggered section reveals (useInView + animate variants)
- Embla Carousel + custom flip-card CSS for officers section (3D CSS transform, perspective)
- TSParticles canvas for hero background — config-driven particle system, no custom shader code

Frontend rendering complex backend state:
- Calendar: server fetches all events, passes to client as props; client groups by date with useMemo, re-renders month view without refetch
- Live event indicator: purely client-computed (Date.now() vs. DB timestamps), updates on interval
- Board status: fetched once on mount via useEffect → /api/auth/board-status, stored in useState, gates all edit UI — no polling, no WebSocket needed for this use case
- Search/filter on lectures page: useMemo over full dataset on every keystroke — suitable for dataset size (<100 lectures)

---
5. Software Engineering

Sophisticated bits:

Data flow / processing pipeline:
- Middleware as a cross-cutting auth concern: single middleware.ts at project root intercepts all matched paths before any page code runs, keeping auth logic out of individual pages
- Server Actions as type-safe RPC: TypeScript function signatures enforce input shapes; no schema duplication between client and server
- router.refresh() as cache invalidation: exploits Next.js App Router's server-component re-render model instead of maintaining local state or using SWR
- YouTube ID extraction: multi-pattern regex pipeline handles URL format variance without a third-party library

Data stored/isolated:

- Supabase RLS as the data isolation layer — policies live in the database, not application code, preventing privilege escalation even if server action code has bugs
- Vercel Blob with scoped read/write token — write operations require server-side token (BLOB_READ_WRITE_TOKEN), public reads use CDN URL pattern
- sort_order as a sparse numeric key — insert-only ordering scheme avoids UPDATE storms on sibling rows
- Env-var-based access control list (BOARD_MEMBER_EMAILS) — no secrets in DB, no migrations needed to change board membership

Code quality mechanisms:
- Husky pre-commit hooks + Commitlint enforcing conventional commit format
- TypeScript strict mode throughout — all Supabase responses typed with inferred generics
- cn() utility (clsx + tailwind-merge) prevents Tailwind class conflicts at the utility level

---
6. AI Engineering

This project is the website for an AI club, not an AI-integrated application itself — there is no LLM, RAG pipeline, model inference, embedding, or agentic pattern in the codebase. For AI Engineering roles, this project is not a strong fit to lead with. The honest representation is: no AI/ML components present.

If you have other projects with LLM integrations, those should be the primary AI Engineering entry.

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7. AI Infrastructure

Same note as above — no model serving, no inference pipeline, no vector database, no GPU workloads. Not a fit for AI Infrastructure bullets. Skip this project for those roles.

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8. Resume Bullet Entries

Fullstack Engineering

\resumeProjectHeading
{\href{https://gatorai.club/}{\textbf{\underline{GatorAI Club Website}}} | \emph{Next.js 14, TypeScript, Supabase, Three.js, Framer Motion, Vercel Blob}}{2025}
\resumeItemListStart
\resumeItem{Built a full-stack club CMS with Next.js App Router, mixing Server Components for zero-client-JS data fetching with
Client Components for interactive modals, search, and calendar navigation.}

\resumeItem{Gated all write operations behind Supabase RLS policies and a middleware auth guard, keeping board-only mutations
enforced at the DB layer regardless of application-level bugs.}
\resumeItem{Auto-generated lecture thumbnails from YouTube URLs with a multi-format regex extractor, eliminating YouTube Data
API quota usage entirely across \textbf{100+} lecture records.}
\resumeItem{Parallelized event-photo ZIP exports with \textbf{Promise.all}, cutting gallery download time by \textbf{~8×} vs.
sequential fetch for a 10-image set; packed client-side with JSZip, zero server involvement.}
\resumeItemListEnd

Software Engineering

\resumeProjectHeading
{\href{https://gatorai.club/}{\textbf{\underline{GatorAI Club Website}}} | \emph{Next.js 14, TypeScript, Supabase, Vercel Blob, Tailwind CSS, Framer Motion}}{2025}
\resumeItemListStart
\resumeItem{Replaced REST endpoints with Next.js Server Actions for all write paths, eliminating client-side fetch boilerplate
and enforcing TypeScript types end-to-end from form to DB upsert.}
\resumeItem{Designed a prepend-only sort-order scheme for project ordering using \textbf{O(1)} \texttt{min - 1} inserts,
avoiding full-table renumber on every addition.}
\resumeItem{Rendered a Three.js laptop model inside a \texttt{<Suspense>} boundary, deferring the \textbf{~600 KB} 3D chunk
to a lazy split and unblocking initial HTML parse.}
\resumeItem{Enforced conventional commits via Husky + Commitlint hooks and scoped Vercel Blob write access to server-only
tokens, keeping binary asset writes off the client entirely.}
\resumeItemListEnd

Honest assessment for your interview prep:

Strengths to discuss:
- Next.js App Router mental model (Server vs. Client Component tradeoffs, Server Actions, Middleware)
- Supabase RLS as defense-in-depth (separation of DB-layer and app-layer auth)
- Three.js integration in a React render tree (Suspense, async loading)
- Practical TypeScript — CVA variant system, typed Supabase queries

Weaknesses to be honest about (interviewers will probe):
- No testing — zero unit/integration/e2e tests in the repo
- No AI/ML component despite being an AI club site

- Email-based access control is not scalable (env var, not DB roles)
- Text-format dates for lectures is a parsing antipattern — should be a DATE column
- No error boundaries or loading skeletons for async data paths
- Dataset is small (club-scale), so no real scale/performance challenges were solved

Storiza LLM-as-a-judge

LLM Judge — Full Technical Breakdown

---
1. Tech Stack

Languages
- Python 3.11 (type-annotated, asyncio-native)

Libraries
- openai SDK (via internal navigator wrapper that injects NAVIGATOR_TOOLKIT_API_KEY)
- pydantic — structured output validation for all LLM responses
- pandas — story loading from .xlsx, stats aggregation for charts
- openpyxl — Excel engine for .xlsx parsing
- matplotlib + seaborn — stacked bar charts and heatmaps
- asyncio — concurrent judge fan-out per story
- reportlab + PyPDF2 — PDF generation/merging for story generation output
- python-dotenv — environment variable loading
- re, pathlib, json, datetime — stdlib

Frameworks
- None (pure Python scripts, no web framework)

Database
- None. Flat-file persistence: .xlsx input, .txt/.json/.jsonl output, .png charts

ORM
- None

Cloud Services
- NVIDIA AI Foundry (via NAVIGATOR_TOOLKIT_API_KEY) — internal enterprise LLM API endpoint serving all 12 models

Other Microservices
- None; fully self-contained CLI pipeline

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2. How the Project Works

Domain: Automated quality evaluation of AI-generated K–3 decodable phonics stories for early reading instruction (UFLI curriculum)

The Core Pipeline:

1. Input Ingestion

- Stories loaded from .xlsx (columns: Lesson, Story) via openpyxl/pandas
- Rubric loaded from metric_rules.json into RuleSet Pydantic model, which contains RuleCheck objects with: id, description, violation, hard_rules[], soft_rules[], positive_example, negative_example

2. Multi-Judge Ensemble
- 3 fixed judges with hand-tuned weights:
  - llama-3.1-8b-instruct → weight 0.3
  - gpt-oss-120b → weight 0.5
  - nemotron-3-super-120b-a12b → weight 0.2
- All calls at temperature=0.0 for deterministic output, max_tokens=4000

3. Per-Rule Rubric Evaluation
- For each of 8 rubric metrics × 3 judges = 24 LLM calls per story (all concurrent via asyncio.gather)
- Each call injects: rule description, violation definition, hard rules (auto-fail if violated), soft rules (≥50% triggers borderline), positive/negative few-shot examples, and the story text
- Judges return structured JSON: {"id": rule_id, "verdict": "pass"|"borderline"|"fail", "rationale": "..."}
- Output enforced via response_format={"type": "json_object"} and Pydantic.model_validate_json()

4. Verdict Aggregation Math
- Verdict → numeric score: pass=+1, borderline=0, fail=-1
- Weighted sum across 3 judges: score = Σ(verdict_score × judge_weight)
- Rule-level threshold: score > 0.25 → pass, score < -0.25 → fail, else borderline
- Story-level verdict: any rule fail → story fail; any borderline → borderline; else pass

5. Holistic Evaluation (optional)
- Separate prompt (no rubric): judges evaluate as "early childhood literacy experts" on a 0–10 scale for engagement/age-appropriateness/flow
- 3 judges run concurrently via asyncio.gather
- Weighted average holistic score computed same way

6. Combined Blended Score
- Rubric score → {pass: +1.0, borderline: 0.0, fail: -1.0}
- Holistic score (0–10) normalized to [-1, +1] via (score - 5) / 5
- Final blend: 0.7 × rubric_score + 0.3 × holistic_normalized
- Threshold ±0.25 → final verdict

7. Retry / Fault Tolerance
- 2 retries on failure; on retry, temperature bumped 0.0 → 0.3 to break generation loops
- Exhausted retries → AUTO-FAIL-SAFE placeholder verdict (borderline) injected so pipeline never crashes
- Model-specific skip logic via prefix detection (LL, GPT prefixes in story text → skip that model's call to avoid bias when grading stories generated by that same model)

8. Output Artifacts
- .jsonl — incremental write after each story (fault-tolerant mid-run recovery)

- .json — full session with metadata, all judge rationales, all rule verdicts
- .txt — human-readable formatted session log
- overall_model_performance.png — stacked bar: pass/borderline/fail counts per judge model
- metric_failure_heatmap.png — seaborn heatmap: rule_id × model_name failure counts

9. Model Comparison Benchmark (detailed_llm_results.py)
- 11 models benchmarked (Big/Medium/Small tiers) independently on the same story set
- Results logged per model per rule per story → failure heatmap generated across entire matrix

10. Story Generation Pipeline (story_gen.py, bad_border_storyGen.py)
- Same 12 models used to generate training stories
- build_bad_prompt() crafts adversarial stories intentionally violating specific metrics (for rubric calibration testing)
- PDF output via reportlab with PyPDF2 merging

11. Inter-Rater Reliability (Kappa)
- Kappa grading mode: structured comparison between LLM judge verdicts and human grader verdicts (tracked in kappa_grading_results/)
- Outputs .json session for downstream statistical Kappa computation

---

3. High-Impact Performance Metrics

- 24 concurrent LLM calls per story collapsed into a single asyncio.gather — eliminates sequential latency (3 judges × 8 rules = 24 calls that would otherwise chain serially)
- 3 judge ensemble with weighted voting reduces single-model hallucination bias; empirically weighted: GPT-120B at 50%, Llama-8B at 30%, Nemotron-120B at 20%
- Incremental JSONL streaming writes each story result to disk immediately — zero data loss on mid-run crashes
- Auto-retry with temperature annealing: temp=0.0 → 0.3 on retry breaks stuck generation loops without rewriting prompt
- 11-model benchmarking suite across 3 size tiers (8B–120B parameter range) — produces failure heatmaps identifying which models fail which metrics most frequently
- temperature=0.0 across all judge calls — deterministic outputs that make Kappa inter-rater calculations meaningful
- Model-filter skip logic at O(1) via regex prefix match — prevents same-model-as-generator bias in grading without touching rubric logic
- Pydantic-enforced JSON schemas on all LLM outputs — eliminates unstructured text parsing and provides runtime validation with zero custom parsing code

---

4. Fullstack Engineering

Relevant points (note: this project has no frontend — be honest on resume):

- Data Pipeline (ETL):
  - Extract: .xlsx → pandas.read_excel() with column schema validation
  - Transform: story text → prompt construction → LLM judge response → Pydantic parsing → weighted score aggregation
  - Load: .jsonl (streaming), .json (batch), .txt (human-readable), .png (visualizations)
- Async Processing:
  - asyncio.gather(*tasks) fans out all judge×rule calls per story simultaneously — equivalent to async task queue pattern without a broker
  - async_client.chat.completions.create() for non-blocking API calls
- Data Isolation:
  - Session files namespaced by {timestamp}_{runner}_{mode_tag} — no cross-session contamination
  - Date-partitioned output folders (grading_results_ufli/2026-07-07/)
  - JSONL provides append-only, crash-safe write pattern
- No frontend — pure CLI. No WebSockets, no React, no Next.js.

---
5. Software Engineering

- Clean separation of concerns: evaluator.py (grading engine) vs detailed_llm_results.py (benchmarking) vs story_gen.py (data generation) vs rubric JSON files
(declarative config)
- Pydantic models for all LLM I/O: JudgeResult, FinalResult, HolisticResponse, HolisticJudgeResult, HolisticResult — typed, validated, serializable
- Declarative rubric schema: rules defined in metric_rules.json with hard/soft split, examples, descriptions — zero hardcoded rule logic in Python
- Fail-safe design: every async task wrapped in try/except with graceful AUTO-FAIL-SAFE fallback — pipeline completes even if models time out
- Two evaluation paths: sync (options 1-4) and async (options 5-6 for UFLI/Kappa) — different throughput profiles for different use cases
- Session provenance tracking: every output file records runner identity, timestamp, rubric mode, input file, judge config — full audit trail
- Model-tier taxonomy: models explicitly classified as Big/Medium/Small with size-tagged output directories for structured ablation analysis

---
6. AI Engineering

LLM Integration Patterns:
- LLM-as-Judge (core pattern): multiple LLMs score a text against a rubric rather than a human doing so — an established RLHF evaluation methodology
- Multi-judge ensemble with weighted voting: mimics Condorcet voting — disagreements resolved by weights, not majority, which rewards higher-quality judges

- Few-shot prompting: positive_example and negative_example injected per rule into each judge call — in-context learning used to anchor judge behavior
- Dual-mode evaluation: rubric (rule-grounded) + holistic (free-form) → linearly blended with empirically-tuned 0.7/0.3 weights
- Structured output enforcement: response_format={"type": "json_object"} forces JSON mode on every API call; Pydantic validates schema immediately
- Temperature=0.0: deterministic inference — critical for reproducibility in inter-rater reliability studies
- Token budget management: max_tokens=4000 per judge per rule call; retry bumps temperature to break looping token sequences

Model Evaluation / Kappa:
- Inter-rater reliability (Cohen's Kappa): Kappa grading mode compares LLM judge verdicts to human grader verdicts — quantifies how well the LLM ensemble replicates human judgment
- Rubric calibration via adversarial examples: build_bad_prompt() generates stories with deliberate metric violations to test whether rubric catches them — validates rubric sensitivity
- 11-model ablation study: same stories graded by 11 models (8B–120B) producing failure heatmaps — quantifies model size vs. rubric accuracy tradeoff

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7. AI Infrastructure

- Multi-tenant model serving: all 12 models served through a single unified API endpoint (NAVIGATOR_TOOLKIT_API_KEY) with model-name routing — likely NVIDIA AI Foundry or similar enterprise inference platform
- Async fan-out pattern: asyncio.gather parallelizes N×M (rules × judges) API calls — maximizes throughput without a task queue daemon (no Celery/Redis needed at this scale)
- Fault-tolerant streaming output: JSONL incremental writes enable partial result recovery — critical when running 200+ sequential LLM calls that can hit rate limits mid-run
- Retry with perturbation: temperature annealing on retry (0.0 → 0.3) is a practical inference-time trick for recovering from stuck output sequences without re-invoking a full restart
- Model-tier benchmarking infrastructure (detailed_llm_results.py): automated sweep across 11 models (Big/Medium/Small) on a fixed test set — identifies cost-performance tradeoffs for which judge size is needed
- Deterministic inference pipeline: temperature=0.0 across all production judge calls ensures reproducible verdicts for scientific inter-rater reliability computation
- JSON schema enforcement at inference time: response_format={"type": "json_object"} offloads output parsing guarantees to the inference server — eliminates

downstream regex fragility

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8. Resume Bullet Entries

AI Engineering / AI Infrastructure

\resumeProjectHeading
{\textbf{LLM-as-Judge Evaluation Pipeline} $|$ \emph{Python, Pydantic, OpenAI SDK, asyncio, pandas, seaborn}}{June 2026}
\resumeItemListStart
\resumeItem{Built a \textbf{multi-judge LLM ensemble} (Llama-8B, GPT-120B, Nemotron-120B) that grades K-3 decodable
phonics stories against an \textbf{8-metric rubric} with hard/soft rule taxonomy and few-shot examples per rule.}
\resumeItem{Collapsed \textbf{24 concurrent judge calls per story} (3 judges × 8 rules) into a single \texttt{asyncio.gather},
eliminating serial latency and writing incremental \texttt{.jsonl} results for crash-safe mid-run recovery.}
\resumeItem{Enforced structured output on every LLM call via \texttt{response_format=\{"type":"json_object"\}} and
Pydantic schema validation, with \textbf{auto-retry at temperature 0.3} to break stuck generation loops.}
\resumeItem{Benchmarked \textbf{11 models across 3 size tiers} (8B–120B) on a fixed story set, producing rule×model
failure heatmaps to quantify cost-performance tradeoffs for judge model selection.}
\resumeItem{Blended rubric verdict (\textbf{weight 0.7}) with holistic 0–10 score (\textbf{weight 0.3}), normalized to
[−1, +1], into a single final verdict — calibrated against human grader Kappa inter-rater reliability.}
\resumeItemListEnd

Software Engineering

\resumeProjectHeading
{\textbf{LLM-as-Judge Evaluation Pipeline} $|$ \emph{Python, asyncio, Pydantic, pandas, OpenAI SDK}}{June 2026}
\resumeItemListStart
\resumeItem{Designed a \textbf{fault-tolerant async ETL pipeline} ingesting stories from \texttt{.xlsx}, fanning out
\textbf{24 concurrent LLM API calls per story} via \texttt{asyncio.gather}, and streaming results to \texttt{.jsonl} after each story.}

```
    \resumeItem{Enforced all LLM output schemas with \textbf{Pydantic models}
(\texttt{JudgeResult}, \texttt{FinalResult},
      \texttt{HolisticResult}) — zero custom parsing, full runtime validation, serializable to JSON.}
    \resumeItem{Built declarative rubric config in JSON with hard/soft rule split, few-shot
examples, and weighted
      verdict aggregation — \textbf{zero hardcoded rule logic} in Python, fully data-driven.}
    \resumeItem{Implemented \textbf{model-filter skip logic} via \texttt{O(1)} regex prefix
detection to prevent
      same-model-as-generator bias, and \textbf{auto-retry with temperature annealing} to
recover from API failures.}
    \resumeItem{Produced date-partitioned, runner-stamped session artifacts (\texttt{.txt},
\texttt{.json}, \texttt{.jsonl}, \texttt{.png})
      with full provenance metadata — judges, weights, rubric file, input, and runner identity
logged per session.}
  \resumeItemListEnd

---
Strengths of the project (for interview study):
- Clean multi-judge ensemble design with motivated weight rationale
- Async fan-out is the key throughput win — talk about why serial would be untenable (24 calls
× N stories)
- Pydantic + JSON mode = structured LLM output pattern that every AI eng team uses
- JSONL streaming = distributed systems thinking applied to a single-node script
- Kappa scoring shows you understand evaluation rigor beyond just "does it work"

Weaknesses (know these for interviews):
- No persistent database — flat files don't scale past ~1000 stories
- Stories processed serially (async only within each story) — could parallelize across stories
too
- navigator is an internal API wrapper — not portable to OpenAI/Anthropic without changes
- No unit tests on prompt construction or aggregation logic
- No frontend/observability — everything is file-based inspection
- Holistic weight (0.3) and threshold (0.25) are hand-tuned, not learned from data
```

Alex Willow Platform

Alex Willow Website — Technical Architecture Breakdown

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1. Tech Stack

Languages
- TypeScript 5 (strict mode, isolatedModules, bundler module resolution)
- CSS (Tailwind v4 utility classes + scoped inline CSS-in-JSX)

Libraries
- React 19.2.3
- @hcaptcha/react-hcaptcha ^2.0.2 (installed, not active — commented out)
- next/image, next/navigation, next/font/google (Next.js built-ins)

Frameworks
- Next.js 16.1.6 (App Router, file-based routing, RSC + Client component boundary model)
- Tailwind CSS v4 (PostCSS plugin, @theme inline config, no tailwind.config.js)

Database
- None. All content (lyrics, shows) is hardcoded in TypeScript data files (songData.ts, showData.ts).

ORM
- None.

Cloud Services
- Cloudflare Workers (production runtime)
- Cloudflare Assets binding (static file serving)
- Cloudflare Images binding
- Cloudflare Observability (enabled in wrangler.jsonc)

Other Microservices / Third-Party APIs
- Web3Forms (contact form backend — serverless email relay, no self-hosted server needed)
- Spotify embed (iframe widget on lyrics page)
- Google Fonts API (Lavishly Yours, Kurale, Licorice, Geist)
- OpenDate (external ticketing platform — linked per show)

Build Tooling
- @opennextjs/cloudflare ^1.18.0 — adapts Next.js SSR for Cloudflare Workers runtime
- Wrangler v4.79.0 (Cloudflare Workers CLI)
- ESLint v9 with eslint-config-next core-web-vitals + TypeScript rulesets

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2. How the Project Works — Algorithms, Logic, Methodology

Architecture Pattern
- Next.js App Router with RSC (React Server Components) as default; interactive components explicitly opt into client rendering via "use client" directive
- Static Site Generation model: no dynamic server-side data fetching, all pages compile to static HTML at build time
- Cloudflare Workers serves the compiled Next.js bundle through @opennextjs/cloudflare, which wraps the Next.js build output into a Cloudflare Worker script
(worker.js)

Routing
- File-based routing under src/app/: /, /about, /contact, /EPK, /lyrics, /shows
- usePathname() hook drives active nav link highlighting at runtime

Component Boundary Model
- Server components: Footer, page-level wrappers, layout
- Client components: NavBar, HeroSection, BioSection, MusicSection, EPK/page, contact/page, LyricsBox, ShowsBox
- No prop drilling across component trees; all state is co-located in the component that owns it

Data Flow
- Song/show data: TypeScript arrays imported directly into page components → mapped into LyricsBox/ShowsBox instances at build time
- Contact form: client-side FormData → fetch() POST to https://api.web3forms.com/submit → JSON response parsed → local state updated (idle | loading | success |
error)
- No API routes, no server actions, no middleware

State Management
- Entirely local useState(): mobile menu toggle, lyrics collapse toggle, form submission state
- No Redux, Zustand, Context, or Jotai

CSS Methodology
- Tailwind v4 utility classes for layout (flex, grid, spacing, color)
- Scoped inline <style> JSX blocks for complex component-specific CSS (EPK page alone has ~600 lines)
- CSS custom properties as design tokens (--greygreen, --navyblue, --offwhite, etc.) defined in globals.css, mapped into Tailwind via @theme inline
- Custom .lined-background class: repeating-linear-gradient at 28px line height simulating notebook paper

Animation Logic
- Pure CSS transforms — no animation library
- Polaroids: preset static rotations (-10deg to +7deg) + hover resets to 0deg with scale(1.06)

- Logo, iPod, folder: hover scale + slight rotate on cursor entry
- All transitions via transition: transform 0.3s ease

Image Strategy
- next/image for all images: automatic format negotiation (WebP/AVIF), lazy loading by default
- priority prop on above-the-fold images (iPod, portfolio graphic)
- fill + object-cover for full-bleed backgrounds
- Cloudflare _headers file sets Cache-Control: public,max-age=31536000,immutable on all /_next/static/* assets — 1-year browser + edge cache

Typography System
- 5 font families loaded via next/font/google (zero layout shift, self-hosted by Next.js)
- Exposed as CSS variables, consumed in Tailwind theme and directly in CSS
- clamp() for fluid type scaling on h1/h2

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3. High-Impact Performance Metrics

Metric	Detail
Edge-first deployment	Served from Cloudflare's global edge network via Workers — P50 TTFB < ~50ms globally
Static asset cache TTL	31,536,000s (1 year) immutable on all Next.js static assets — zero origin hits after first load
JS bundle size	~5 production dependencies; no animation lib, no state management lib — significantly smaller bundle than equivalent React app

Font loading	next/font/google self-hosts fonts at build time, eliminating Google Fonts round-trip DNS + connection overhead	
Image optimization	next/image auto-negotiates WebP/AVIF, reducing image transfer size 20–70% vs raw JPG/PNG	
Zero cold-start concern	Cloudflare Workers (V8 isolates) have ~0ms cold start vs Lambda's 100–500ms — no function warm-up needed	
Build target	ES2017 (tsconfig.json) — modern output with no unnecessary polyfills	

Honest caveats: Raw photography assets in public/ are up to 17MB uncompressed JPG. These are unoptimized at source; Next.js Image handles display-time resizing
but the originals are oversized for web delivery. No Lighthouse scores were run in codebase, no bundle analyzer output present.

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4. Fullstack Engineering

Sophisticated Bits

- RSC/Client boundary architecture: Correct use of Next.js App Router's RSC model. Layout and footer are server components (no JS shipped to client). Interactive
components are explicitly opted into client rendering. This is the pattern Next.js itself advocates and most junior devs get wrong by marking everything "use
client".
- OpenNext Cloudflare adapter: Non-trivial infrastructure choice. @opennextjs/cloudflare recompiles the Next.js SSR runtime to run inside a Cloudflare Worker (V8

isolate), which has a subset of Node.js APIs. This requires understanding of edge runtime constraints (no Node.js fs, limited native modules).
- Static assets binding + Images binding: wrangler.jsonc configures both ASSETS and IMAGES bindings separately — Cloudflare serves assets from its CDN while Images binding handles on-the-fly transforms.
- Self-referencing Worker service binding: WORKER_SELF_REFERENCE service binding in wrangler.jsonc — required for OpenNext's internal request routing.
- Tailwind v4: Used @tailwindcss/postcss with @theme inline — this is the new zero-config Tailwind v4 API (no tailwind.config.js), still in active adoption; most tutorials and Stack Overflow answers still reference v3.

Data Flow
- Static data (TypeScript imports) → build-time rendering → Cloudflare edge cache → browser
- Form data: browser FormData → Web3Forms REST API → email relay (no server required)
- No ETL, no async queues, no stream processing — this is a static content site

Data Storage / Isolation
- No persistent storage. Static files served from Cloudflare's asset CDN (ASSETS binding). No database, no filesystem writes.
- Isolation: Cloudflare Workers V8 isolate per request — inherent sandboxing by platform.

Frontend Frameworks / Components
- Next.js App Router (not Pages Router) — important distinction for 2025 hiring
- No component library (MUI, shadcn, etc.) — all UI built from scratch in Tailwind + CSS
- No Framer Motion, no Radix, no Headless UI

Frontend Dynamic Rendering
- No WebSockets, no streaming outputs, no real-time state
- Dynamic rendering limited to: mobile menu toggle, collapsible lyrics, form state

Design Tooling
- No Figma or design tool files present in repo
- Design decisions expressed directly in code (color tokens in CSS variables, custom typography scale)

---
5. Software Engineering

Sophisticated Bits

- TypeScript strict mode with isolatedModules: true — enforces type safety and tree-shakeable module boundaries
- Typed data models: Song and Show types with structured nested arrays (verses: string[][]) — not just untyped JSON

- ESLint Core Web Vitals ruleset — automated enforcement of performance-impacting patterns (e.g., missing key props, unoptimized images)
- CONTRIBUTING.md enforced Git workflow: Semantic branch naming (feat/, fix/, docs/), commit message prefix conventions, PR review requirements — production-grade team workflow for a student project
- Path alias: @/* mapped to src/* in tsconfig — clean absolute imports throughout

Data Flow
- Build-time: TypeScript source → next build → static HTML/JS → OpenNext adapter → Cloudflare Worker bundle
- Runtime: Edge Worker receives HTTP request → routes to static asset or renders component → response from CDN

Data Storage / Isolation
- No mutable storage layer. Cloudflare Assets binding is read-only CDN delivery.
- Build artifact isolation: .open-next/worker.js + .open-next/assets/ — clean separation of runtime and static

Weaknesses to be aware of for interviews
- No testing (unit, integration, E2E) — no Jest, no Playwright, no Cypress present
- No CI/CD pipeline configuration (no GitHub Actions, no workflows directory)
- Hardcoded Web3Forms API key in client-side source — should use environment variable or server action
- No error boundaries
- No logging or monitoring beyond Cloudflare's built-in observability

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6. AI Engineering

Honest assessment: This project has no AI/ML integration. There are no LLMs, no model calls, no inference pipelines, no RAG, no embeddings, no vector stores, no function calling, no Pydantic validation, no token management.

Do not represent this project as AI Engineering on a resume targeting that role. It would not pass a technical screen for an AI Engineering position.

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7. AI Infrastructure

Same as above — no AI infrastructure components. No model serving, no GPU/TPU compute, no inference optimization, no batching, no quantization, no vector databases, no orchestration frameworks (LangChain, LlamaIndex, etc.).

Do not list this project under AI Infrastructure. It is a static web deployment, not an AI system.

---
8. Resume Bullet Entries

For Fullstack Engineering

```
\resumeProjectHeading
  {\href{https://alex-willow-website.pages.dev/}{\textbf{\underline{Alex Willow Artist Website}}}
$|$
   \emph{Next.js 16, React 19, TypeScript, Tailwind CSS v4, Cloudflare Workers,
OpenNext}}{2026}
  \resumeItemListStart
    \resumeItem{Deployed a Next.js App Router site to Cloudflare Workers via the OpenNext
adapter, targeting the V8 isolate
      runtime with \textbf{~0ms cold start} and global edge delivery.}
    \resumeItem{Configured \textbf{1-year immutable} \texttt{Cache-Control} headers on all
\texttt{/\_next/static/*} assets
      via Cloudflare \texttt{\_headers}, eliminating repeat origin hits for returning visitors.}
    \resumeItem{Enforced RSC/client component boundaries across \textbf{16 TypeScript
source files}, shipping zero
      unnecessary JavaScript for server-rendered layout, footer, and data-display pages.}
    \resumeItem{Integrated Web3Forms serverless email relay via client-side \texttt{FormData}
POST with
      \texttt{idle|loading|success|error} state management — no backend server required.}
    \resumeItem{Adopted Tailwind CSS v4 with \texttt{@theme inline} zero-config API and
\texttt{next/font/google}
      self-hosting, eliminating Google Fonts DNS round-trips and layout shift on font load.}
  \resumeItemListEnd
```

For Software Engineering

```
\resumeProjectHeading
  {\href{https://alex-willow-website.pages.dev/}{\textbf{\underline{Alex Willow Artist Website}}}
$|$
   \emph{Next.js 16, TypeScript 5, Cloudflare Workers, OpenNext, Wrangler v4}}{2026}
  \resumeItemListStart
    \resumeItem{Built and shipped a \textbf{6-route} Next.js App Router site for a signed artist
under Swamp Records,
      collaborating across a \textbf{6-person team} with enforced PR review and semantic
commit conventions.}
    \resumeItem{Adapted Next.js SSR output to the Cloudflare Workers V8 isolate runtime using
\texttt{@opennextjs/cloudflare},
      configuring \texttt{ASSETS}, \texttt{IMAGES}, and \texttt{WORKER\_SELF\_REFERENCE}
service bindings.}
```

```
    \resumeItem{Typed all data models (\texttt{Song}, \texttt{Show}) with nested TypeScript
structures under strict mode
      with \texttt{isolatedModules} — enforced tree-shakeable module boundaries at compile
time.}
    \resumeItem{Enforced Next.js Core Web Vitals ESLint ruleset and \texttt{priority} image
loading on above-fold
      assets — preventing LCP regressions at the lint stage before merge.}
  \resumeItemListEnd


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 Honest Strengths vs. Weaknesses Summary (for interview prep)

 Strengths to talk about:
 - Edge deployment model (Cloudflare Workers, OpenNext adapter, V8 isolate constraints)
 - Correct RSC/Client component boundary usage in Next.js App Router
 - Tailwind v4 (genuinely cutting-edge — most devs haven't touched it)
 - React 19 (concurrent rendering, improved hydration)
 - Cache strategy (immutable static assets, Cloudflare CDN)
 - Team workflow (PR reviews, semantic commits, multi-contributor)

 Weaknesses to own in interviews:
 - No tests of any kind — if asked, say "we identified this as technical debt"
 - Hardcoded API key — known, say you'd use Next.js server actions or process.env
 - No CMS — content changes require code deploys; Contentful or Sanity would be the
upgrade path
 - No analytics or observability beyond Cloudflare's built-in dashboard
 - Images in public/ are 10–17MB raw JPGs — you'd use a CDN with format negotiation or
Cloudflare Images transforms in production at scale

 What to say if asked "what would you do differently?"
 ▎"I'd move song and show content into a headless CMS like Sanity so the artist can update
their own data without a code deploy. I'd add Playwright E2E tests for
  the contact form flow and route navigation. I'd also move the Web3Forms key into a Next.js
server action to keep it off the client bundle entirely."
```